



US 20250286036A1

(19) **United States**(12) **Patent Application Publication**
JANG et al.(10) **Pub. No.: US 2025/0286036 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **SEMICONDUCTOR PACKAGE HAVING
STACKED SEMICONDUCTOR CHIPS****H01L 23/48** (2006.01)**H01L 23/498** (2006.01)(71) Applicant: **SAMSUNG ELECTRONICS CO.,
LTD.**, Suwon-si (KR)(52) **U.S. Cl.**CPC **H01L 25/18** (2013.01); **H01L 23/481**
(2013.01); **H01L 23/49822** (2013.01); **H01L**
24/05 (2013.01); **H01L 24/08** (2013.01); **H01L**
2224/05647 (2013.01); **H01L 2224/08146**
(2013.01)(72) Inventors: **Aenee JANG**, Seoul (KR); **Seungduk**
BAEK, Hwaseong-si (KR)(73) Assignee: **SAMSUNG ELECTRONICS CO.,
LTD.**, Suwon-si (KR)

(57)

ABSTRACT(21) Appl. No.: **19/219,758**(22) Filed: **May 27, 2025****Related U.S. Application Data**(63) Continuation of application No. 17/819,231, filed on
Aug. 11, 2022, now Pat. No. 12,347,819.(30) **Foreign Application Priority Data**

Nov. 3, 2021 (KR) 10-2021-0149953

Publication Classification(51) **Int. Cl.****H01L 25/18** (2023.01)**H01L 23/00** (2006.01)

A semiconductor package includes a first semiconductor chip including a first semiconductor substrate having an active surface and an inactive surface and a plurality of first through electrodes passing through the first semiconductor substrate. A plurality of second semiconductor chips each includes a second semiconductor substrate having an active surface and an inactive surface. A plurality of second through electrodes passes through the second semiconductor substrate. A plurality of coupling pads is disposed between the first semiconductor chip and the second semiconductor chips. A plurality of chip coupling insulation layers is disposed between the first semiconductor chip and the second semiconductor chips and at least partially surrounds the coupling pads. At least one supporting dummy substrate is stacked on the second semiconductor chips. At least one supporting coupling insulation layer is disposed on a bottom surface of the at least one supporting dummy substrate.

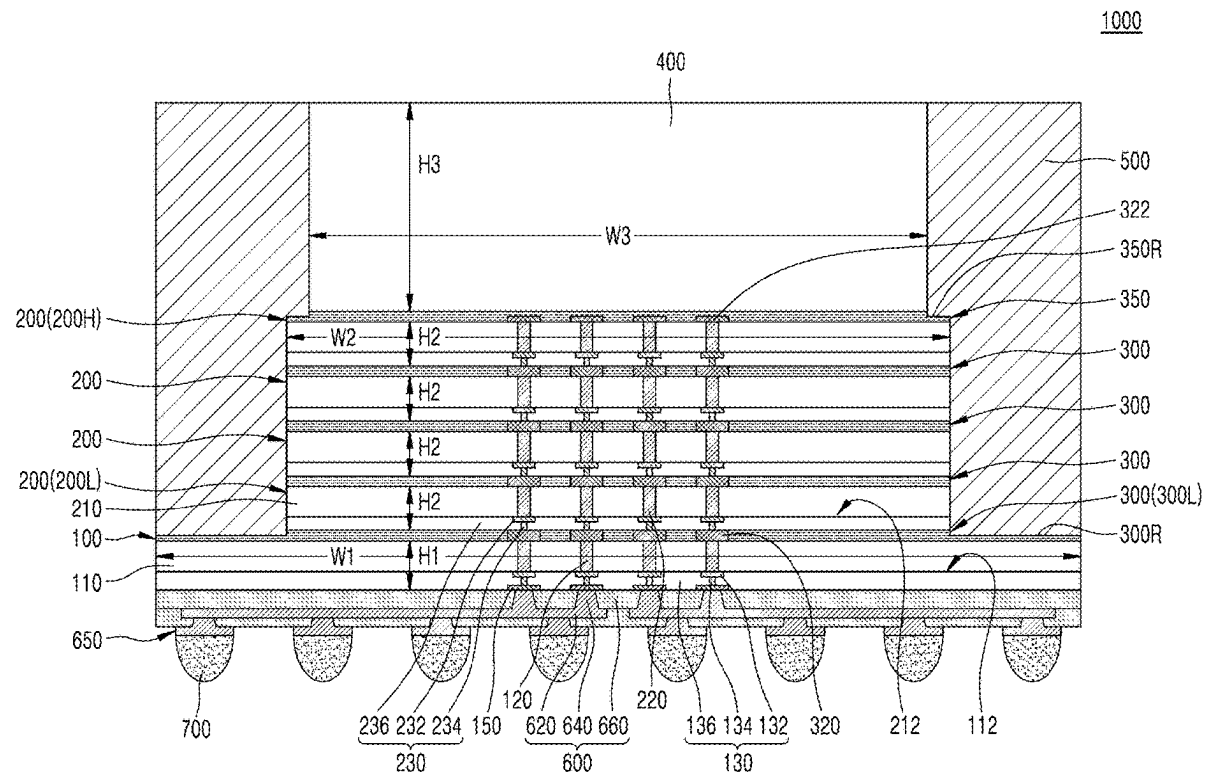


FIG. 1A

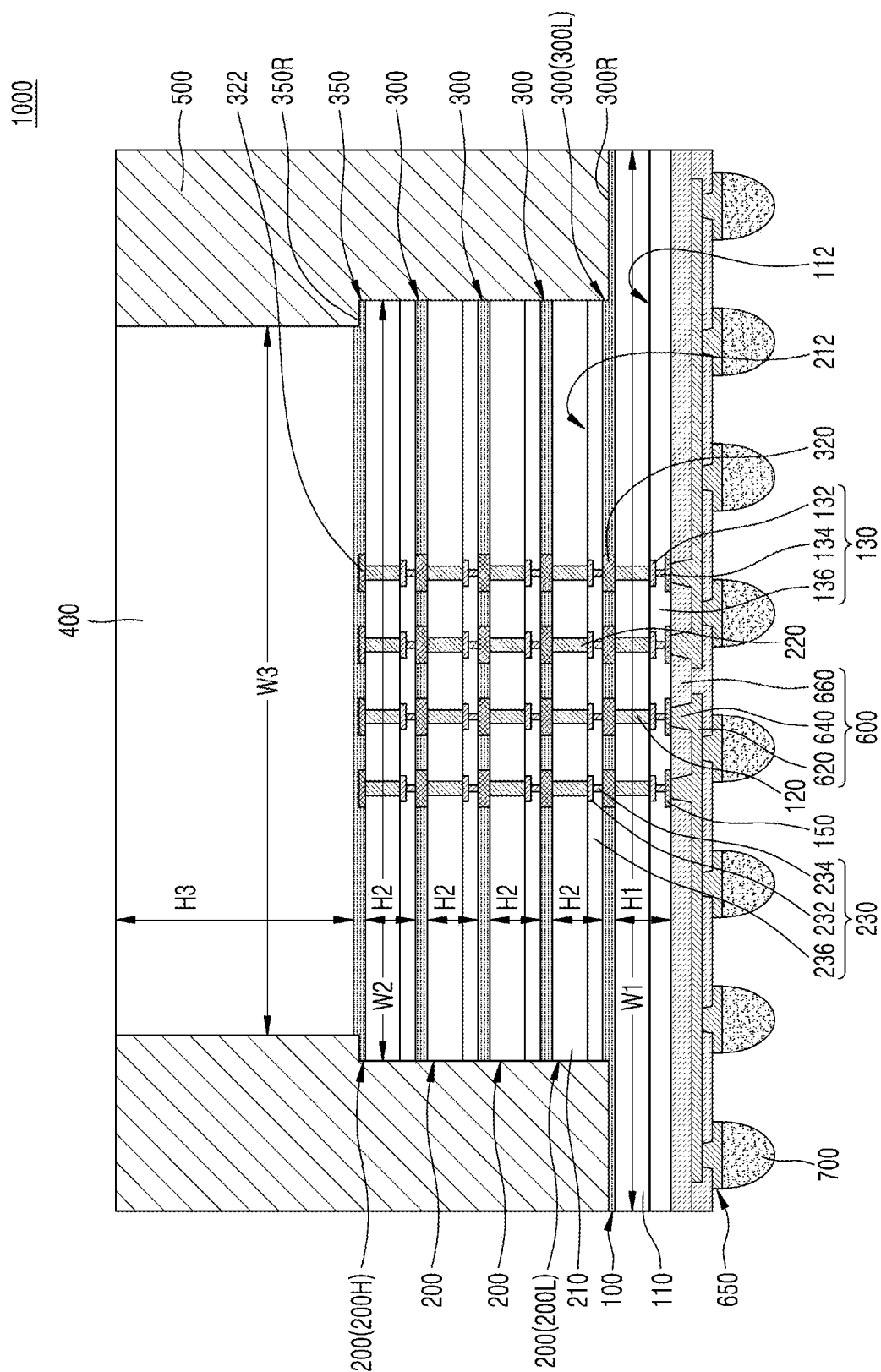


FIG. 1B.

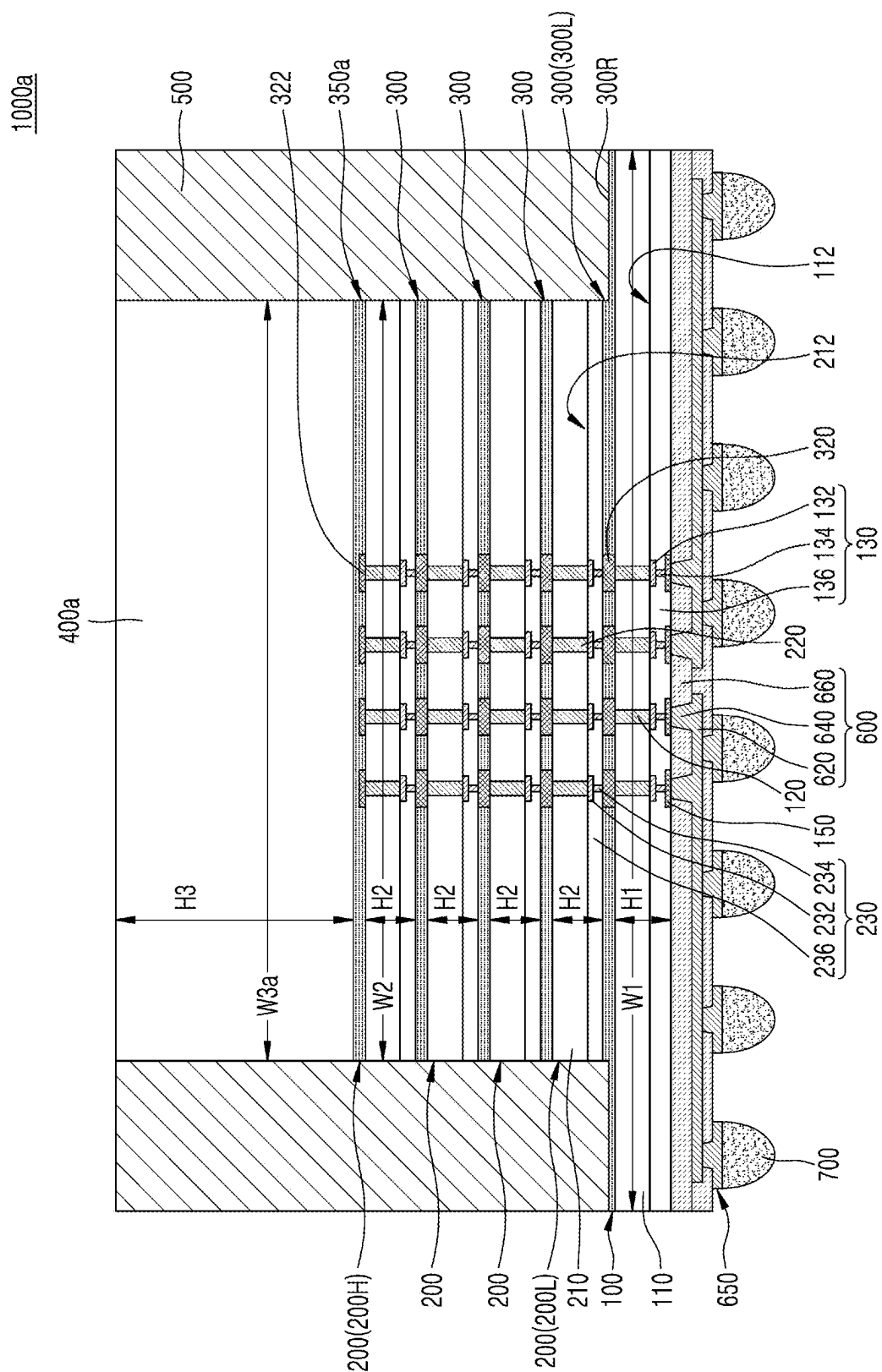


FIG. 1C

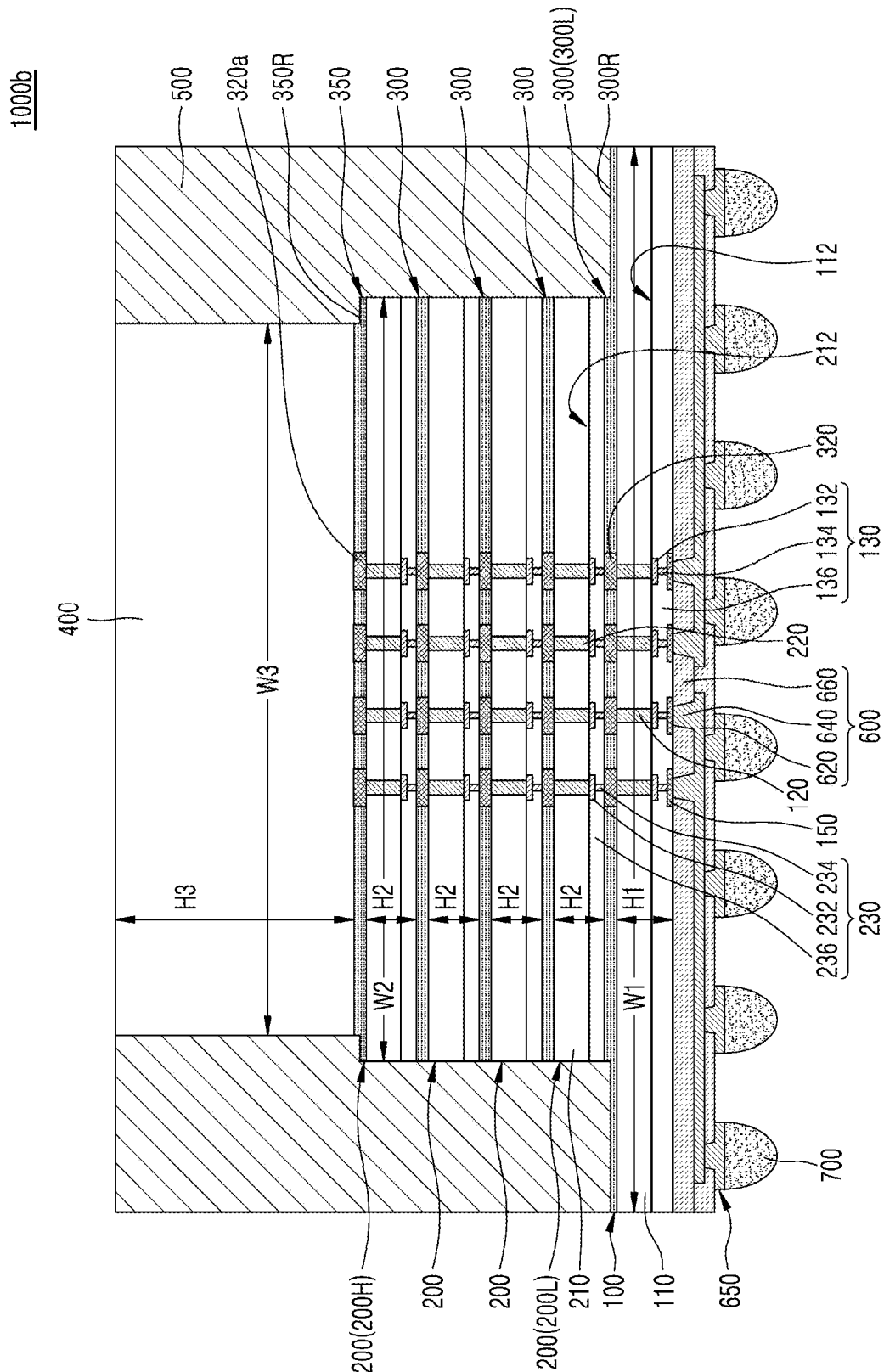


FIG. 1D

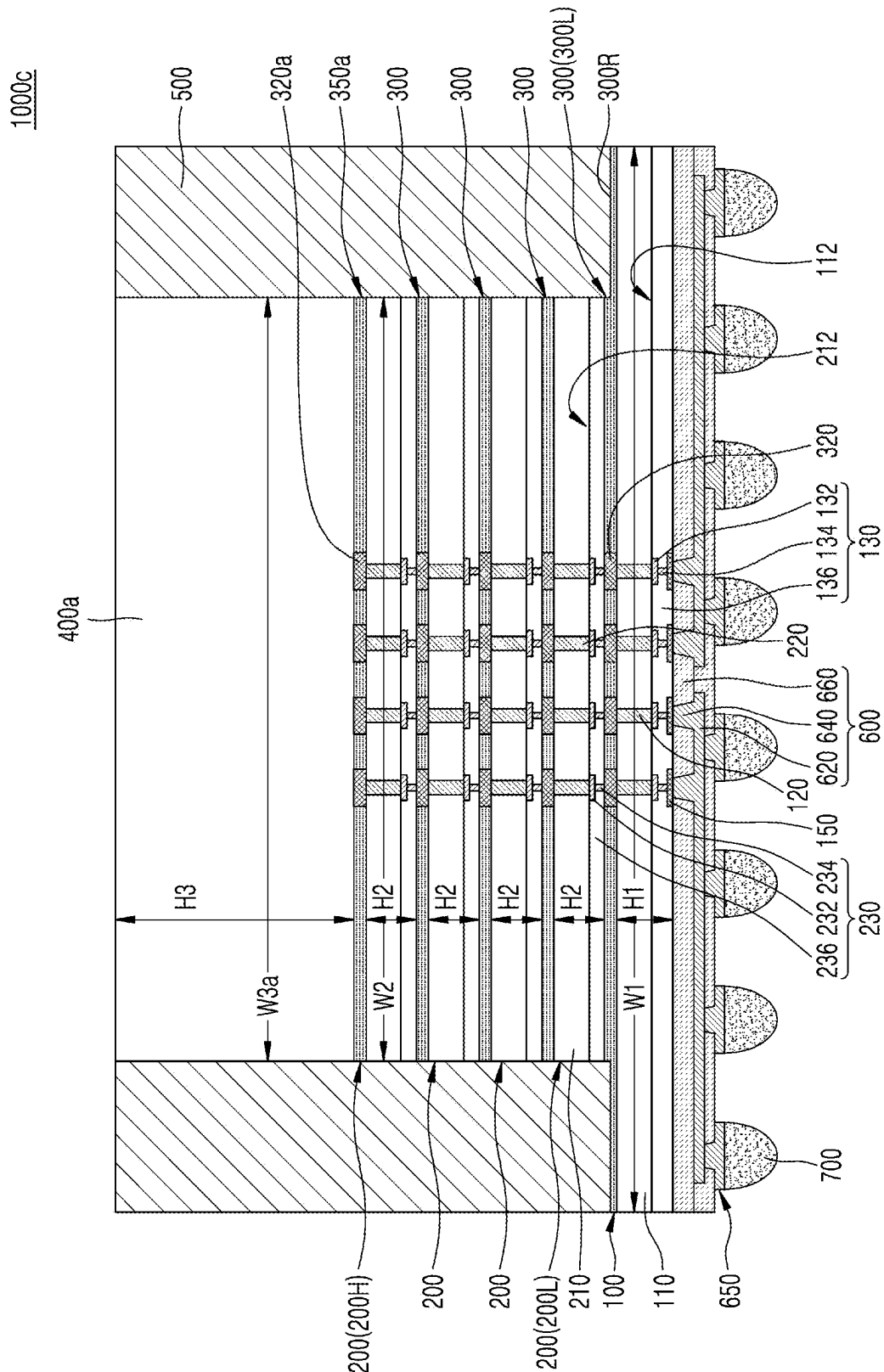
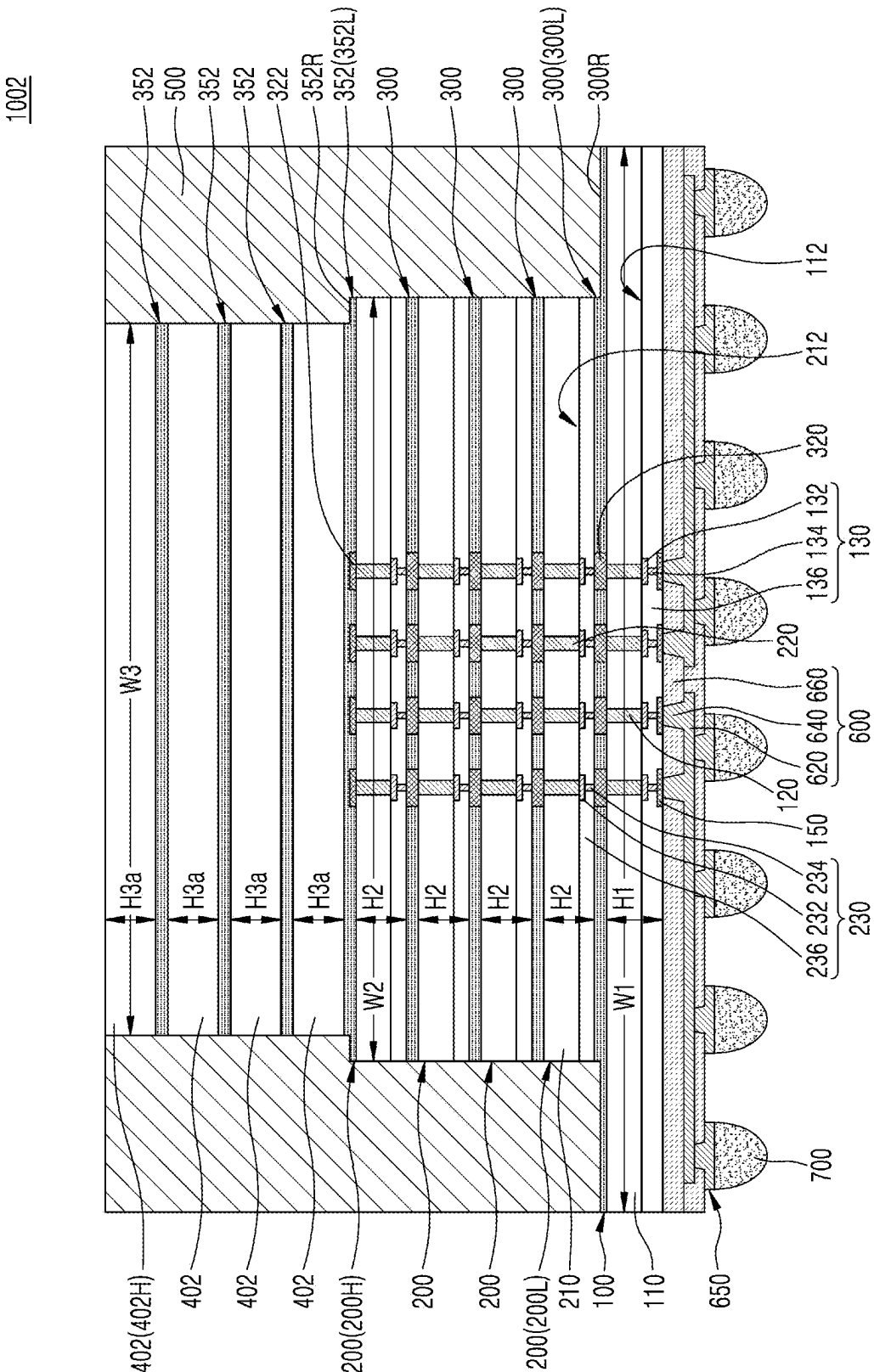


FIG. 2A



1002

FIG. 2B

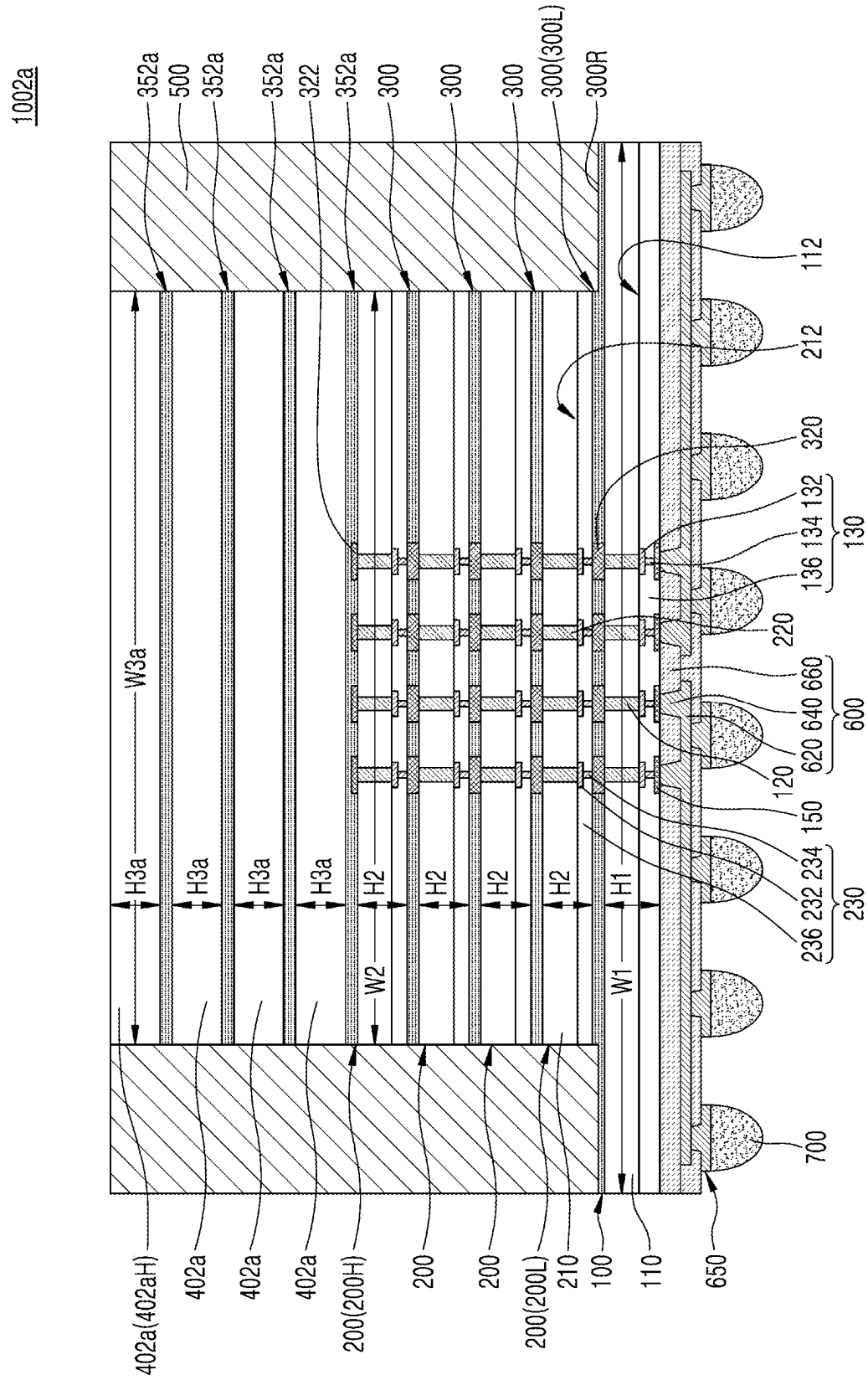


FIG. 2C

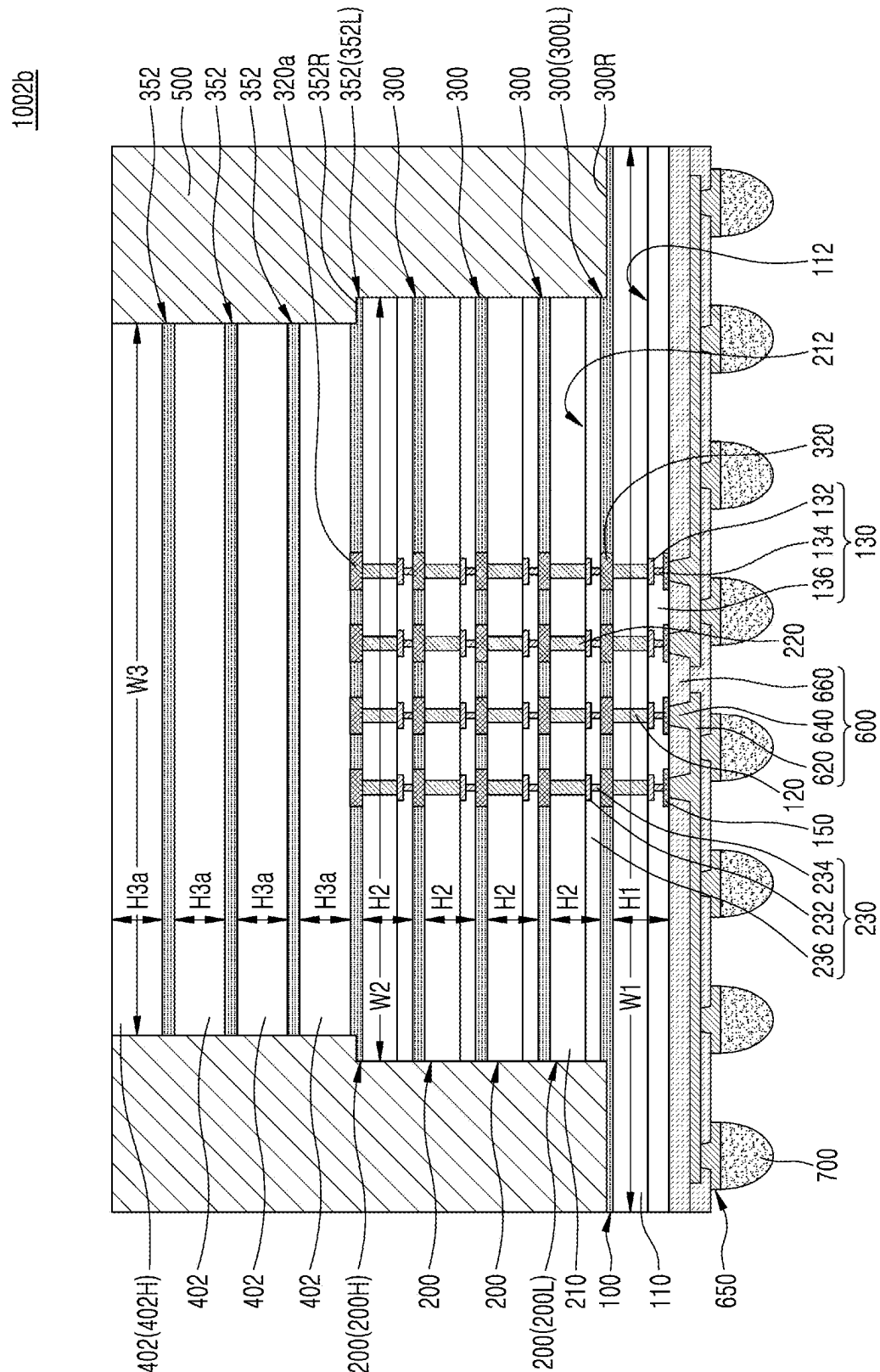


FIG. 2D

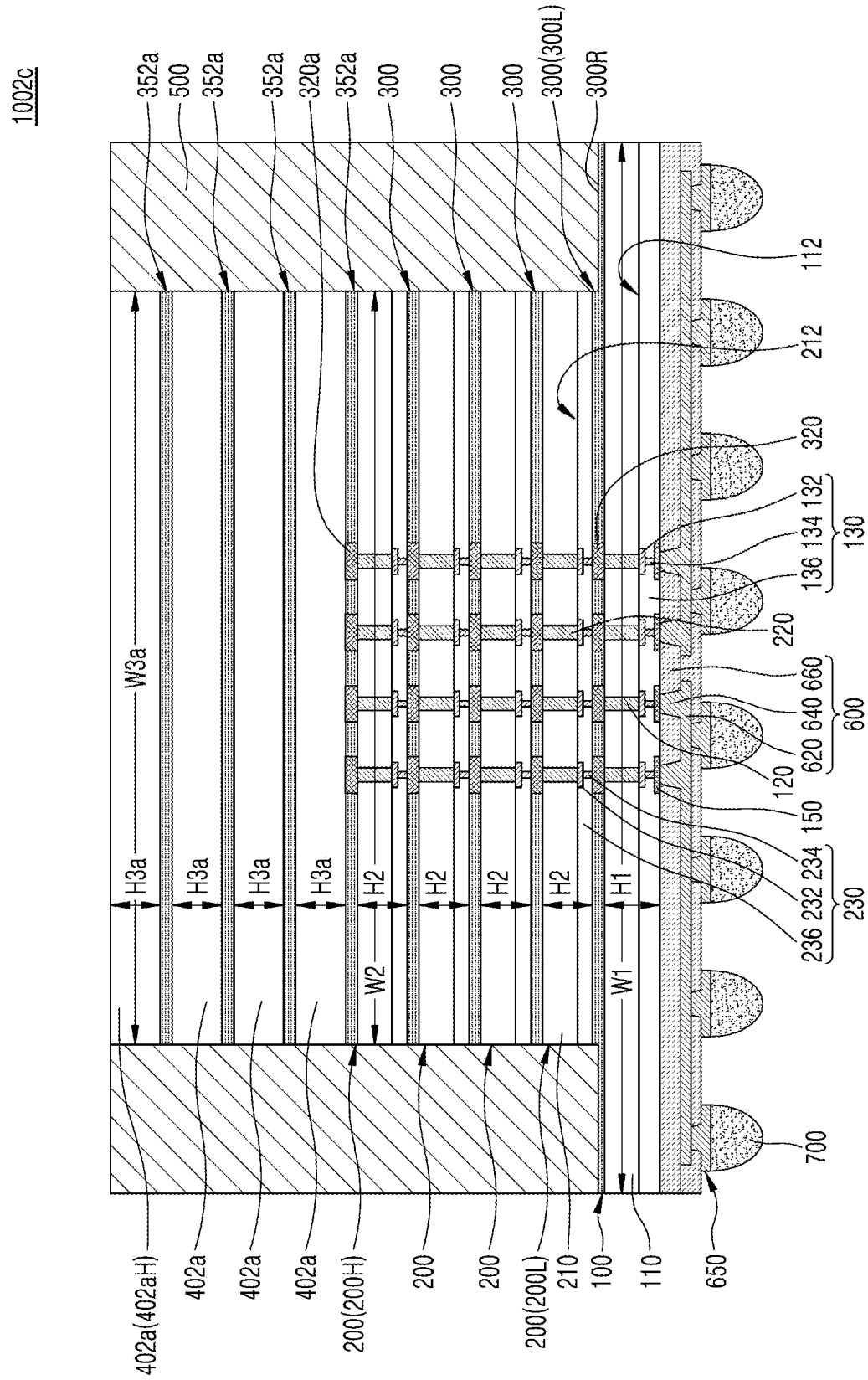


FIG. 2E

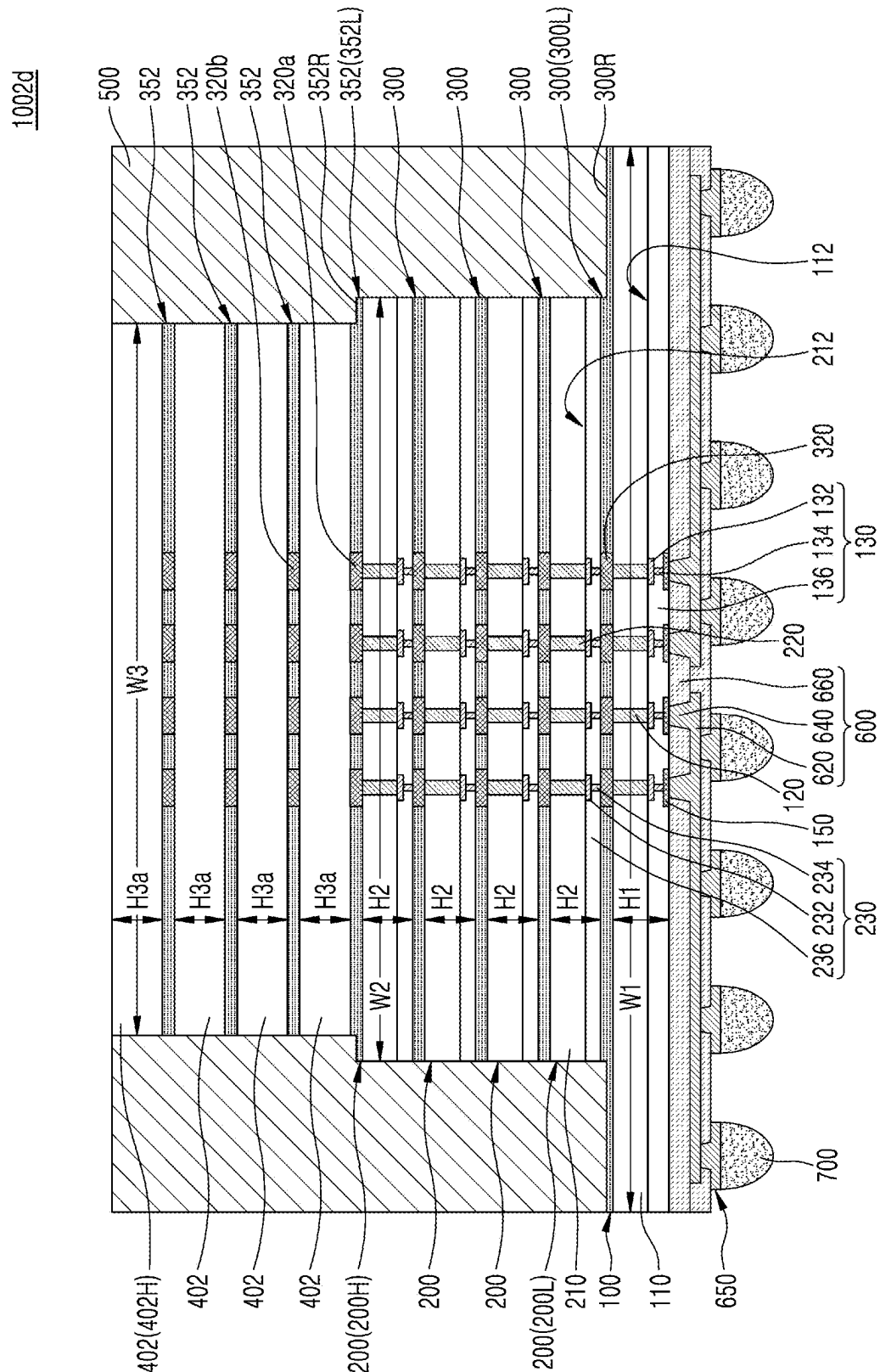


FIG. 2F

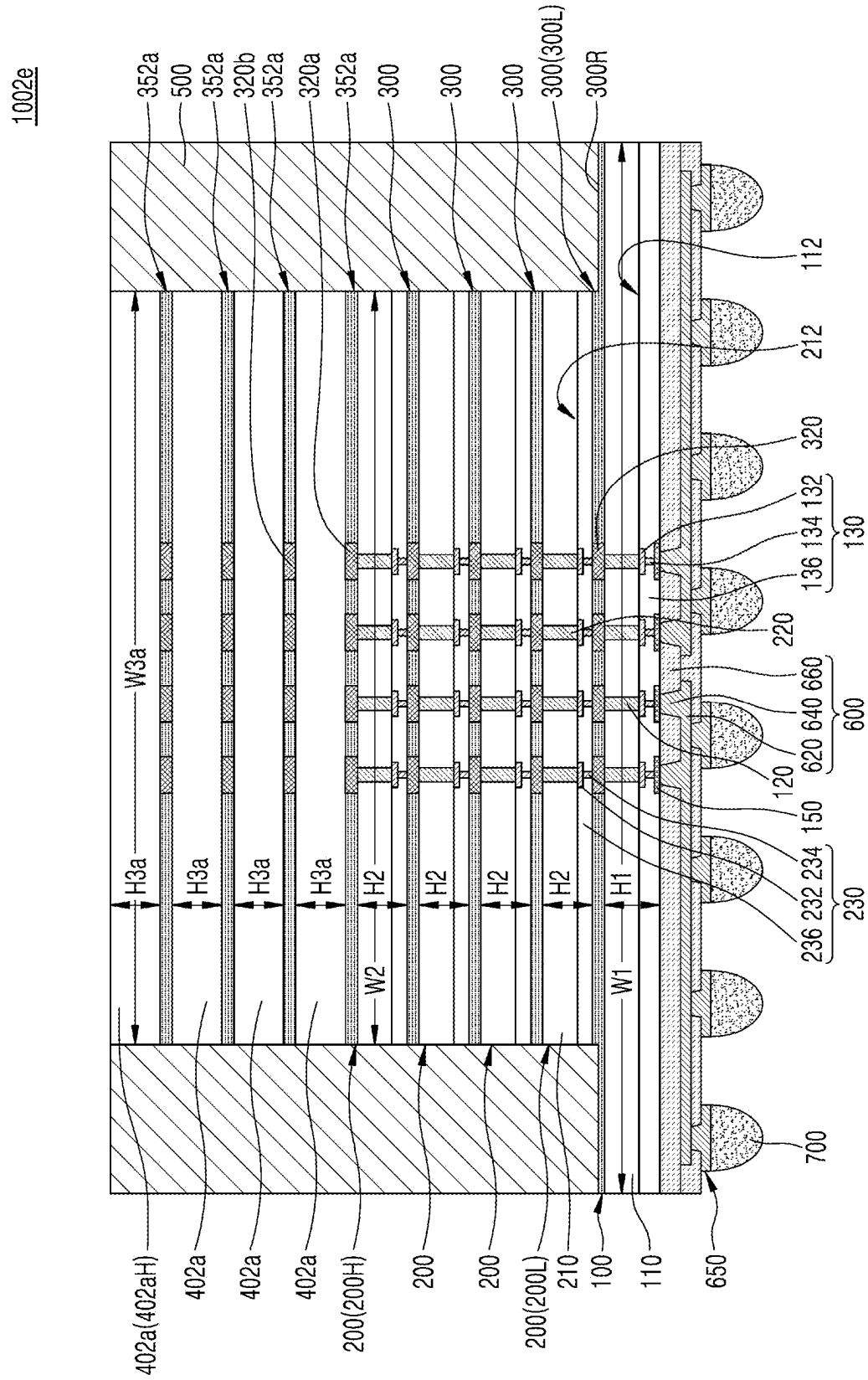


FIG. 2H

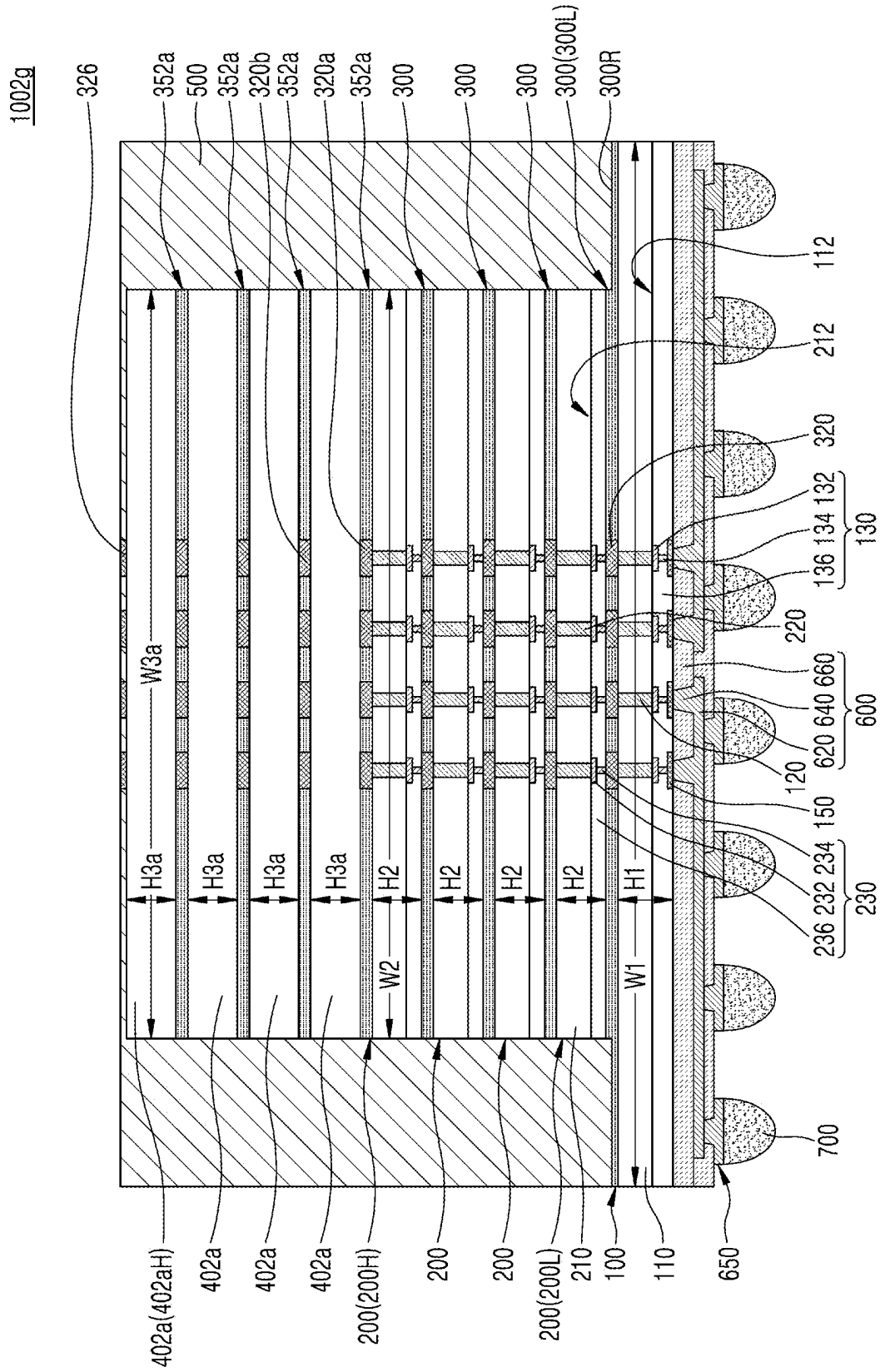


FIG. 3A

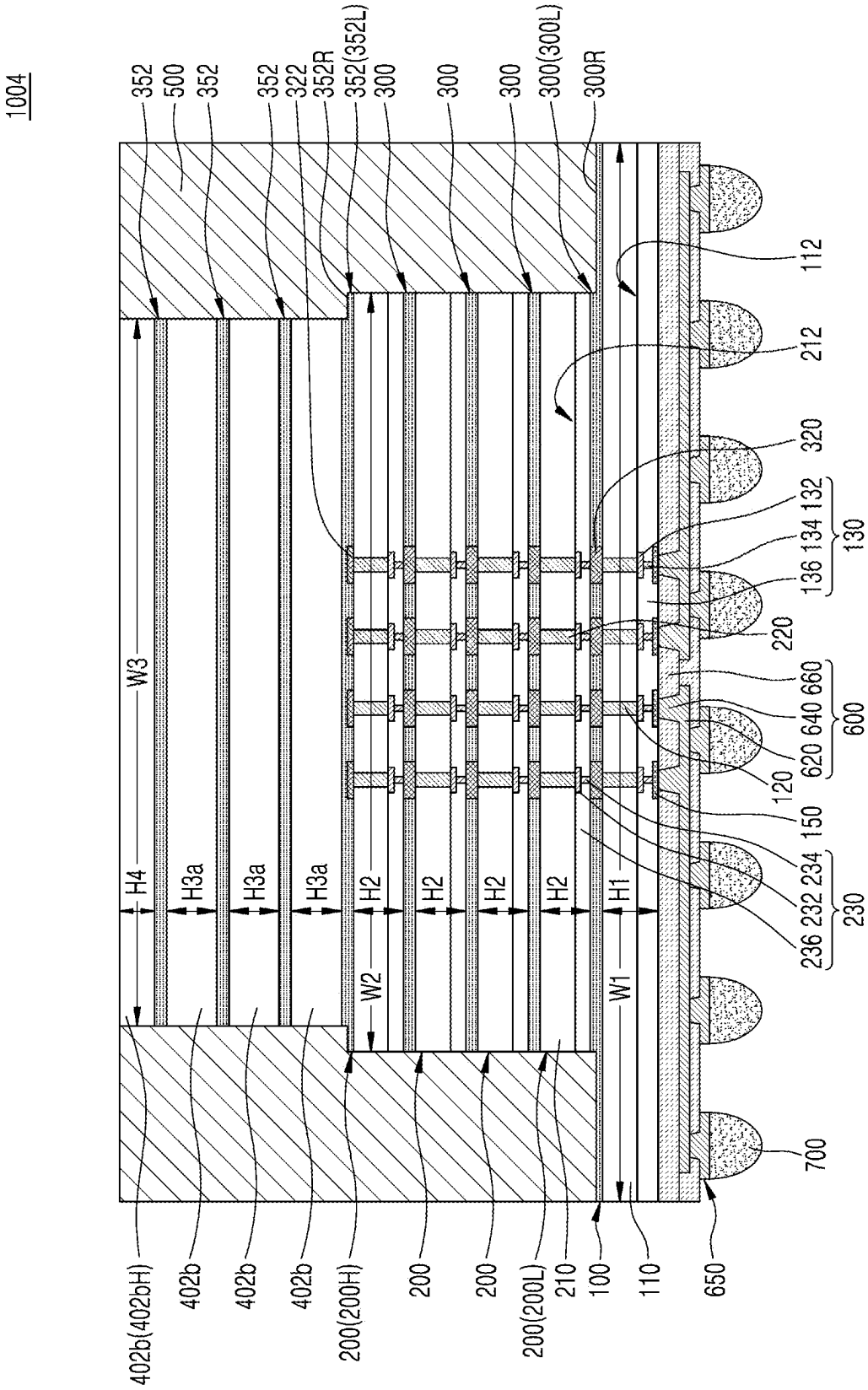


FIG. 3B

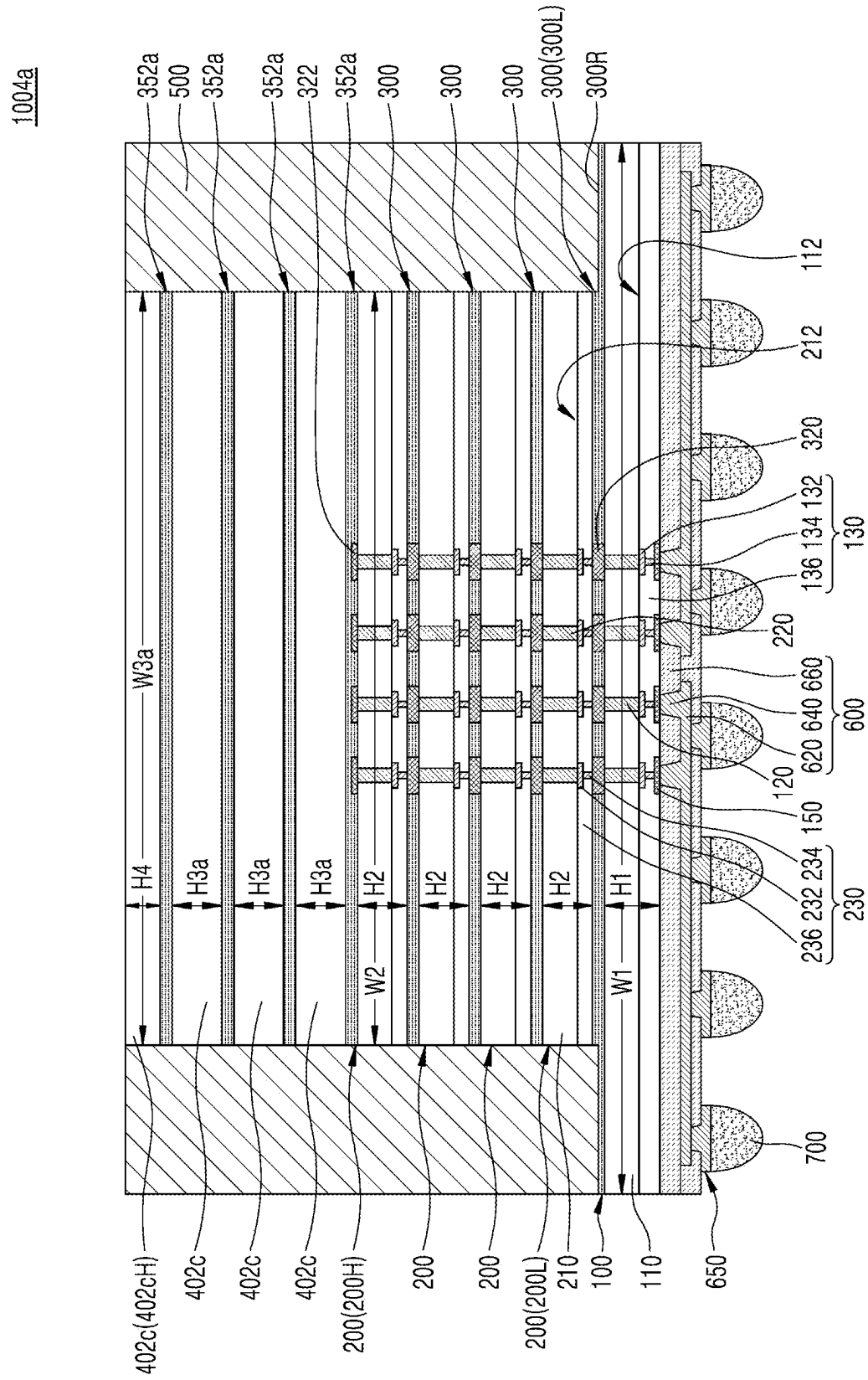


FIG. 3C

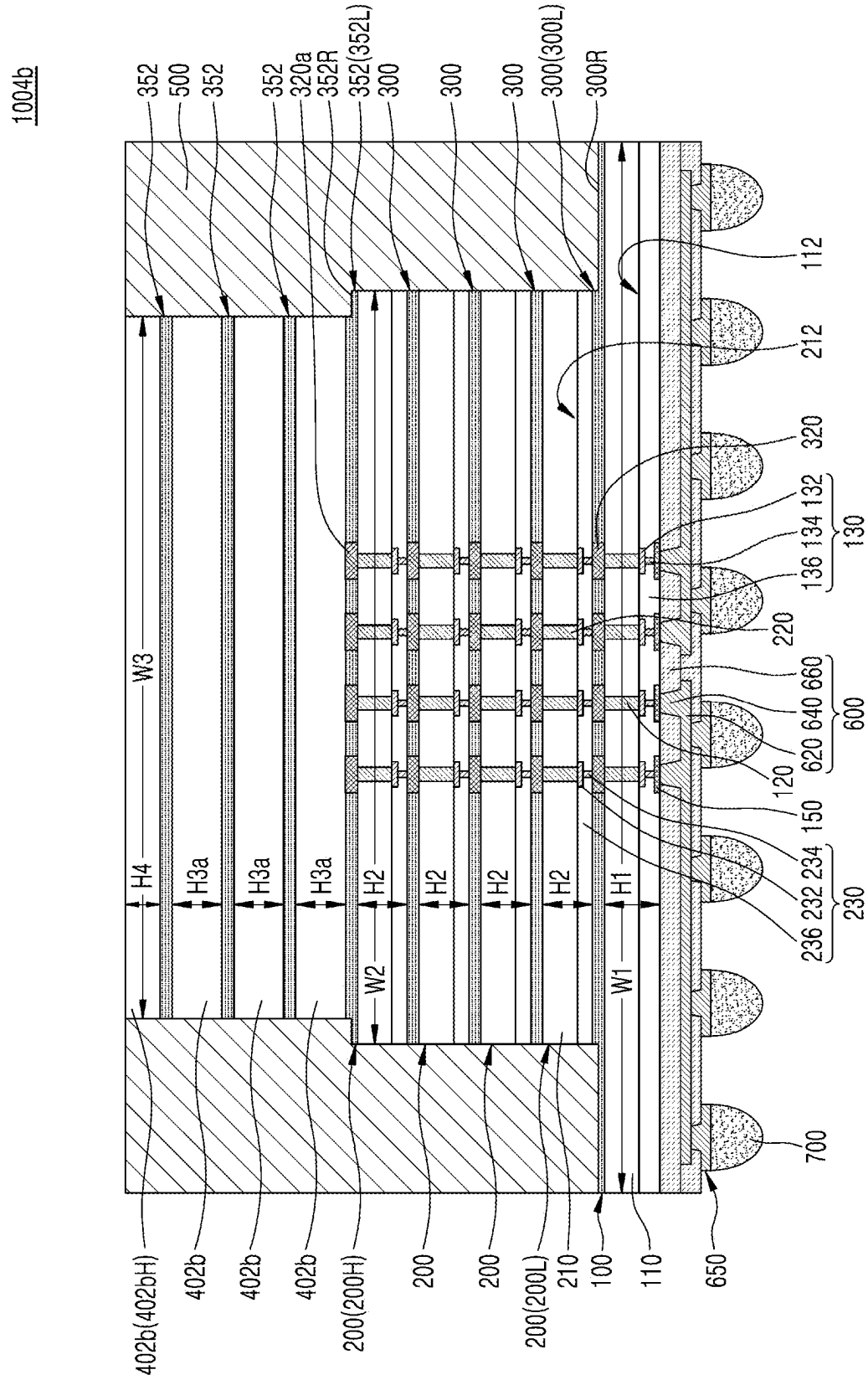


FIG. 3D

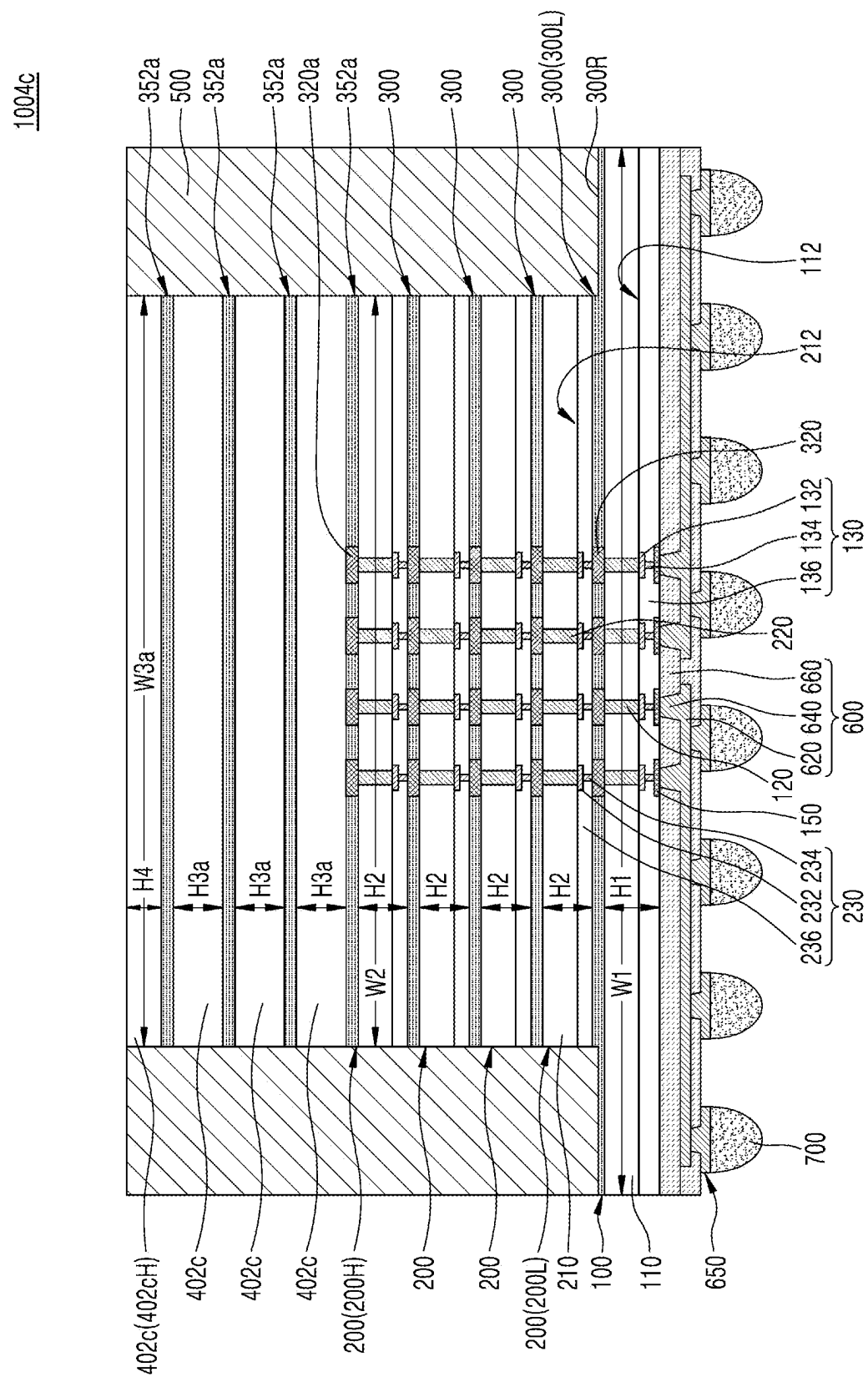


FIG. 3E

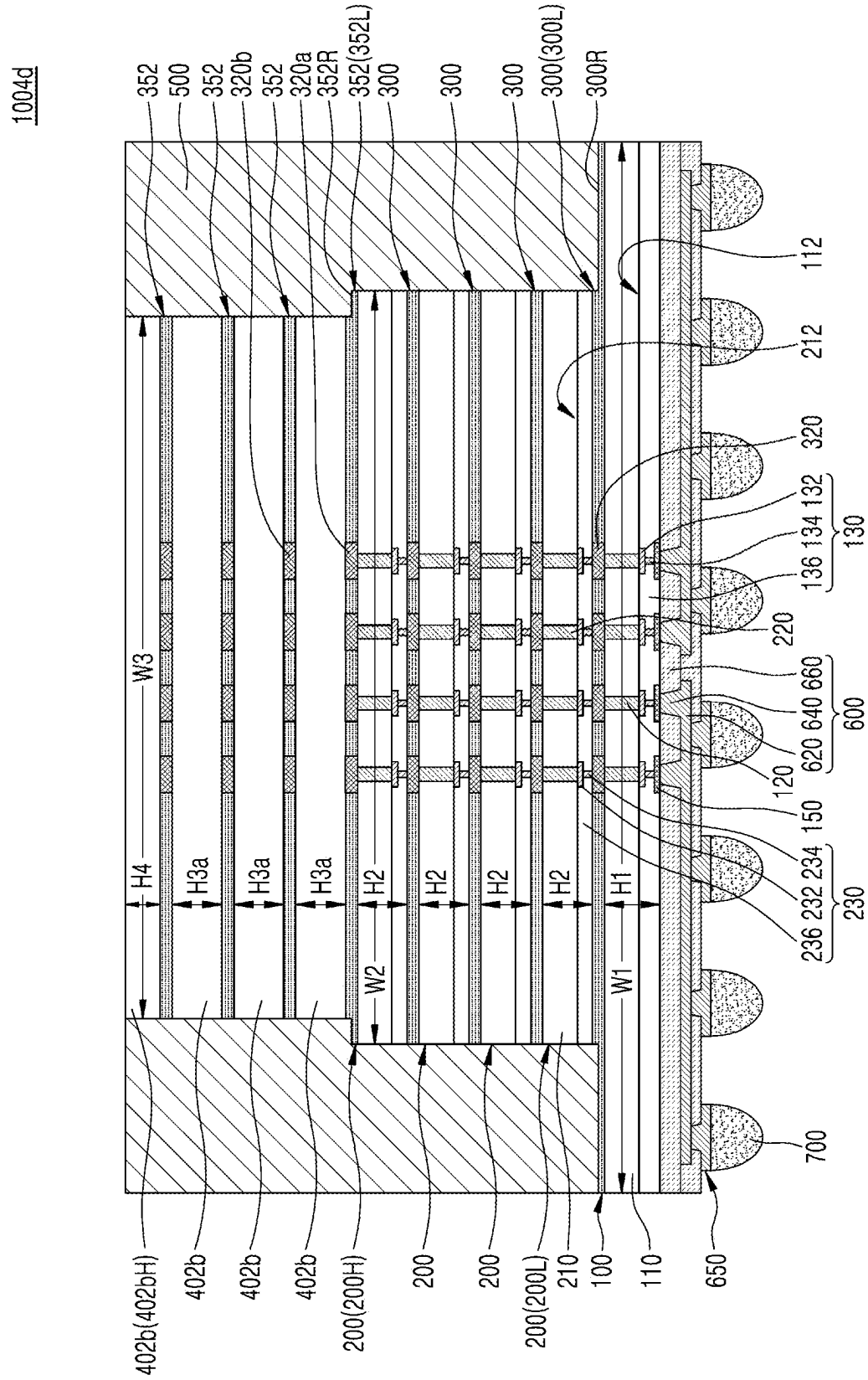


FIG. 3F

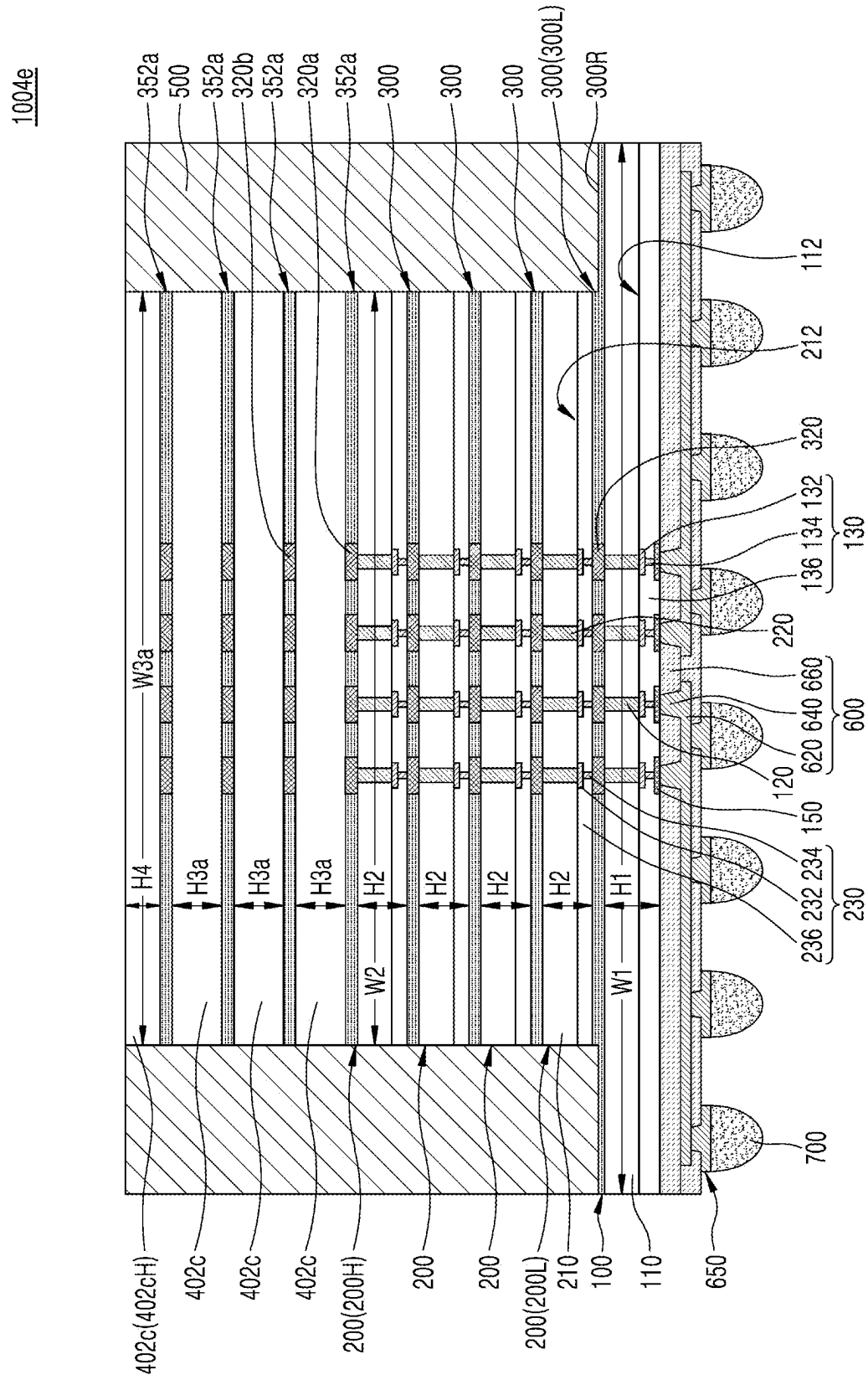


FIG. 4A

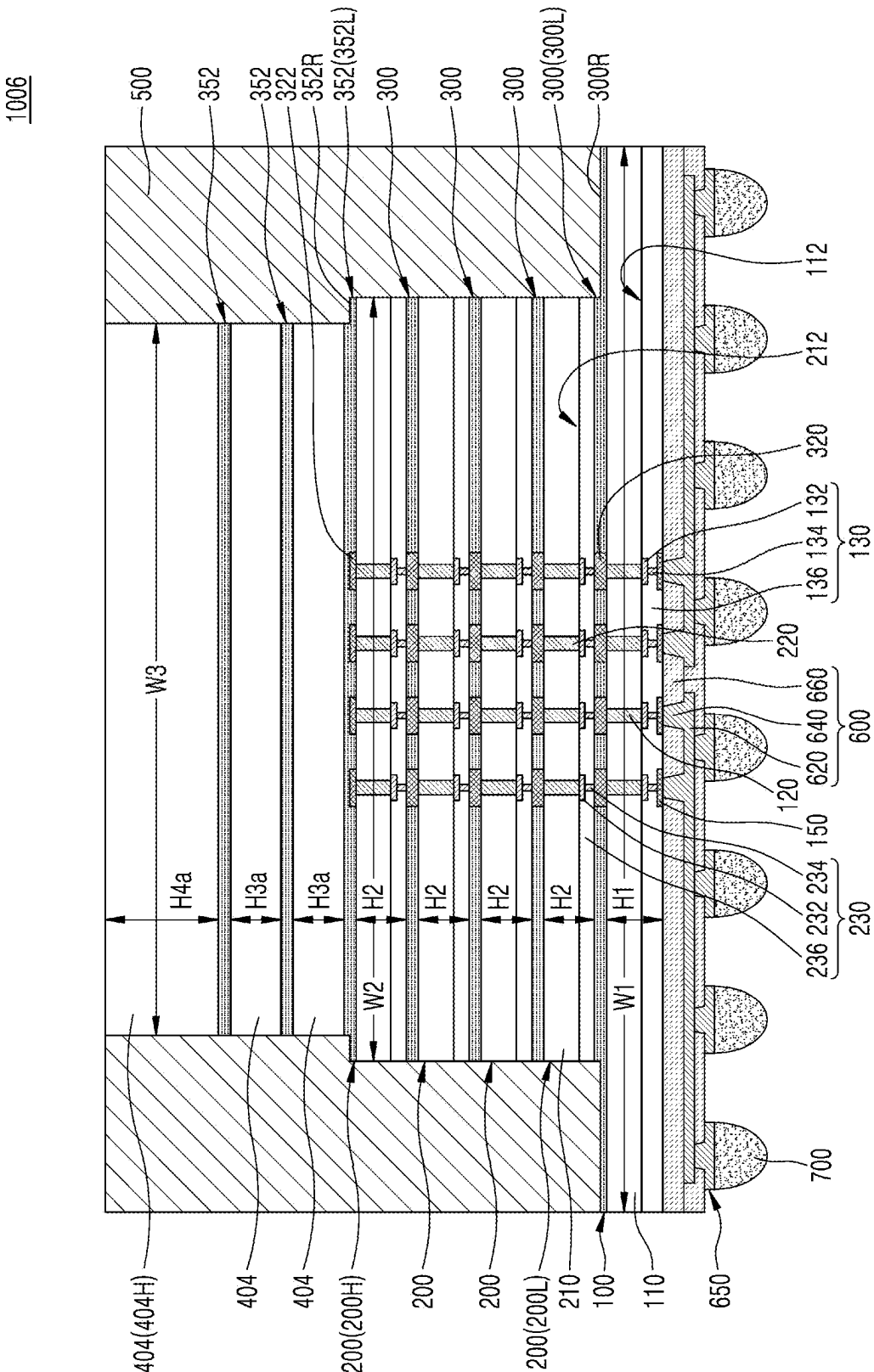


FIG. 4B

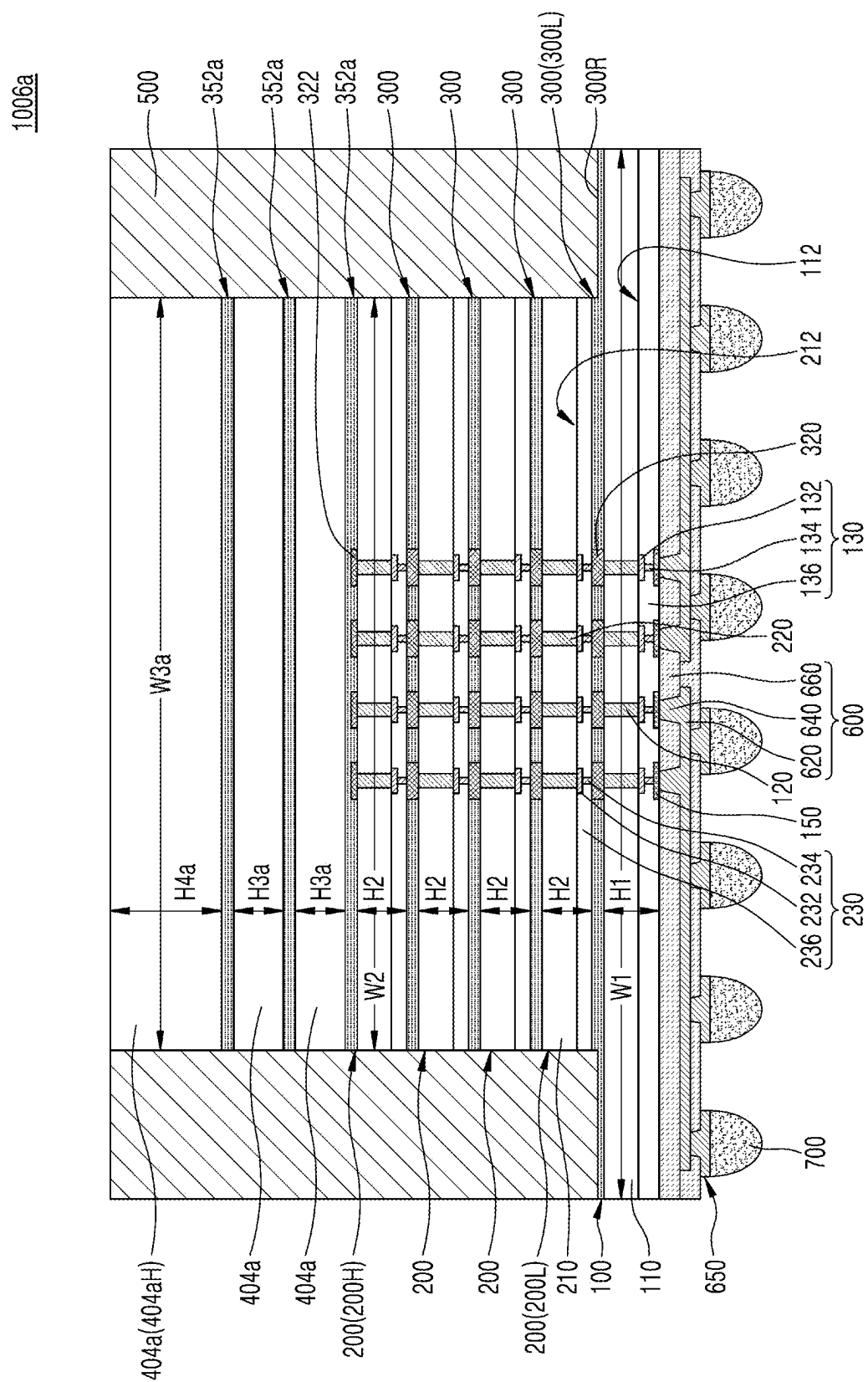


FIG. 4C

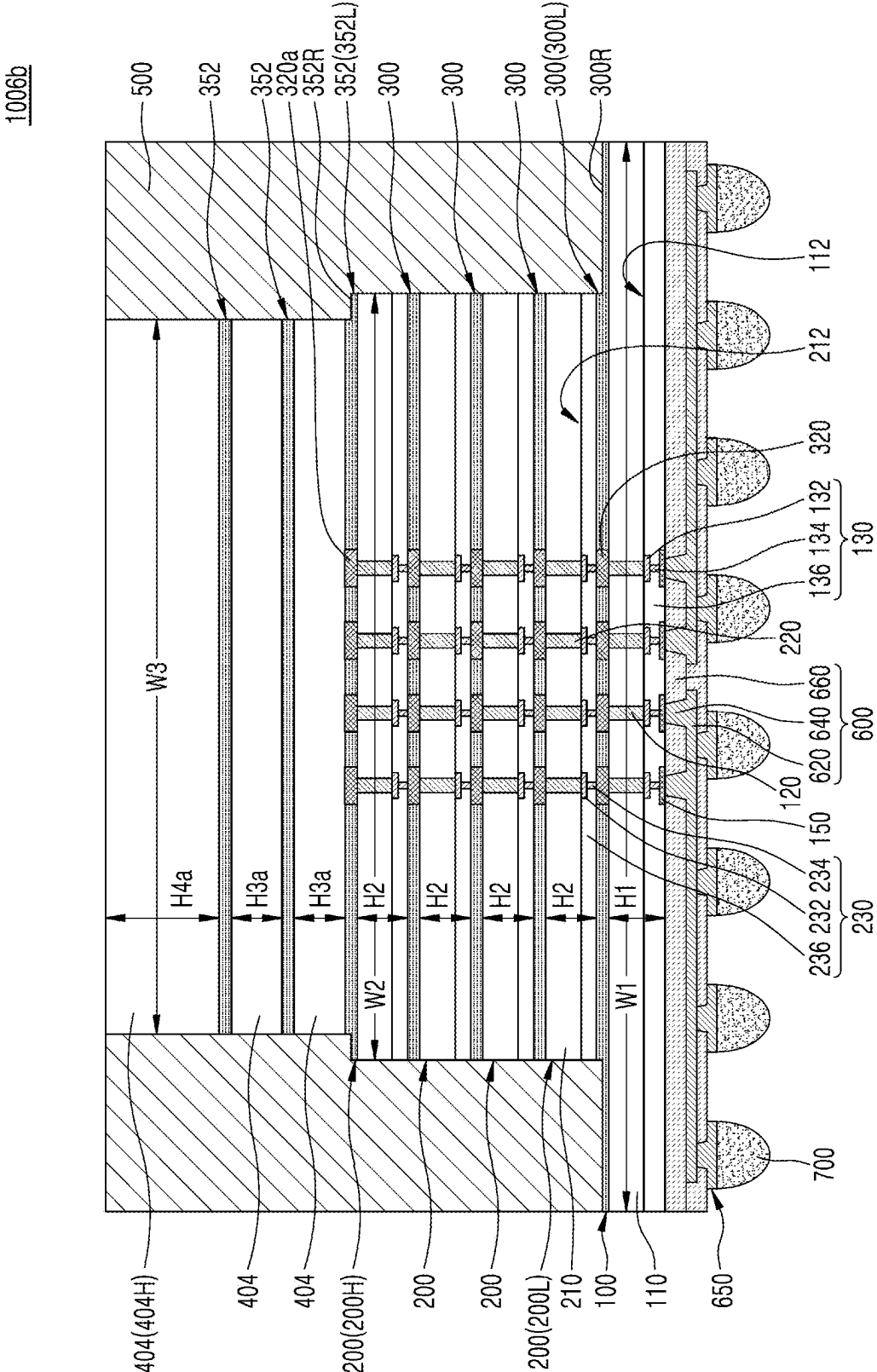


FIG. 4D

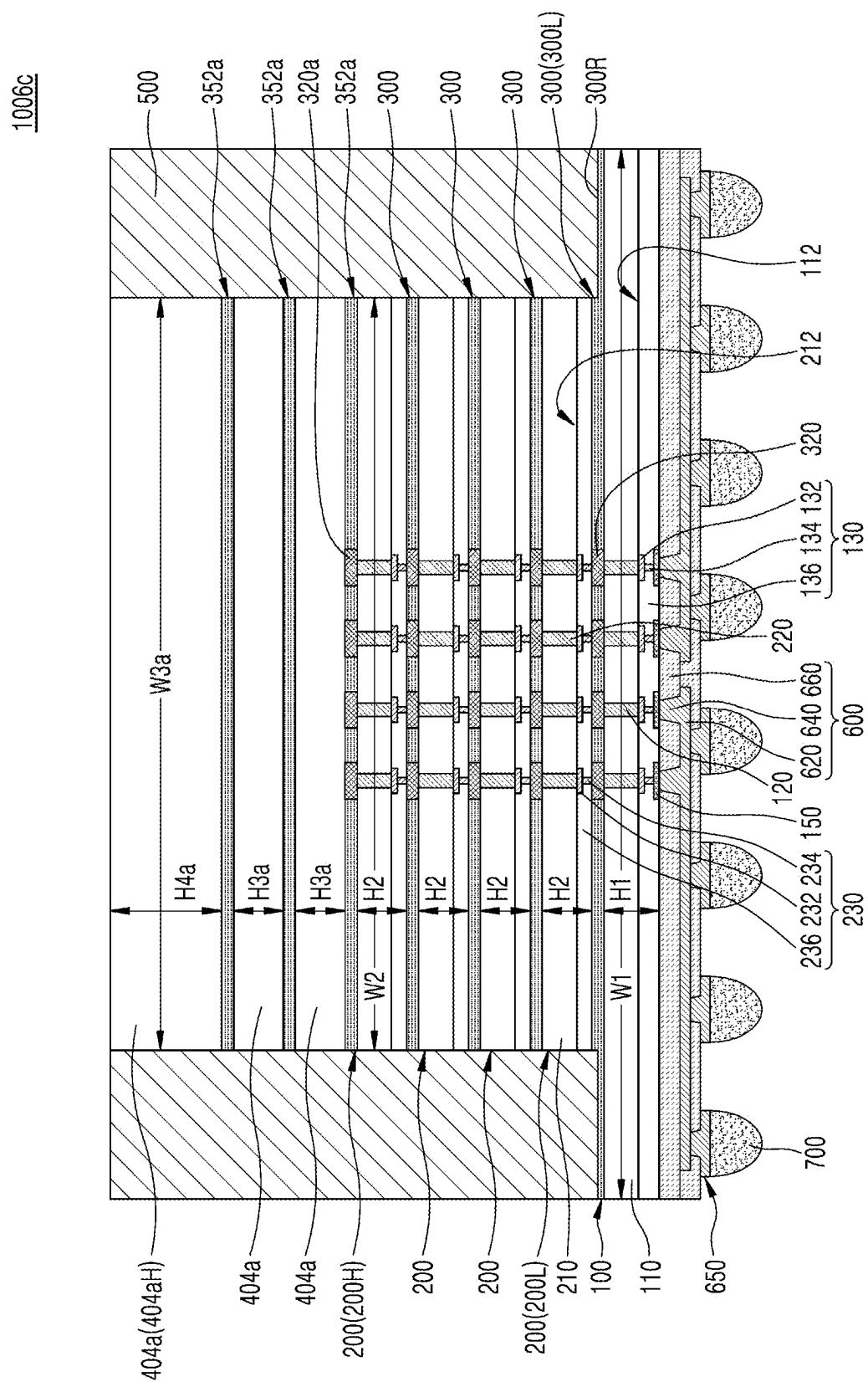


FIG. 4E

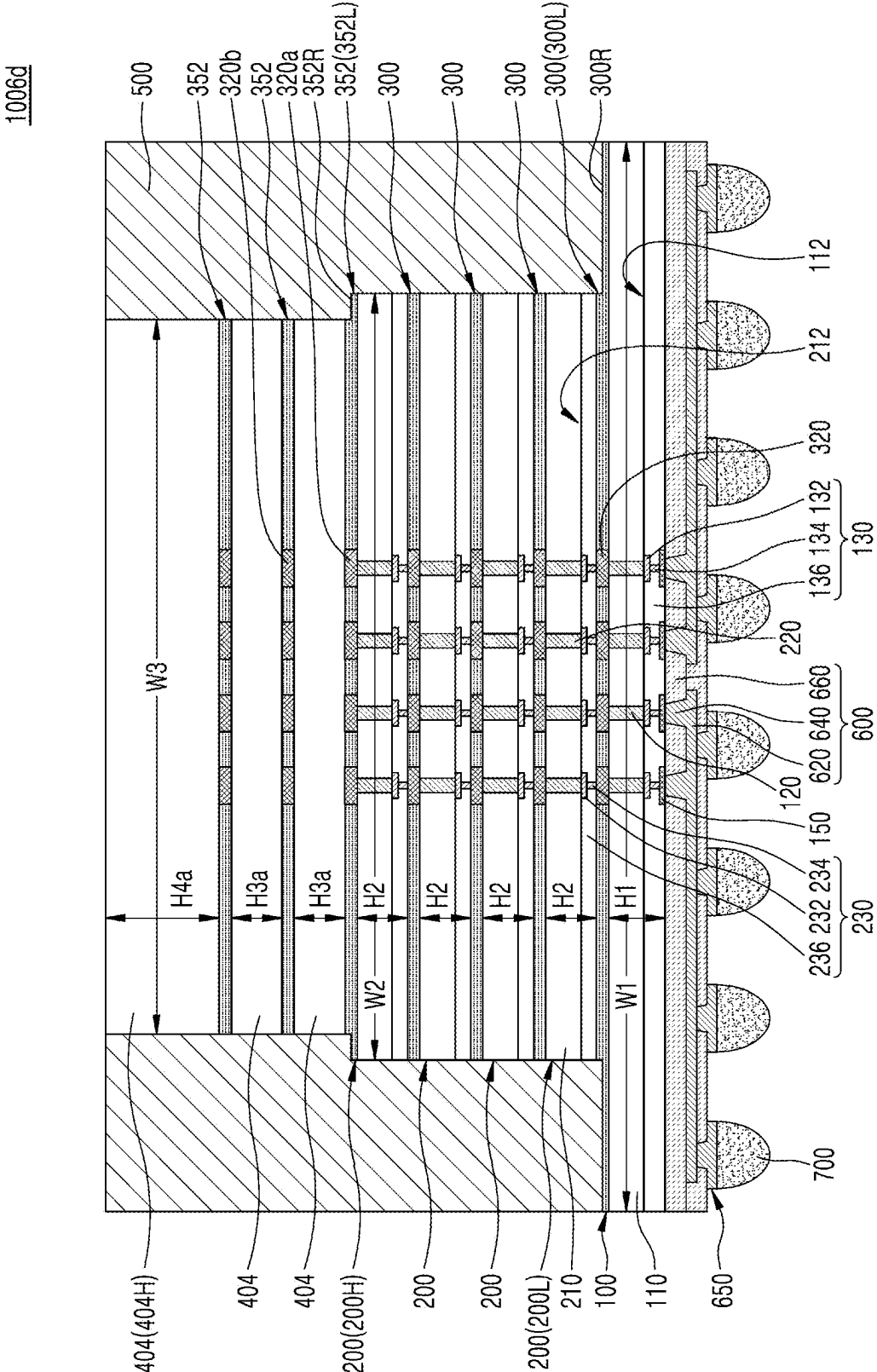


FIG. 4F

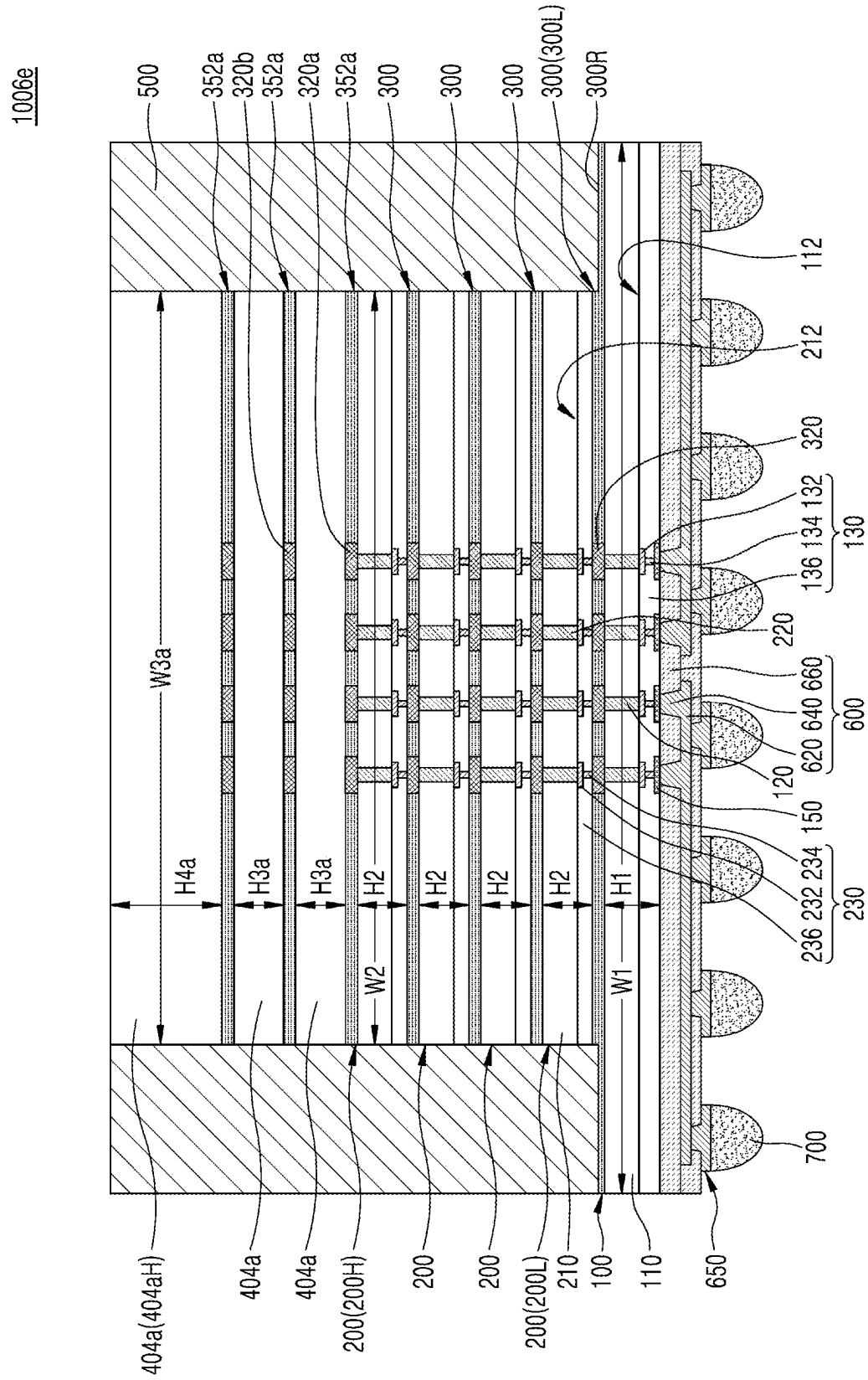


FIG. 4G

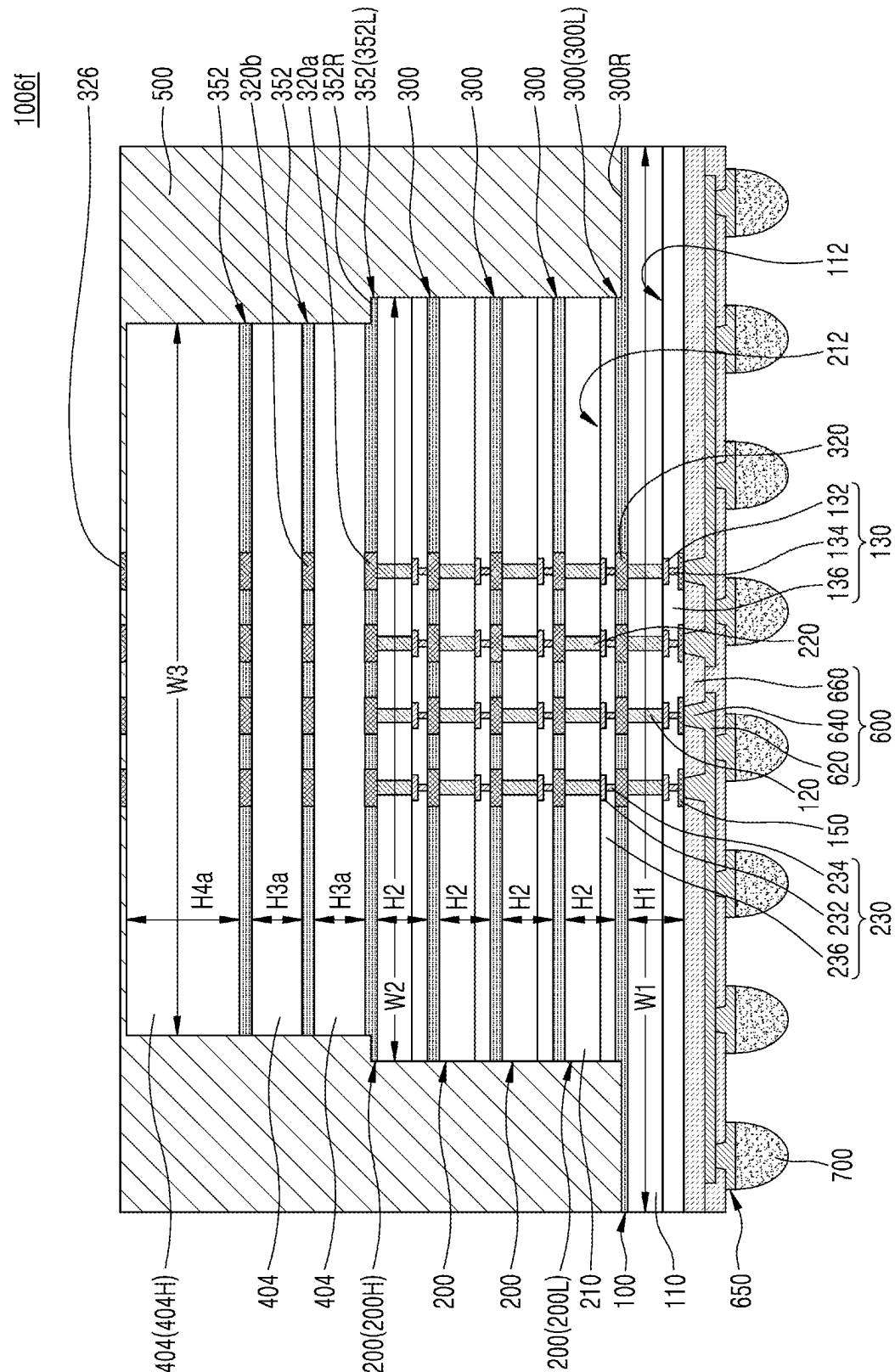


FIG. 4H

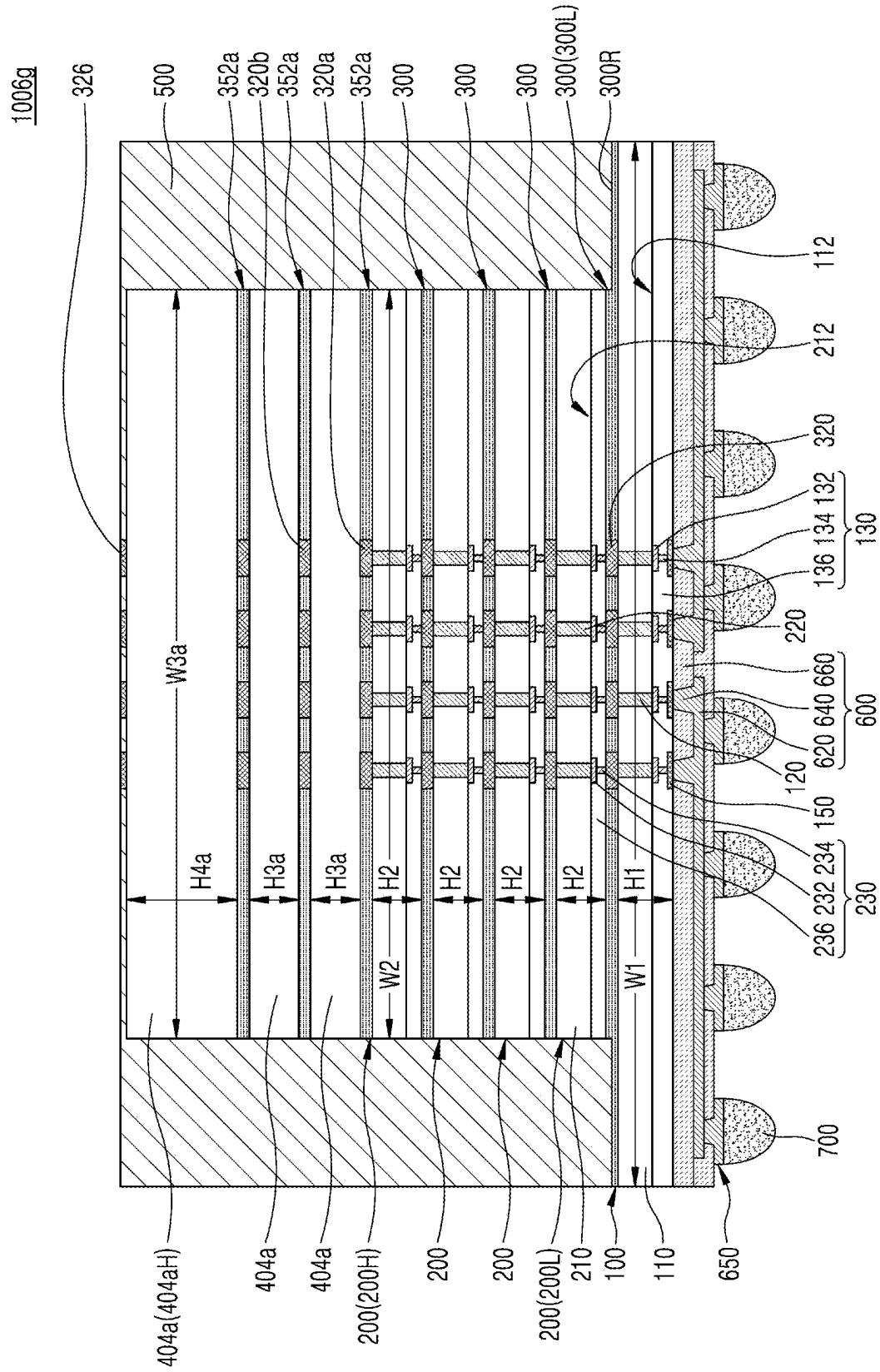


FIG. 5A

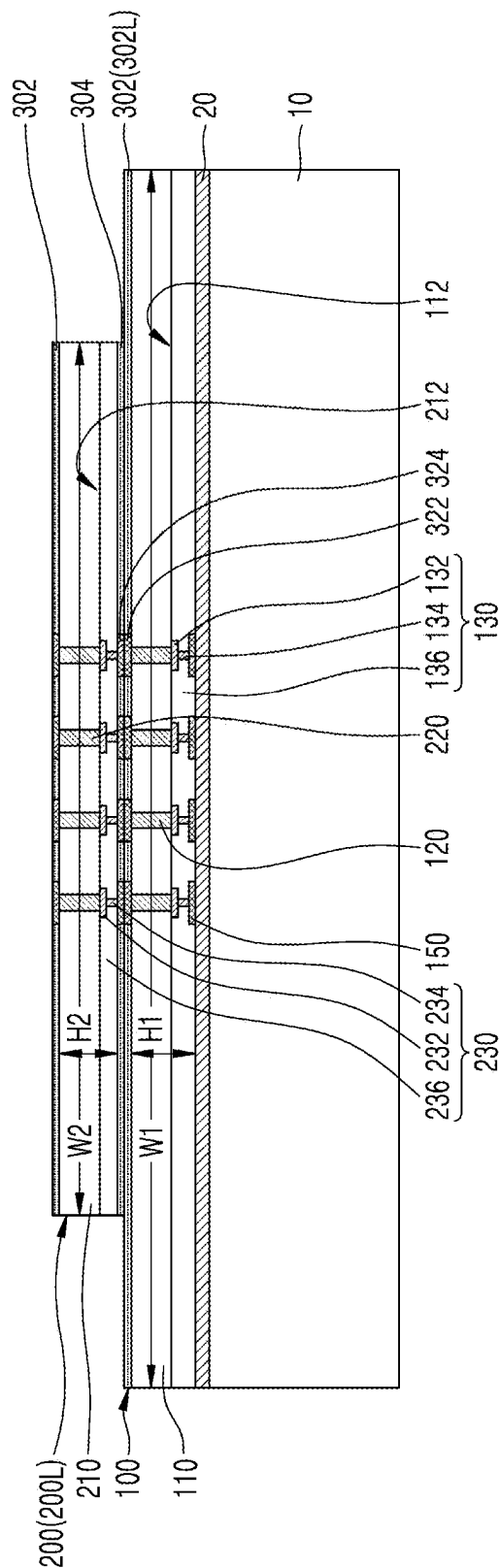


FIG. 5B

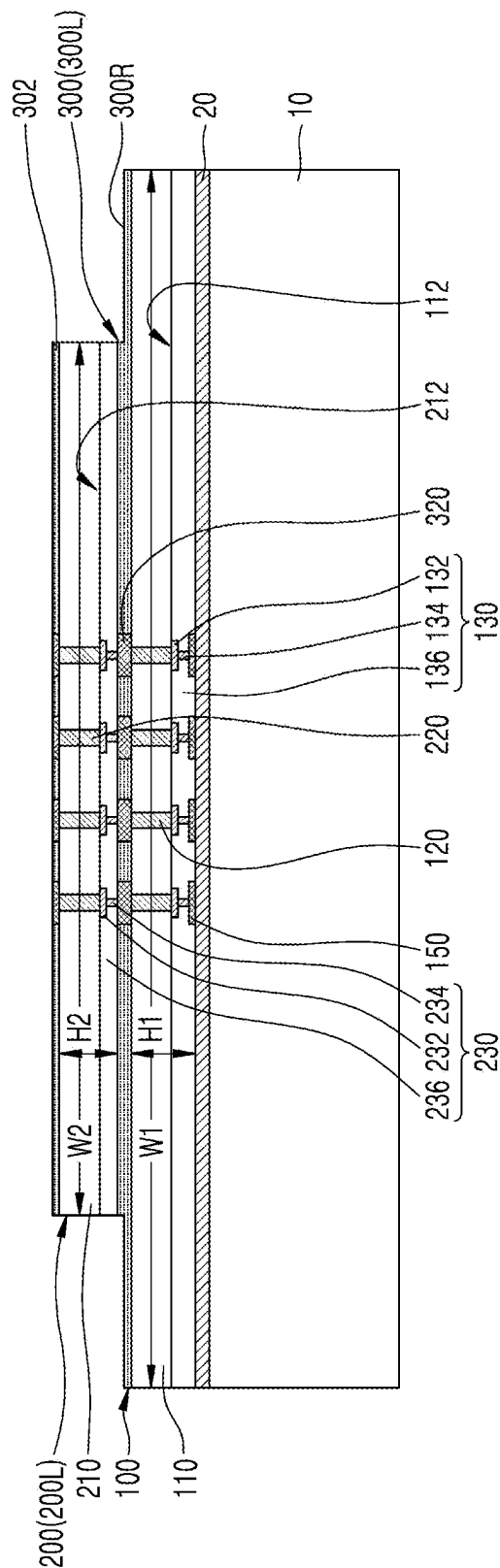


FIG. 5C

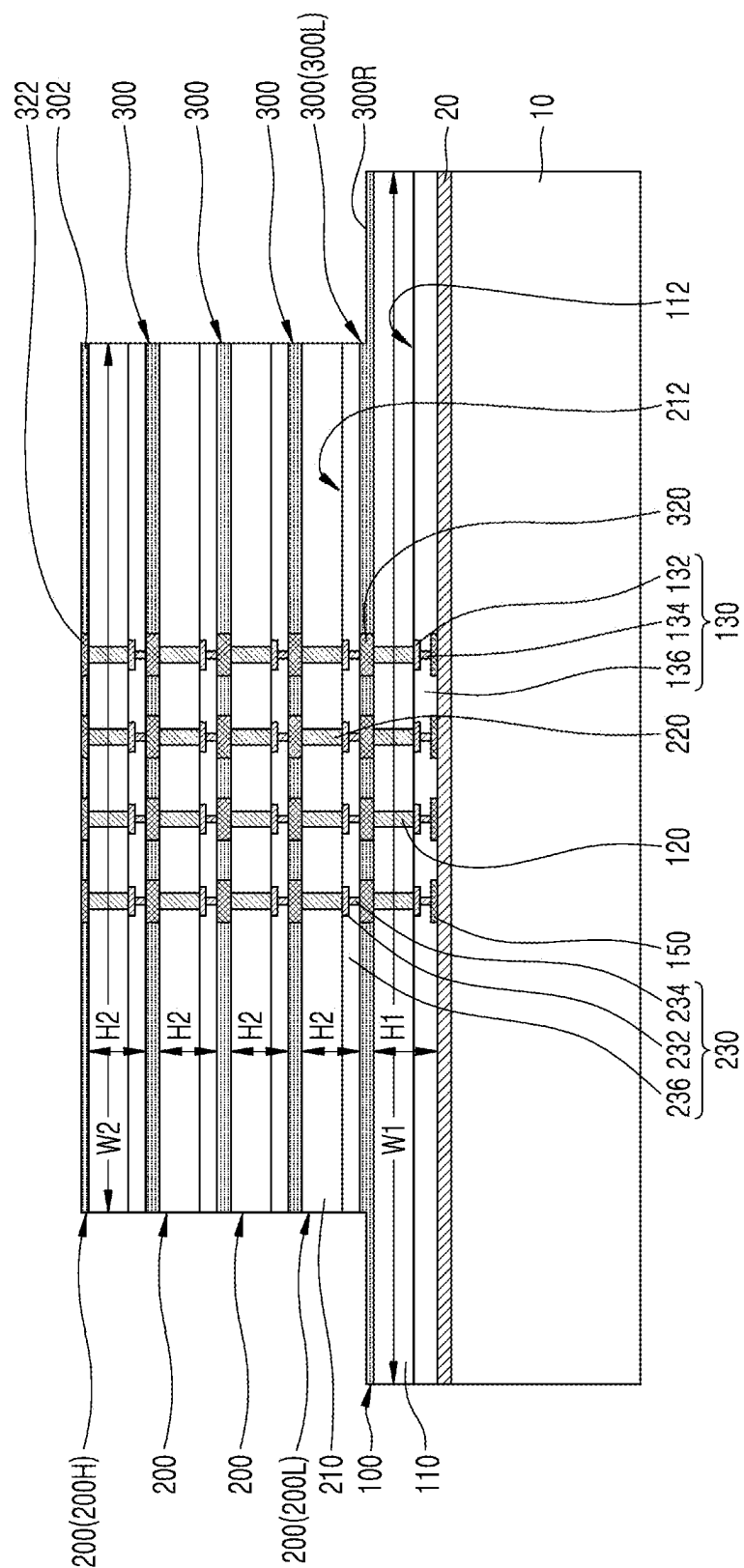


FIG. 5D

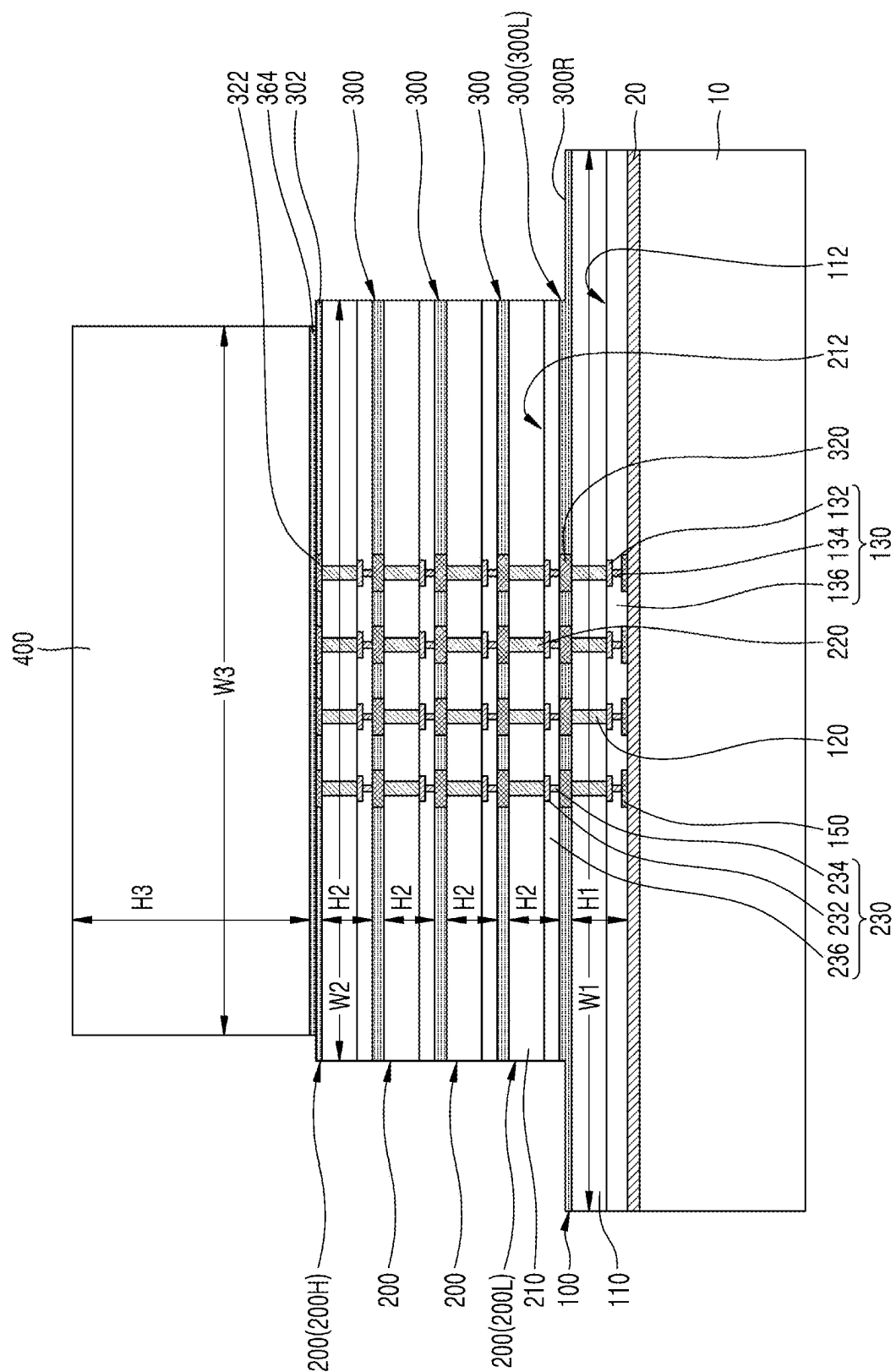


FIG. 5F

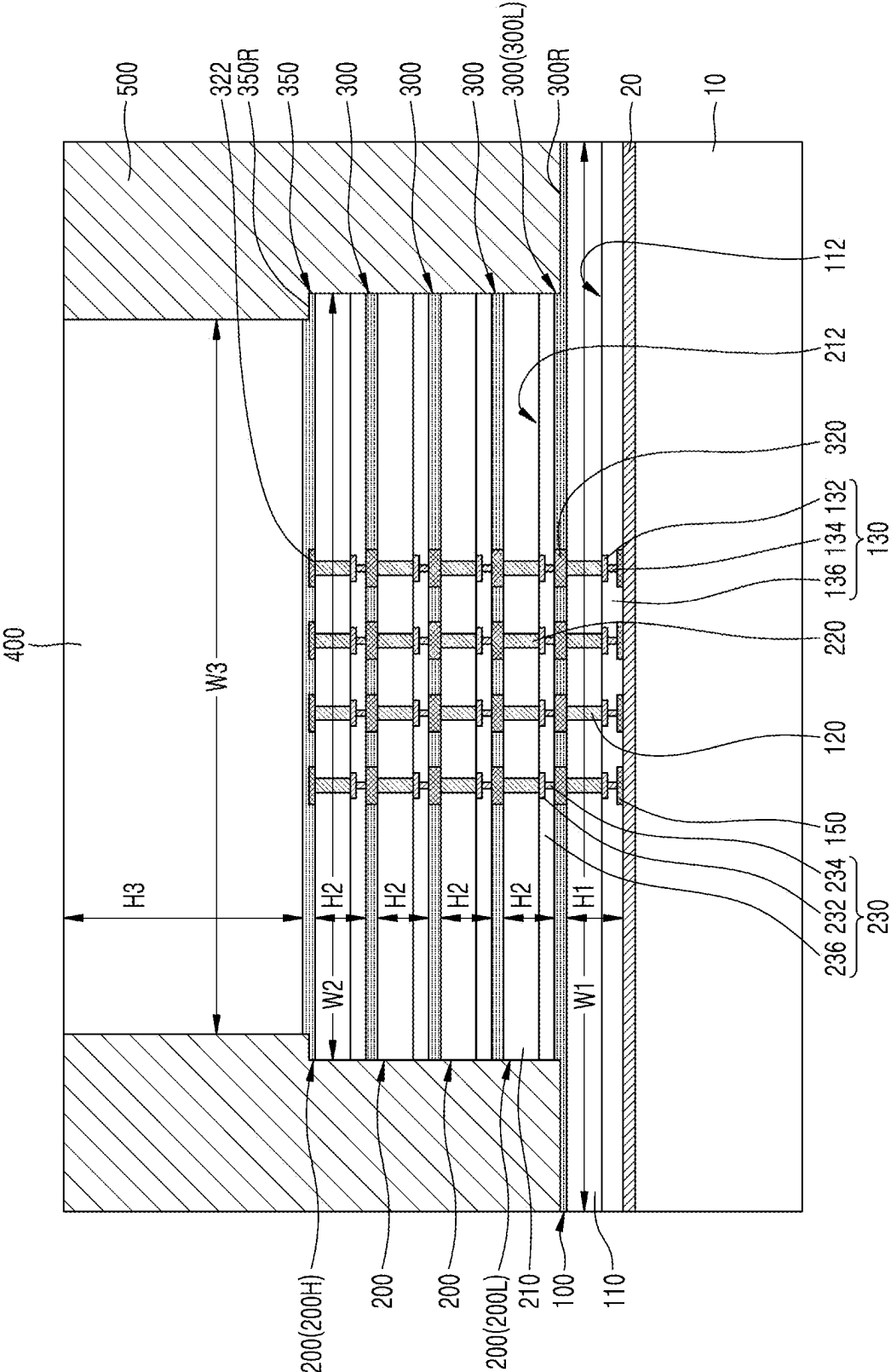


FIG. 5G

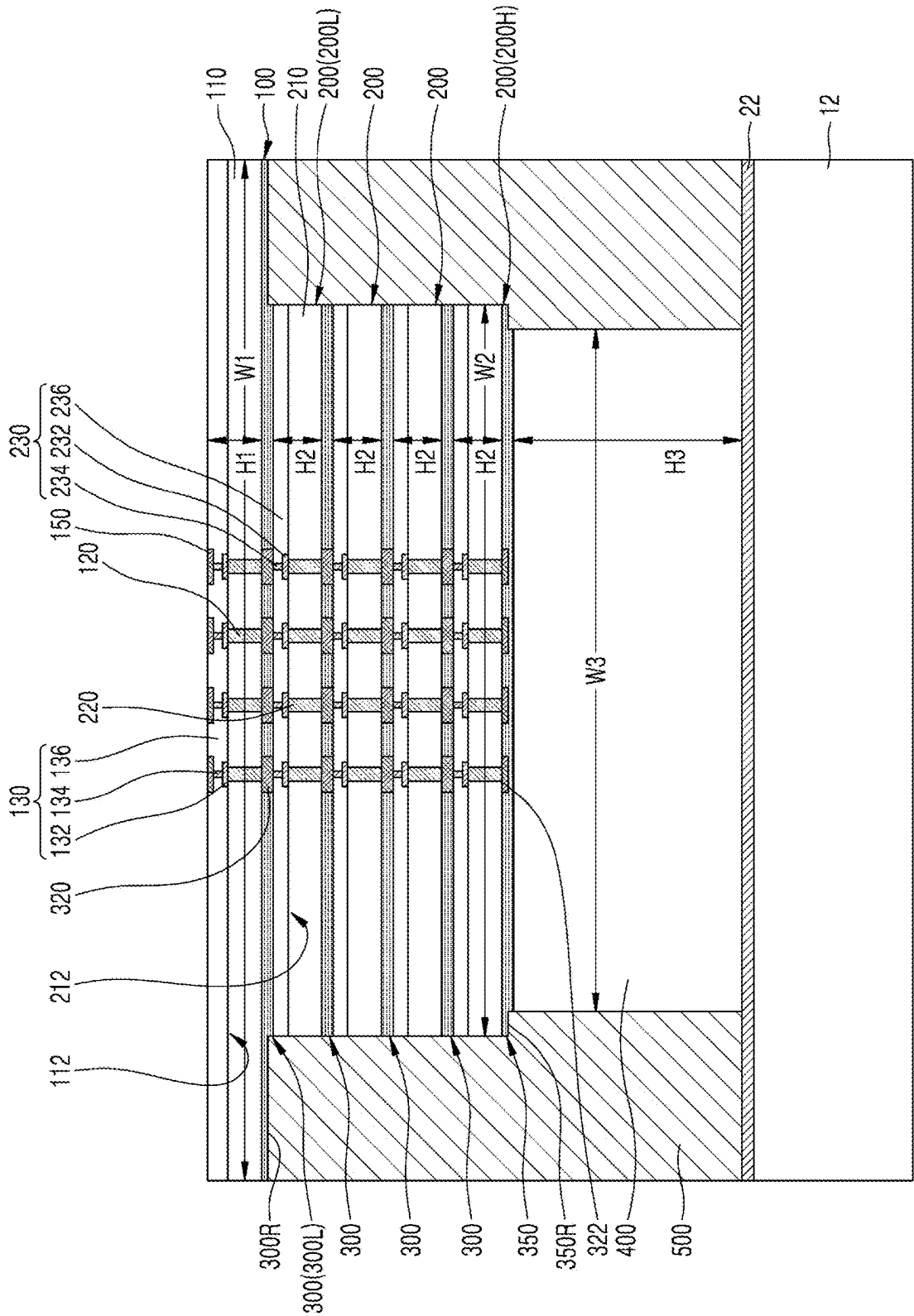


FIG. 5H

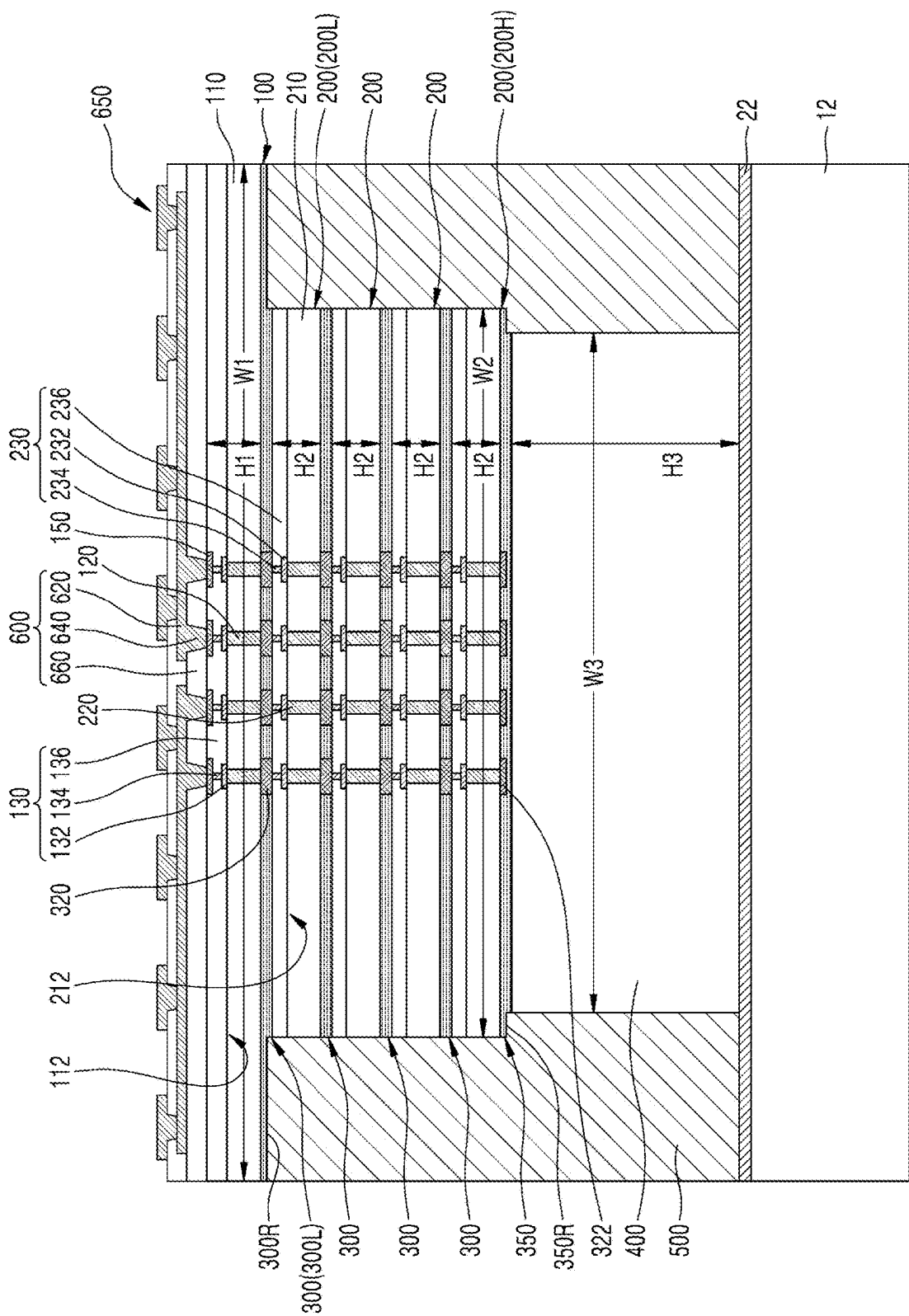


FIG. 5I

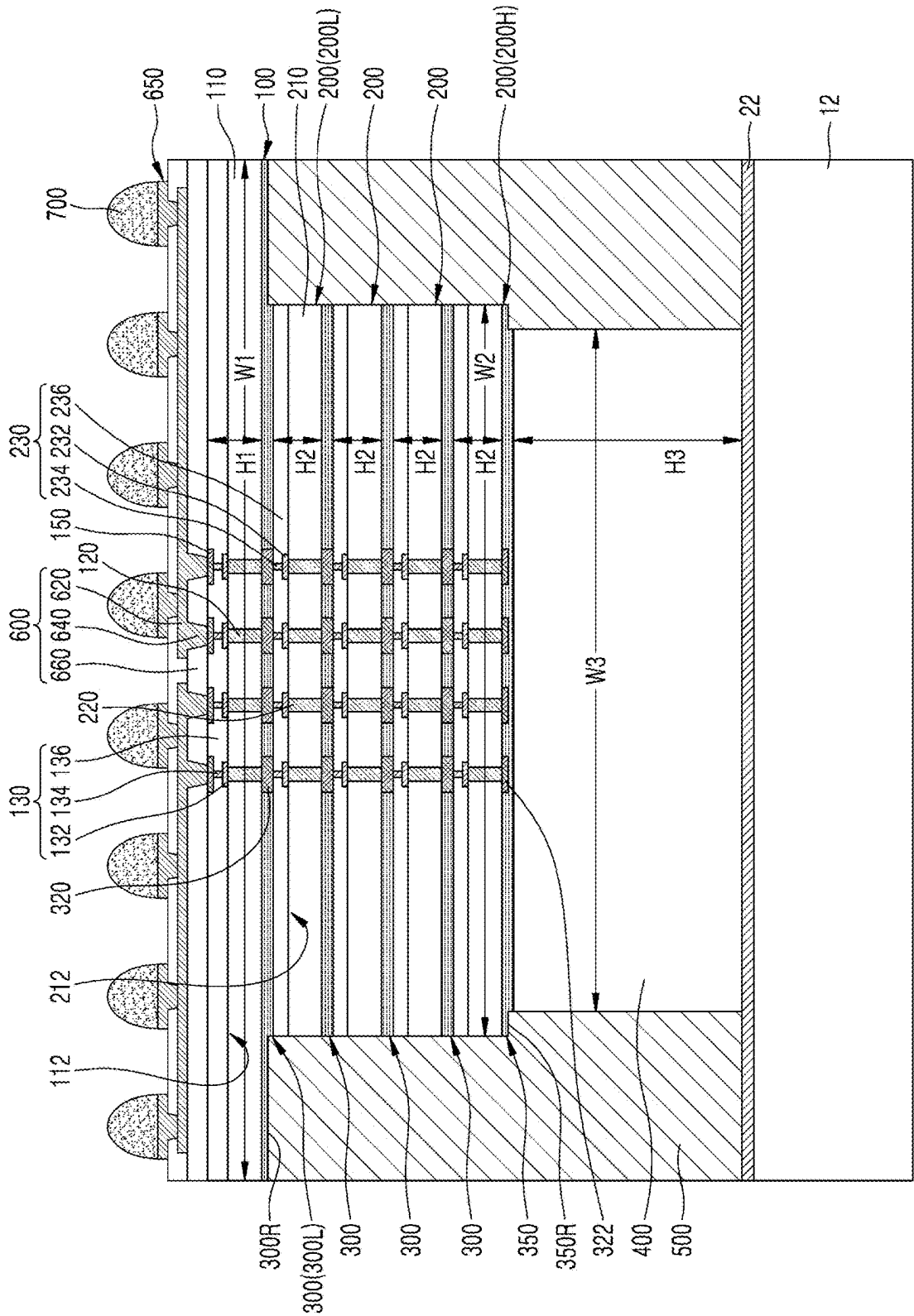


FIG. 6A

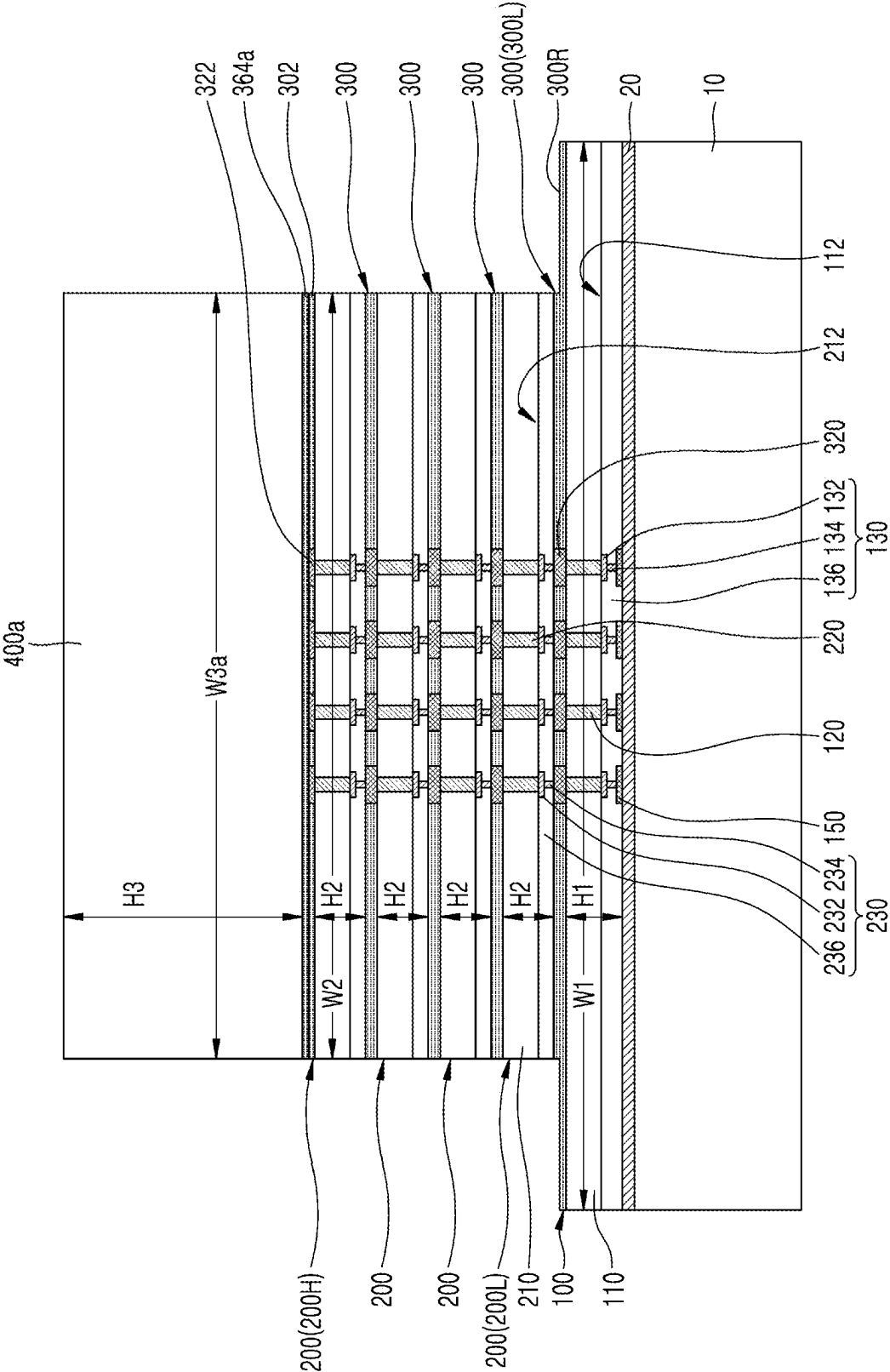


FIG. 6B

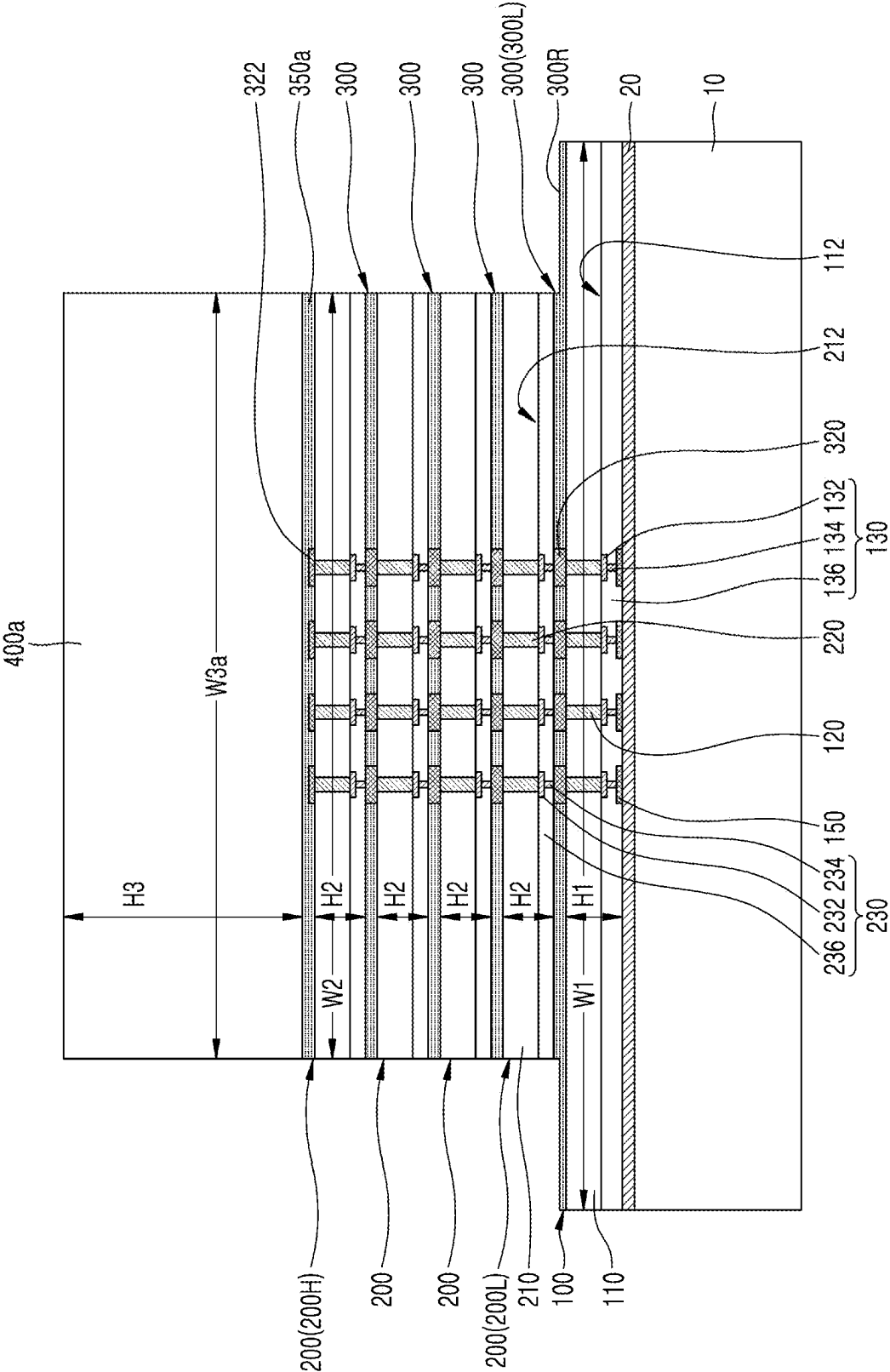


FIG. 7A

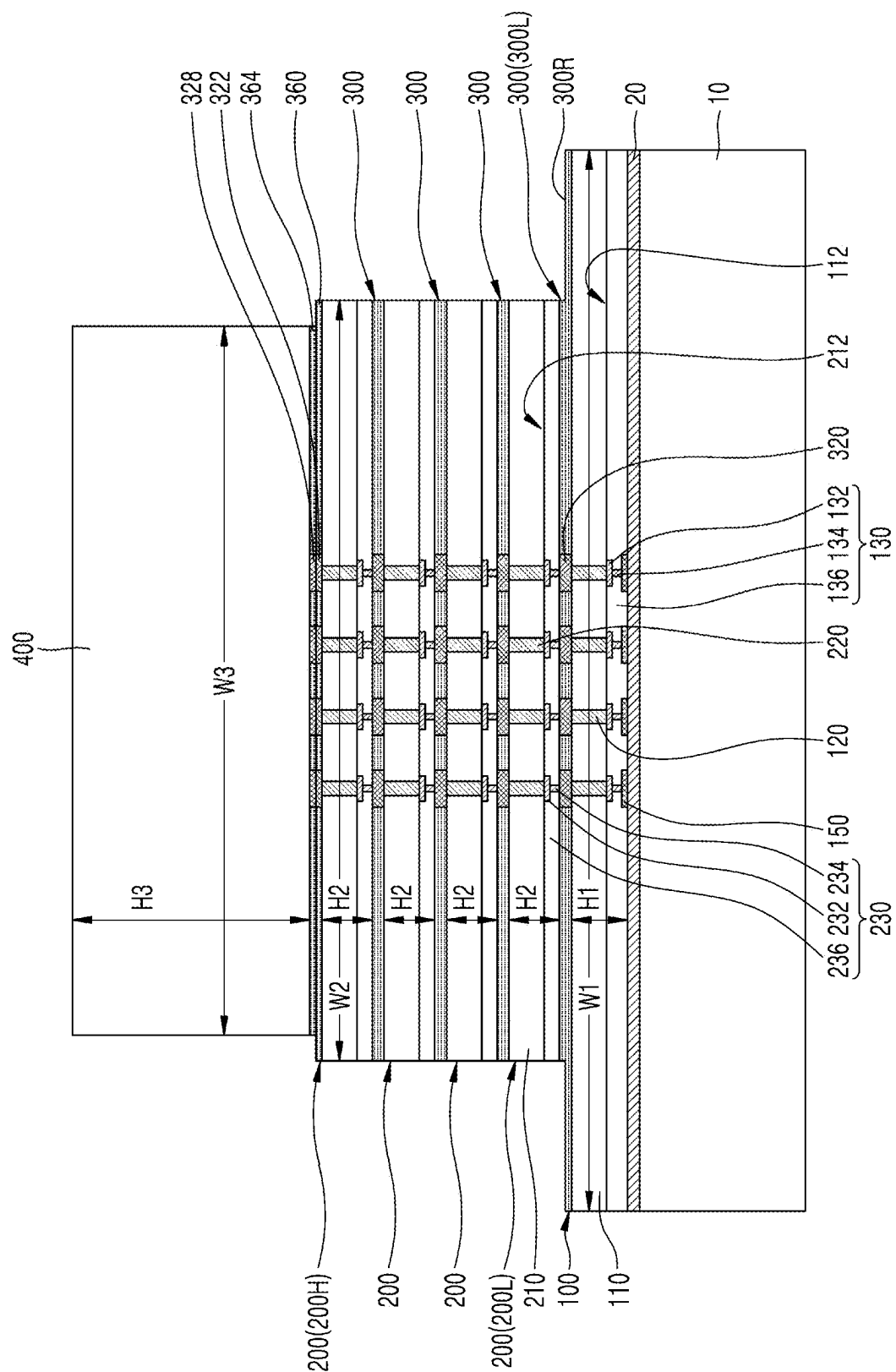


FIG. 7B

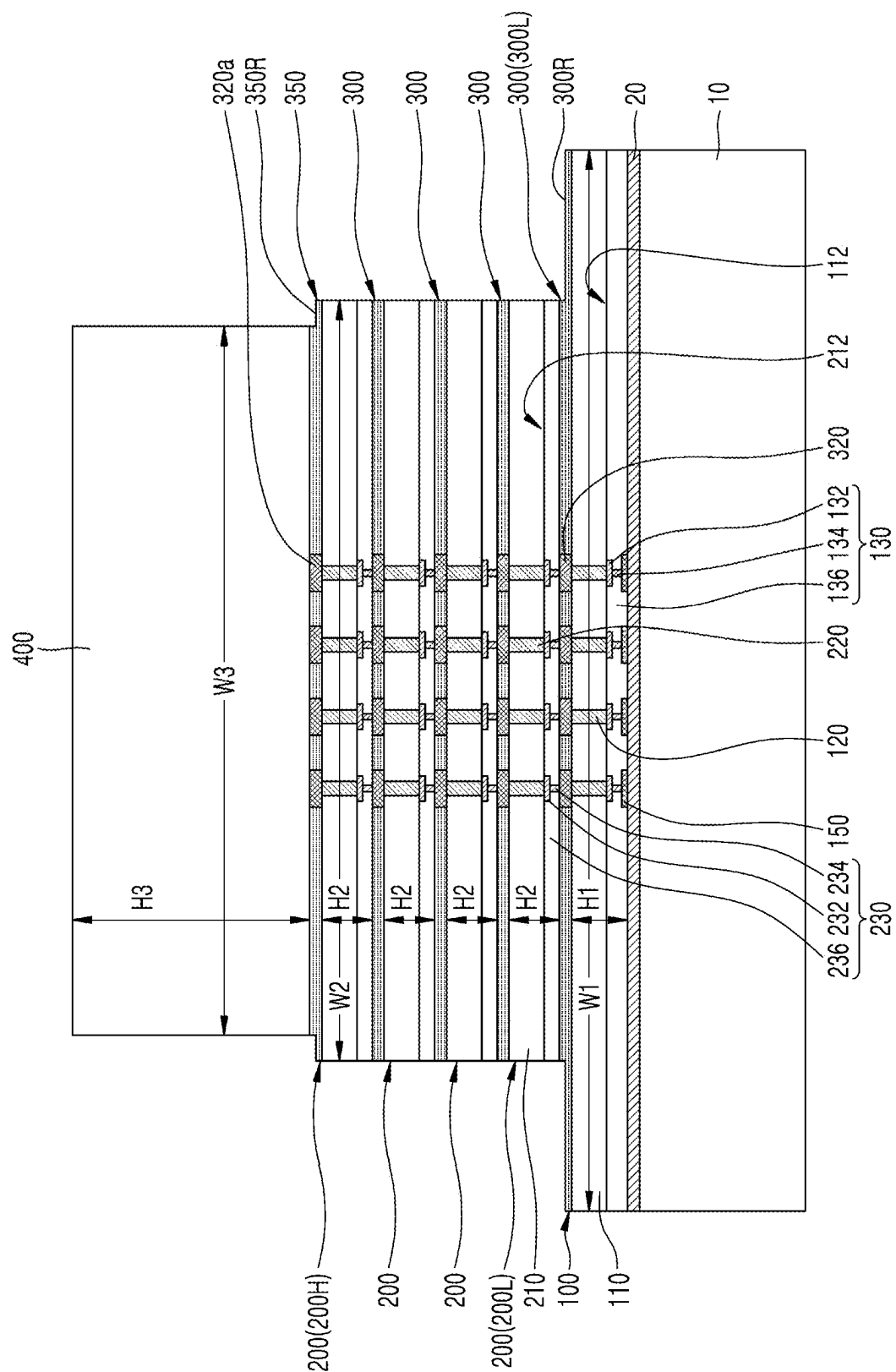


FIG. 8A

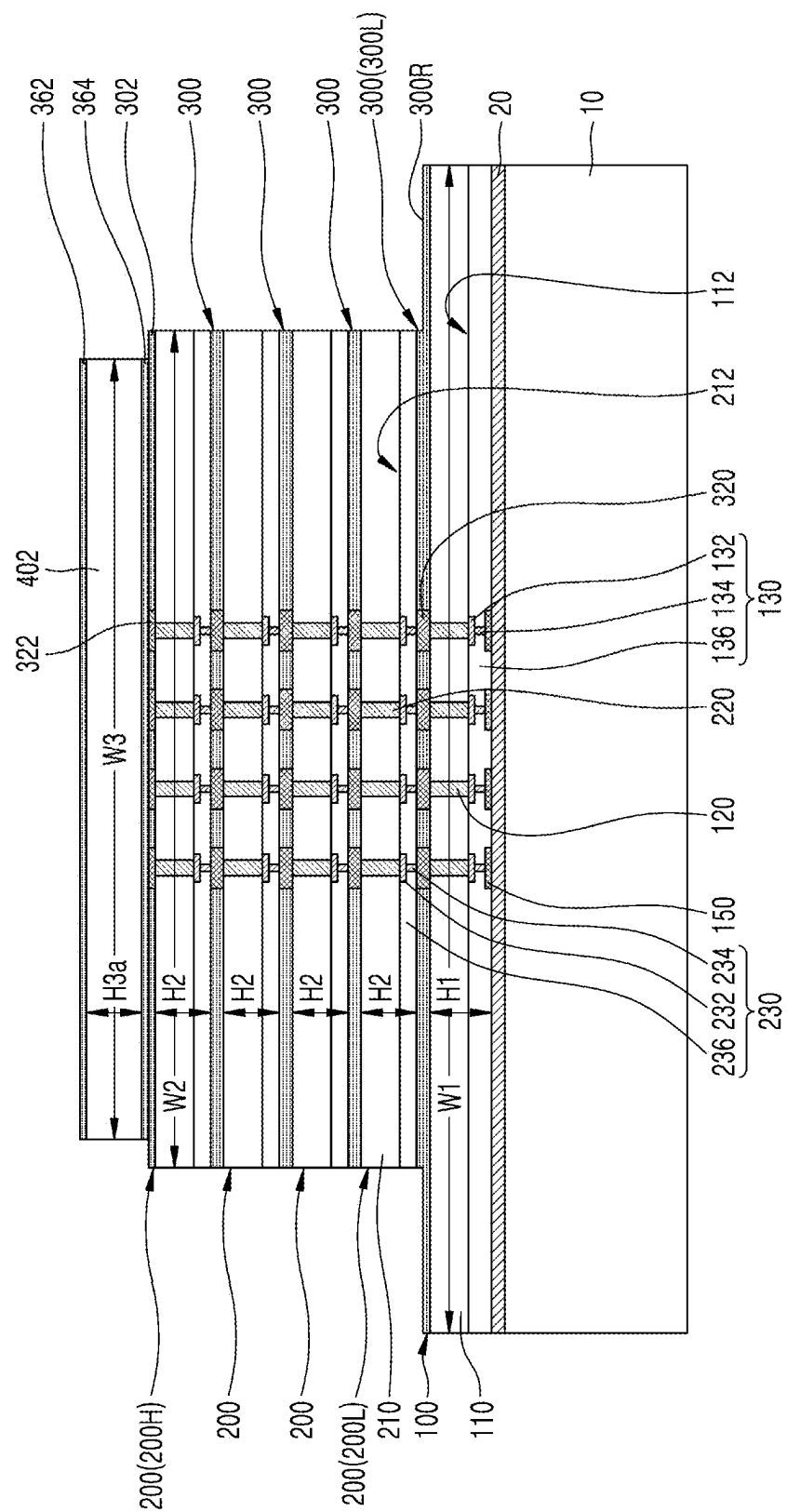


FIG. 8B

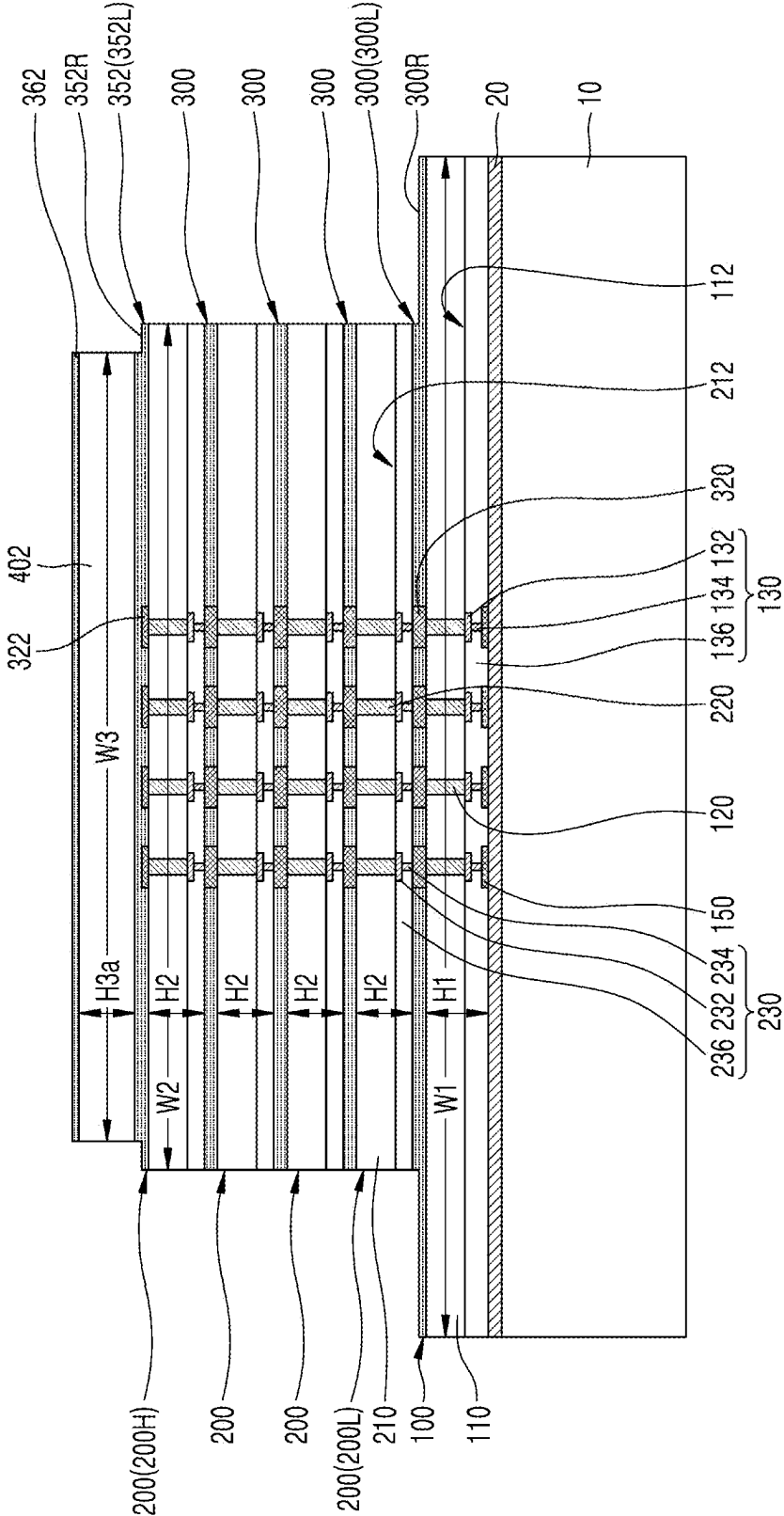


FIG. 8D

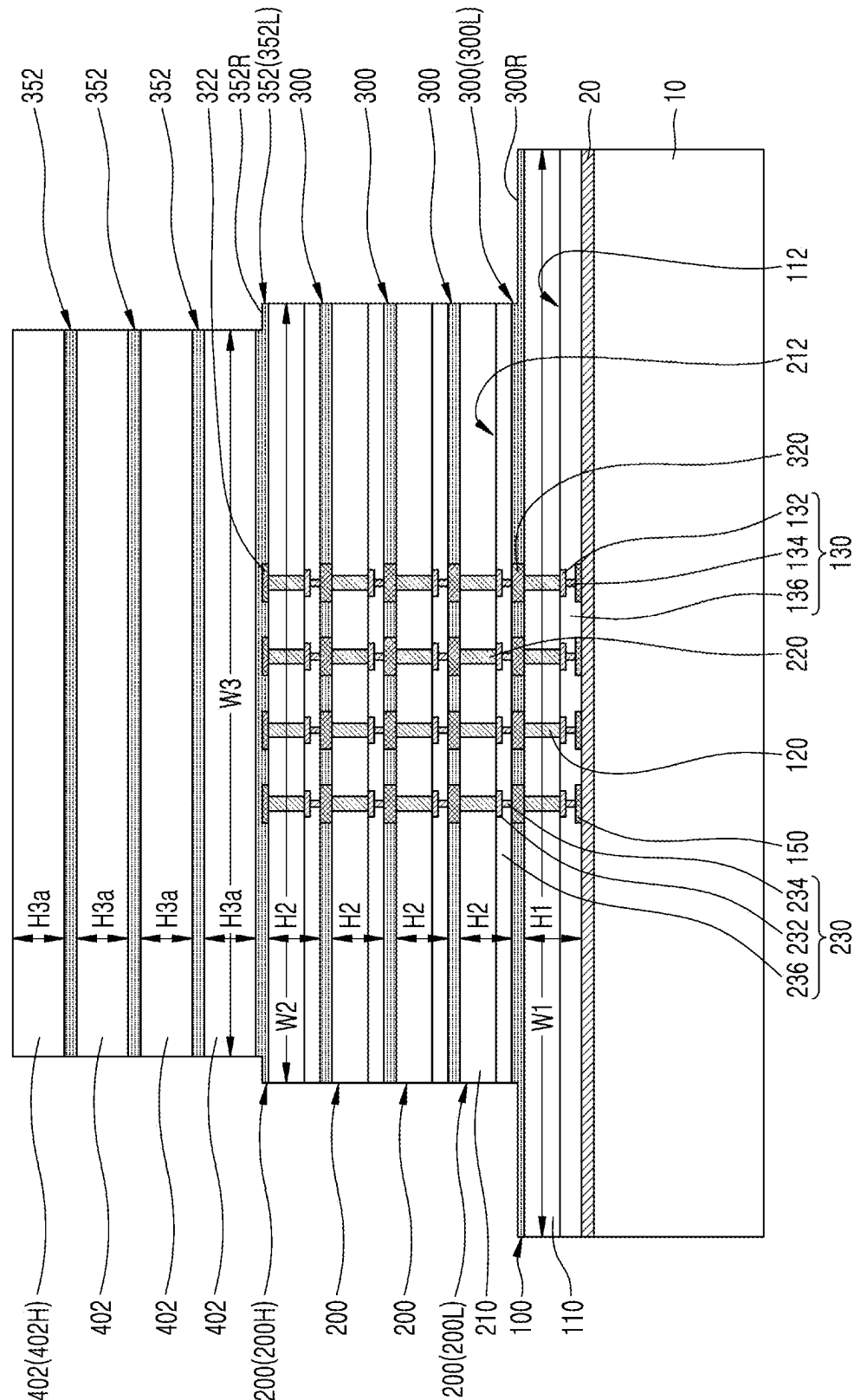


FIG. 9A

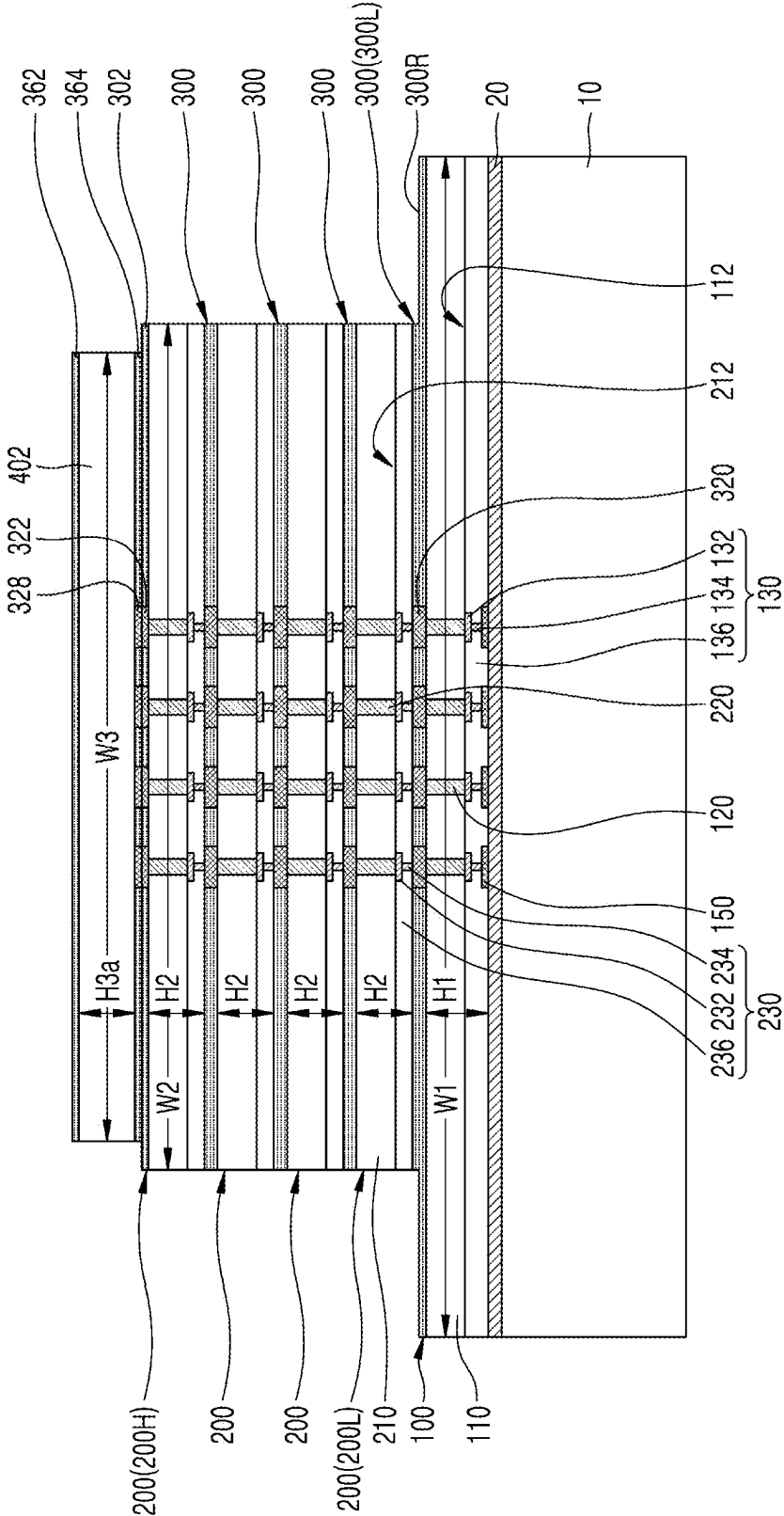


FIG. 9B

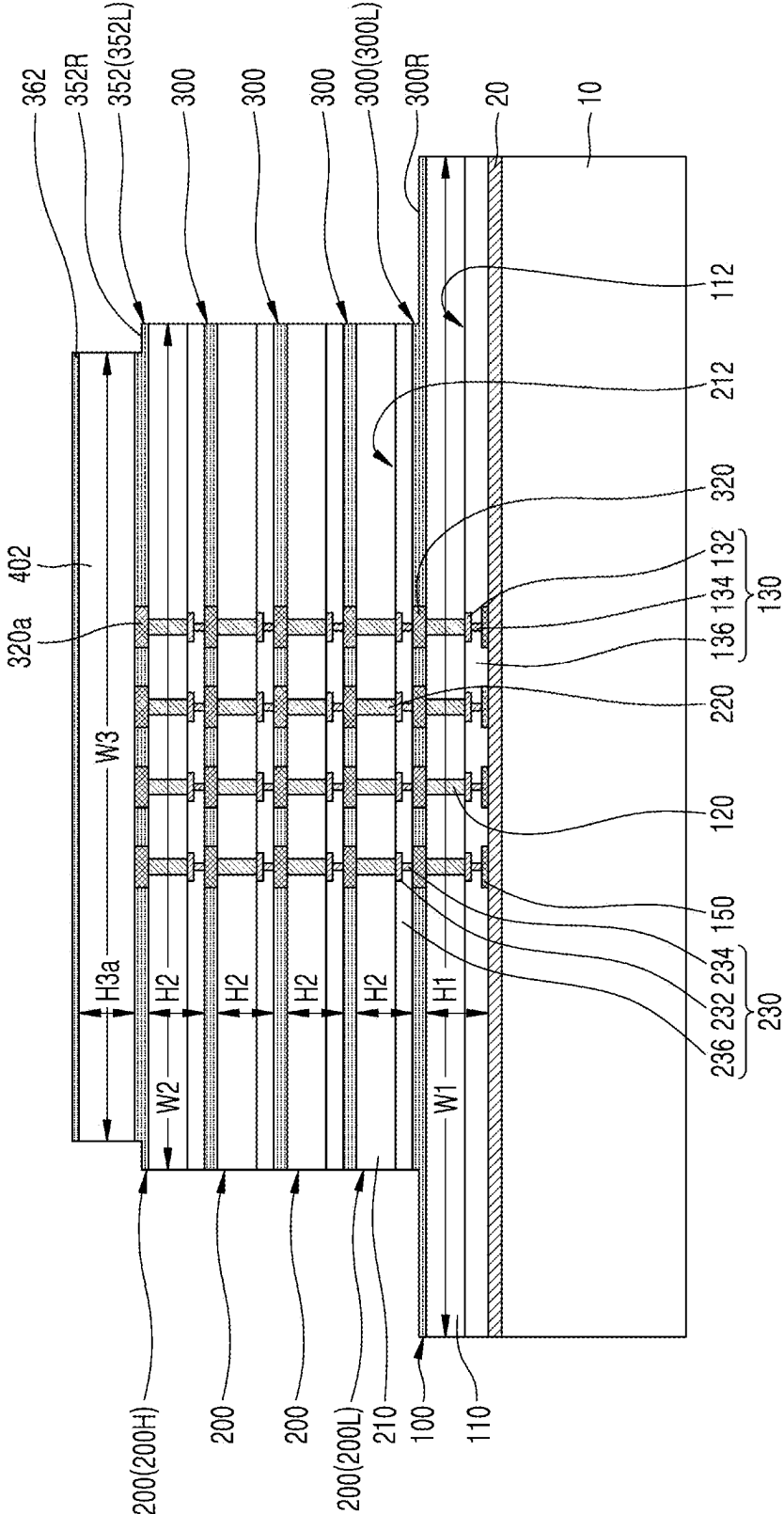


FIG. 10A

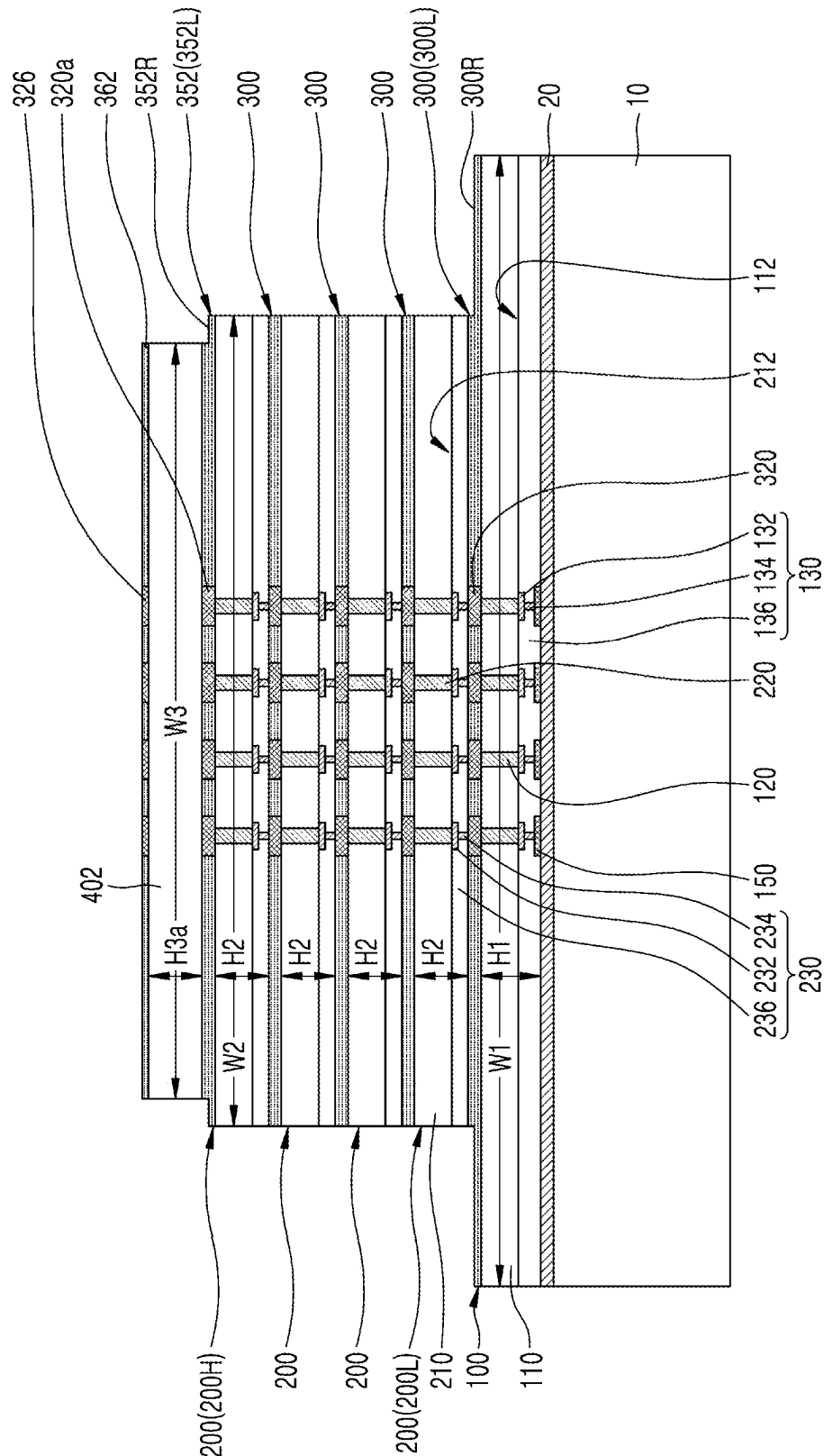


FIG. 10C

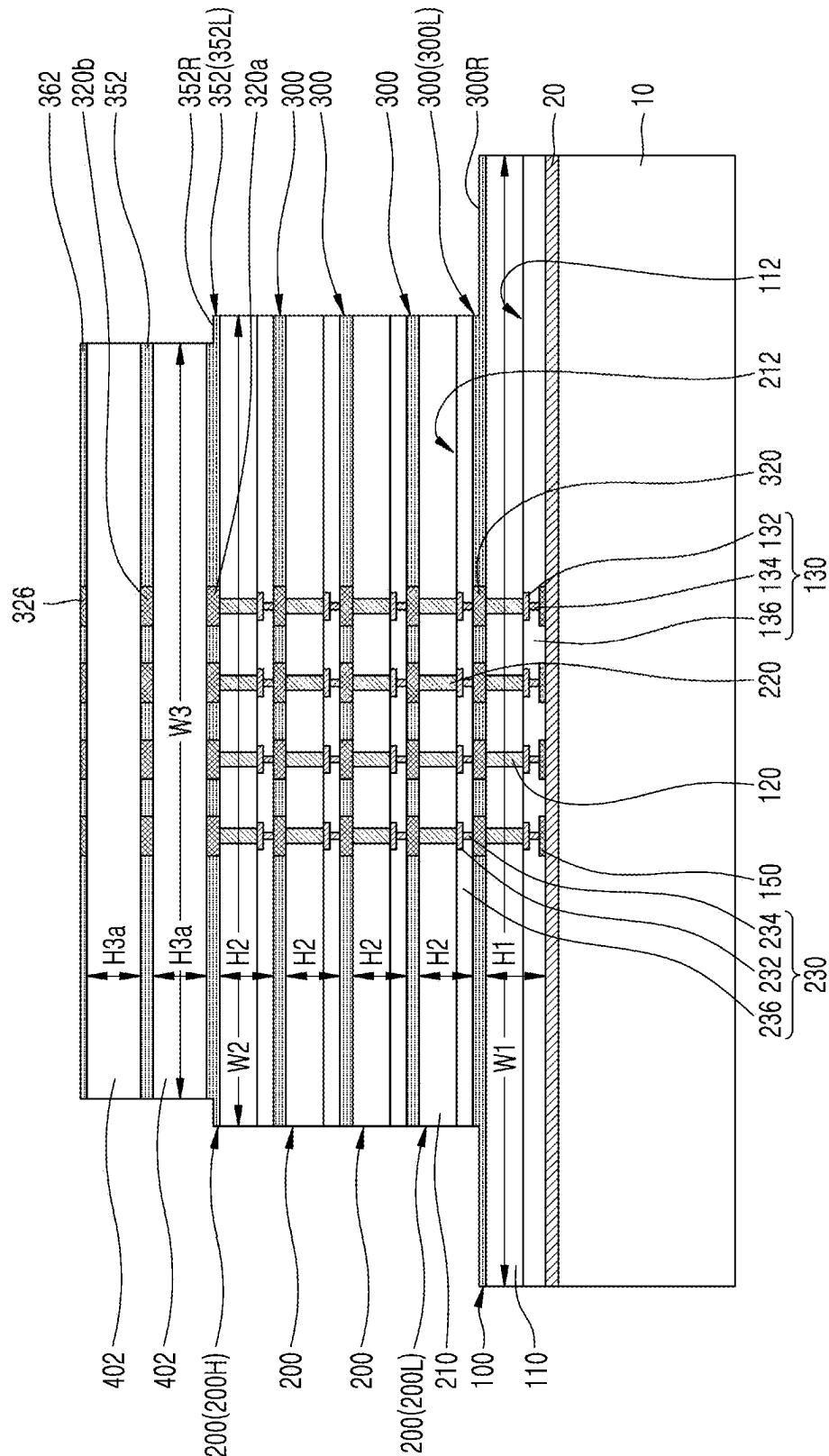


FIG. 11A

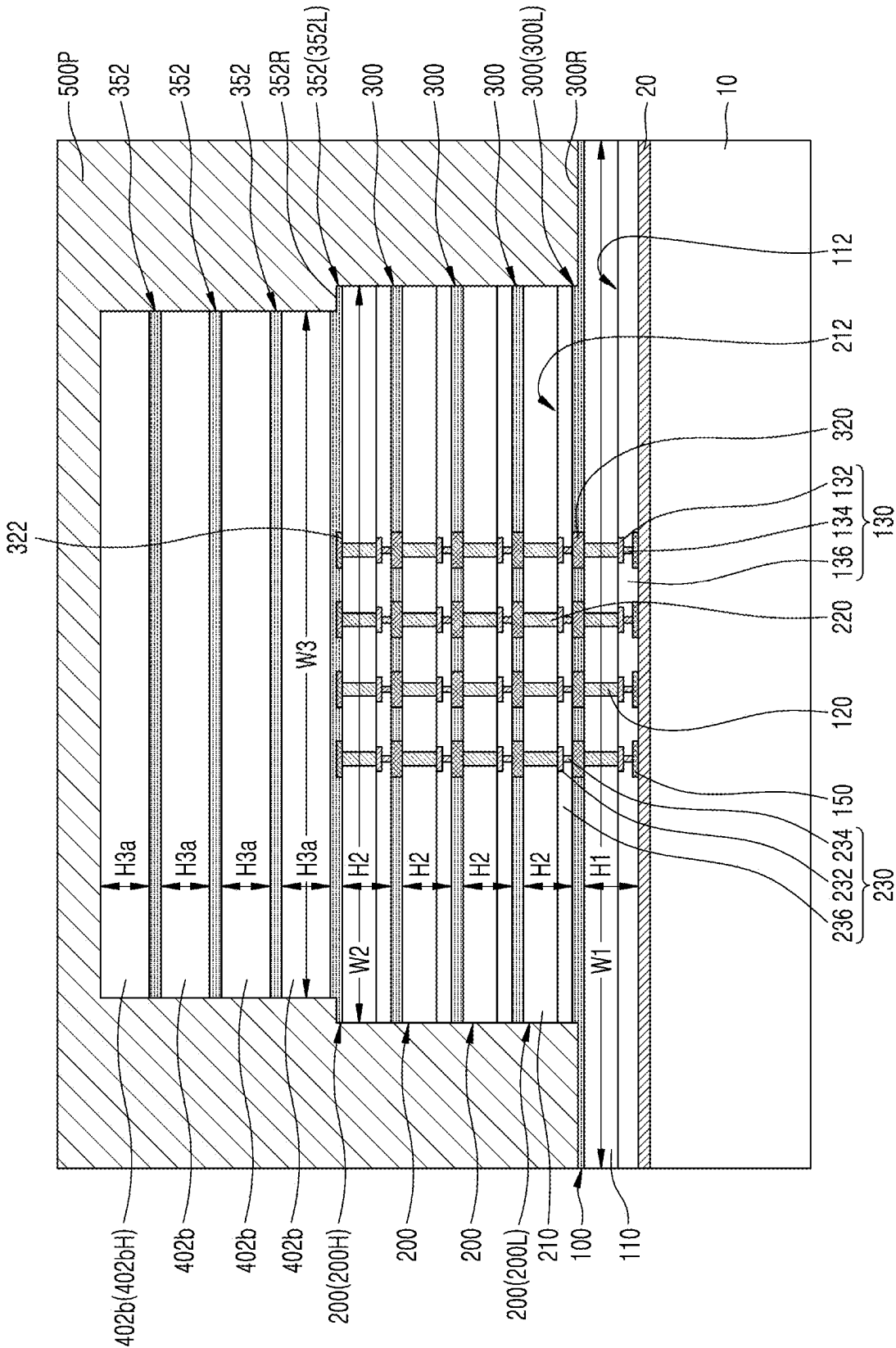


FIG. 11B

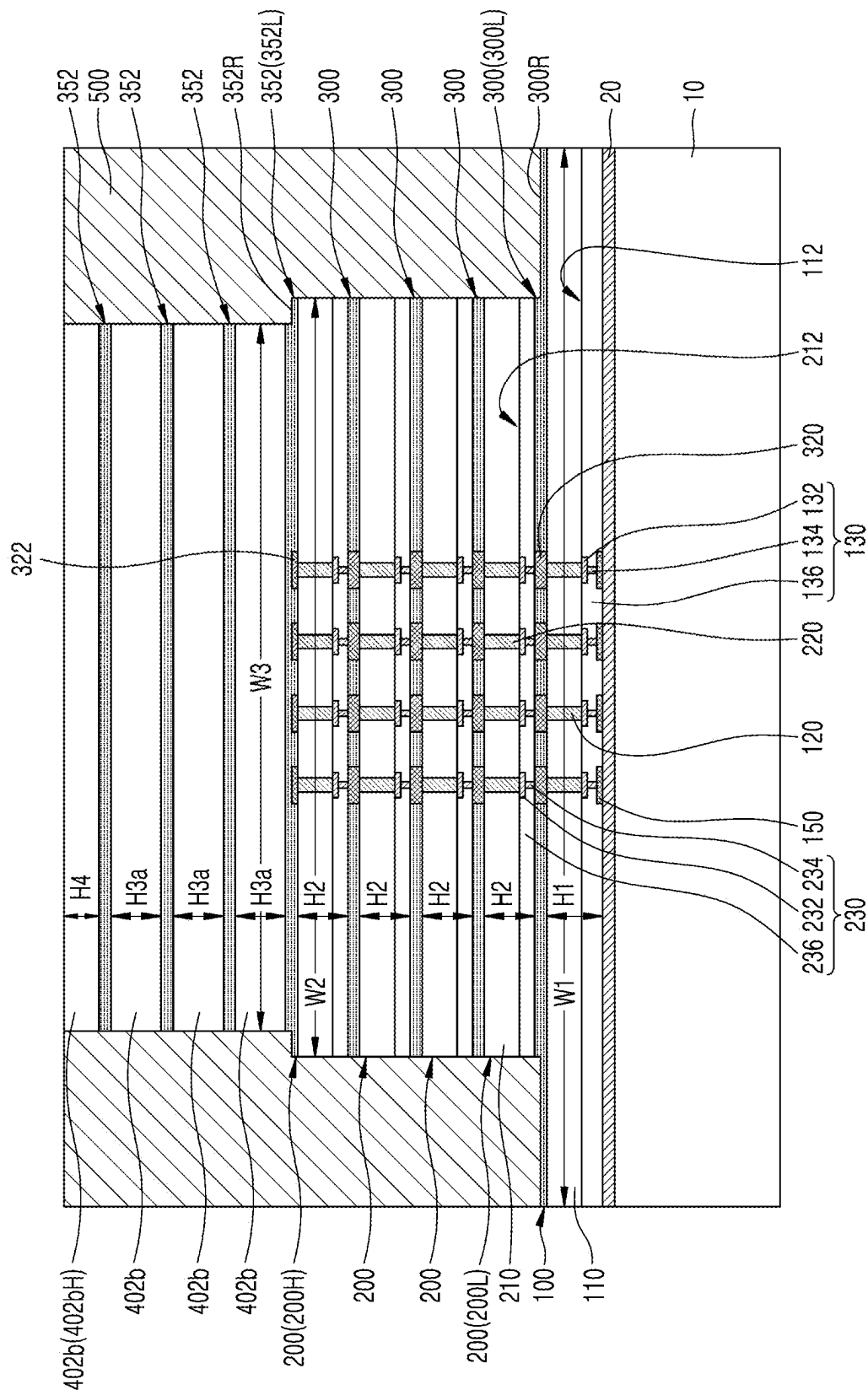


FIG. 12

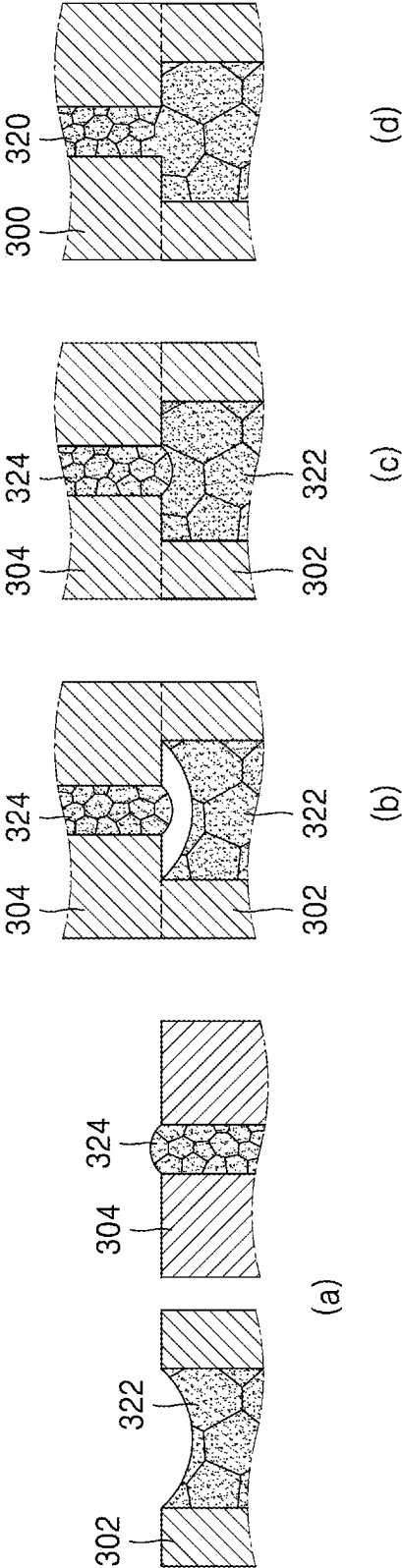


FIG. 13

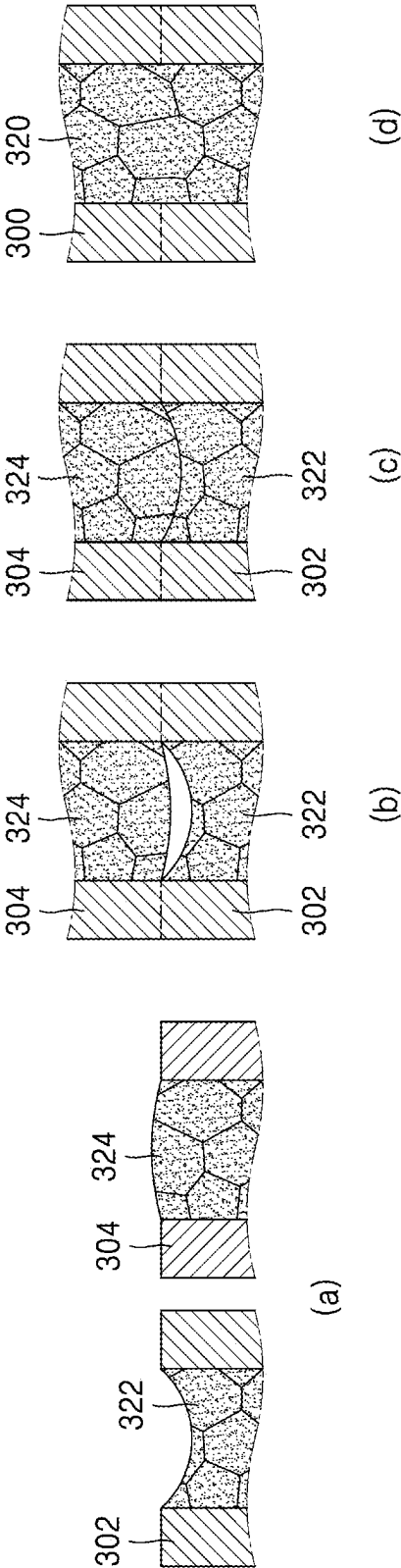
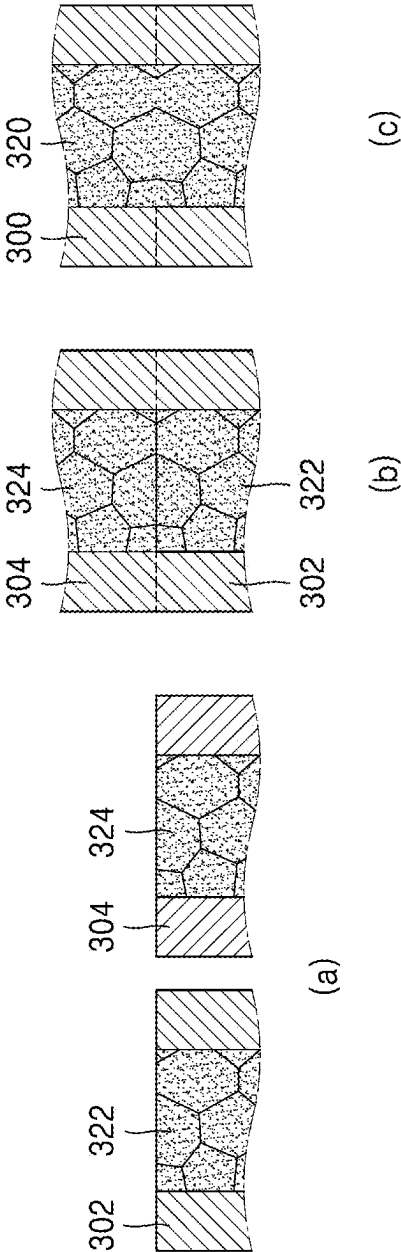


FIG. 14



SEMICONDUCTOR PACKAGE HAVING STACKED SEMICONDUCTOR CHIPS

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application is a Continuation of U.S. patent application Ser. No. 17/819,231, filed on Aug. 11, 2022, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0149953, filed on Nov. 3, 2021 in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entirety.

TECHNICAL FIELD

[0002] The inventive concept relates to a semiconductor package, and more particularly, to a semiconductor package having stacked semiconductor chips.

DISCUSSION OF THE RELATED ART

[0003] Semiconductor packages are becoming more compact while at the same time, performance such as capacity and speed are increasing. One approach for increasing the performance of packaged semiconductor chips while reducing package size is to stack multiple semiconductor chips within a single semiconductor package.

SUMMARY

[0004] The semiconductor package includes a first semiconductor chip including a first semiconductor substrate and a plurality of first through electrodes. The first semiconductor substrate has an active surface and an inactive surface. The plurality of first through electrodes pass through the first semiconductor substrate. A plurality of second semiconductor chips each include a second semiconductor substrate and a plurality of second through electrodes. The second semiconductor substrate has an active surface and an inactive surface. The plurality of second through electrodes pass through the second semiconductor substrate. The plurality of second semiconductor chips is stacked on the first semiconductor chip and each of the plurality of second semiconductor chips has a same vertical height. The active surface of the second semiconductor substrate faces the inactive surface of the first semiconductor substrate. A plurality of coupling pads is disposed between the first semiconductor chip and the plurality of second semiconductor chips and is configured to electrically connect the plurality of first through electrodes to the plurality of second through electrodes. A plurality of chip coupling insulation layers is disposed between the first semiconductor chip and the plurality of second semiconductor chips and at least partially surrounds the plurality of coupling pads. At least one supporting dummy substrate is stacked on the plurality of second semiconductor chips. At least one supporting coupling insulation layer is disposed on a bottom surface of the at least one supporting dummy substrate. Each of the plurality of second semiconductor chips has a same warpage shape bulging in one direction.

[0005] A semiconductor package includes a high-bandwidth memory (HBM) controller die including a first semiconductor substrate and a plurality of first through electrodes. The first semiconductor substrate has an active surface and an inactive surface. The plurality of first through electrodes pass through at least a portion of the first semi-

conductor substrate. A plurality of dynamic random access memory (DRAM) dies each include a second semiconductor substrate and a plurality of second through electrodes. The second semiconductor substrate has an active surface and an inactive surface. The plurality of second through electrodes pass through the second semiconductor substrate. The plurality of DRAM dies is stacked on the HBM controller die and each of the plurality of DRAM dies has a same vertical height. The active surface of the second semiconductor substrate faces the inactive surface of the first semiconductor substrate. A plurality of coupling pads is disposed between the HBM controller die and the plurality of DRAM dies and is configured to electrically connect the plurality of first through electrodes to the plurality of second through electrodes. A plurality of chip coupling insulation layers is disposed between the HBM controller die and the plurality of DRAM dies and at least partially surround the plurality of coupling pads. A supporting dummy substrate is stacked on the plurality of DRAM dies. A supporting coupling insulation layer is disposed on a bottom surface of the supporting dummy substrate. A plurality of top chip connection pads is disposed on a top surface of an uppermost DRAM die among the plurality of DRAM dies. The plurality of top chip connection pads is in contact with the plurality of second through electrodes of the uppermost DRAM die and is at least partially surrounded by the supporting coupling insulation layer covering the top surface of the uppermost DRAM die.

[0006] A semiconductor package includes a base redistribution layer including a plurality of package redistribution line patterns. A plurality of package redistribution vias are respectively in contact with and are connected to some of the plurality of package redistribution line patterns. A package redistribution insulation layer at least partially surrounds the plurality of package redistribution line patterns and the plurality of package redistribution vias. An HBM controller die includes a first semiconductor substrate and a plurality of first through electrodes. The first semiconductor substrate has a first active surface and a first inactive surface. The plurality of first through electrodes pass through the first semiconductor substrate. The HBM controller die is disposed on the base redistribution layer. The first active surface faces the base redistribution layer. A plurality of DRAM dies each include a second semiconductor substrate and a plurality of second through electrodes. The second semiconductor substrate has a second active surface and a second inactive surface. The plurality of second through electrodes pass through the second semiconductor substrate. The plurality of DRAM dies are stacked on the HBM controller die and have a same vertical height as each other and a horizontal width that is less than a horizontal width of the HBM controller die. The second active surface of the second semiconductor substrate faces the first inactive surface of the first semiconductor substrate. A plurality of coupling pads are disposed between the HBM controller die and the plurality of DRAM dies and are configured to electrically connect the plurality of first through electrodes to the plurality of second through electrodes. A plurality of chip coupling insulation layers are disposed between the HBM controller die and the plurality of DRAM dies and at least partially surround the plurality of coupling pads. A supporting dummy substrate is stacked on an uppermost DRAM die among the plurality of DRAM dies. A plurality of top chip connection pads are disposed on a top surface of

the uppermost DRAM die among the plurality of DRAM dies. The plurality of top chip connection pads are in contact with the plurality of second through electrodes of the uppermost DRAM die. A supporting coupling insulation layer covers the top surface of the uppermost DRAM die, side and top surfaces of the plurality of top chip connection pads, and a bottom surface of the supporting dummy substrate and fills between the uppermost DRAM die and the supporting dummy substrate. A package molding layer is disposed on the HBM controller die. The package molding layer covers a top surface of the HBM controller die, side surfaces of the plurality of DRAM dies, and side surfaces of the supporting dummy substrate and exposes a top surface of the supporting dummy substrate. The HBM controller die and the plurality of DRAM dies have a same warpage shape bulging in one direction.

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Embodiments of the inventive concept will be more clearly understood from the following detailed description taken in conjunction with the accompanying drawings, in which:

[0008] FIGS. 1A, 1B, 1C, 1D, 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 3A, 3B, 3C, 3D, 3E, 3F, 4A, 4B, 4C, 4D, 4E, 4F, 4G, and 4H are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0009] FIGS. 5A to 5I are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0010] FIGS. 6A and 6B are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0011] FIGS. 7A and 7B are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0012] FIGS. 8A to 8D are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0013] FIGS. 9A and 9B are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0014] FIGS. 10A to 10C are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0015] FIGS. 11A and 11B are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments;

[0016] FIGS. 12 to 14 are conceptual cross-sectional views of a process of forming a bonding pad in a method of manufacturing a semiconductor package, according to an embodiment; and

[0017] FIG. 15 is a conceptual diagram of the shape of a semiconductor package according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0018] FIGS. 1A, 1B, 1C, 1D, 2A, 2B, 2C, 2D, 2E, 2F, 2G, 2H, 3A, 3B, 3C, 3D, 3E, 3F, 4A, 4B, 4C, 4D, 4E, 4F, 4G, and 4H are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. In FIGS. 1A to 4H, like reference numerals may denote like elements throughout the specification and disclosure, and to the extent that a description of an element is omitted, it may

be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0019] Referring to FIG. 1A, a semiconductor package 1000 may include a first semiconductor chip 100 and a plurality of second semiconductor chips 200. Although it is illustrated in FIG. 1A that the semiconductor package 1000 includes four second semiconductor chips 200, the present invention is not necessarily limited thereto. For example, the semiconductor package 1000 may include two or more second semiconductor chips 200. In some embodiments, the semiconductor package 1000 may include a multiple of 4 second semiconductor chips 200. The second semiconductor chips 200 may be sequentially stacked on the first semiconductor chip 100. For convenience of description, the second semiconductor chip 200 at the bottom, among the second semiconductor chips 200, may be referred to as a lowermost second semiconductor chip 200L, and the second semiconductor chip 200 at the top, among the second semiconductor chips 200, may be referred to as an uppermost second semiconductor chip 200H.

[0020] The first semiconductor chip 100 and the second semiconductor chips 200 of the semiconductor package 1000 may be electrically connected to each other through a plurality of coupling pads 320 and may thus exchange signals with each other and provide power and ground. For example, the coupling pads 320 may be disposed between the first semiconductor chip 100 and the lowermost second semiconductor chip 200L and between two adjacent second semiconductor chips 200.

[0021] For example, the coupling pads 320 may include a material including copper (Cu). A coupling pad 320 disposed between the first semiconductor chip 100 and the lowermost second semiconductor chip 200L may be referred to as a first coupling pad, and a coupling pad 320 disposed between two adjacent second semiconductor chips 200 may be referred to as a second coupling pad.

[0022] The first semiconductor chip 100 may include a first semiconductor substrate 110 having an active surface and an inactive surface, a first semiconductor device 112 disposed on the active surface of the first semiconductor substrate 110, a first wiring structure 130 disposed on the active surface of the first semiconductor substrate 110, and a plurality of first through electrodes 120 connected to the first wiring structure 130 and passing through at least a portion of the first semiconductor chip 100. The first semiconductor chip 100 may further include a plurality of chip pads 150 disposed in the bottom surface of the first semiconductor chip 100. The chip pads 150 are electrically connected to a first wiring pattern 132 and/or a first wiring via 134. The chip pads 150 may be electrically connected to the first semiconductor device 112 and the first wiring structure 130 through the first wiring pattern 132 and/or the first wiring via 134.

[0023] The first semiconductor chip 100 of the semiconductor package 1000 may be arranged such that the active surface of the first semiconductor substrate 110 faces downwards and the inactive surface of the first semiconductor substrate 110 faces upwards. Accordingly, unless particularly mentioned, the top surface of the first semiconductor chip 100 of the semiconductor package 1000 refers to a side that the inactive surface of the first semiconductor substrate 110 faces, and the bottom surface of the first semiconductor chip 100 refers to a side that the active surface of the first

semiconductor substrate **110** faces. However, in the descriptions based on the first semiconductor chip **100**, the bottom surface of the first semiconductor chip **100** that the active surface of the first semiconductor substrate **110** faces may be referred to as a front surface of the first semiconductor chip **100**, and the top surface of the first semiconductor chip **100** that the inactive surface of the first semiconductor substrate **110** faces may be referred to as a back surface of the first semiconductor chip **100**.

[0024] Each of the second semiconductor chips **200** may include a second semiconductor substrate **210** having an active surface and an inactive surface, a second semiconductor device **212** on the active surface of the second semiconductor substrate **210**, and a second wiring structure **230** on the active surface of the second semiconductor substrate **210**.

[0025] Each second semiconductor chip **200** may further include a plurality of second through electrodes **220** connected to the second wiring structure **230** and passing through at least a portion of the second semiconductor chip **200**. Among the second semiconductor chips **200**, the uppermost second semiconductor chip **200H**, which is the farthest from the first semiconductor chip **100** and is at the top of the semiconductor package **1000**, may also include a plurality of second through electrodes **220**.

[0026] In some embodiments, a plurality of top chip connection pads **322** may be disposed on the top surface of the uppermost second semiconductor chip **200H**. The top chip connection pads **322** may be arranged on the top surface of the uppermost second semiconductor chip **200H** and may be connected to the second through electrodes **220**. In some embodiments, a plurality of top chip connection pads **322** might not be disposed on the top surface of the uppermost second semiconductor chip **200H**.

[0027] In some embodiments, the vertical height, i.e., the thickness, of the uppermost second semiconductor chip **200H**, may be substantially the same as that of the other second semiconductor chips **200**.

[0028] In the semiconductor package **1000**, the second semiconductor chips **200** may be sequentially stacked on the first semiconductor chip **100** in a vertical direction such that the active surface of each of the second semiconductor chips **200** faces downwards, i.e., the first semiconductor chip **100**. Accordingly, unless particularly mentioned, the top surface of each second semiconductor chip **200** of the semiconductor package **1000** refers to a side that the inactive surface of the second semiconductor substrate **210** faces, and the bottom surface of the second semiconductor chip **200** refers to a side that the active surface of the second semiconductor substrate **210** faces. However, in the descriptions based on the second semiconductor chip **200**, the bottom surface of the second semiconductor chip **200** that the active surface of the second semiconductor substrate **210** faces may be referred to as a front surface of the second semiconductor chip **200**, and the top surface of the second semiconductor chip **200** that the inactive surface of the second semiconductor substrate **210** faces may be referred to as a back surface of the second semiconductor chip **200**.

[0029] For example, the first semiconductor substrate **110** and the second semiconductor substrate **210** may each include a semiconductor material such as silicon (Si). Alternatively, the first semiconductor substrate **110** and the second semiconductor substrate **210** may each include a semiconductor material such as germanium (Ge). The first

semiconductor substrate **110** and the second semiconductor substrate **210** may each have an active surface and an inactive surface. The first semiconductor substrate **110** and the second semiconductor substrate **210** may each include a conductive region, e.g., an impurity-doped well. The first semiconductor substrate **110** and the second semiconductor substrate **210** may each have various isolation structures including a shallow trench isolation (STI) structure.

[0030] The first semiconductor device **112** and the second semiconductor device **212** may each include various kinds of individual devices. The individual devices may include various microelectronic devices, e.g., a metal-oxide-semiconductor field effect transistor (MOSFET) such as complementary metal-oxide-semiconductor (CMOS) transistor, a system large scale integration (LSI), an image sensor such as a CMOS image sensor (CIS), a micro-electro-mechanical system (MEMS), an active element, and a passive element. The individual devices may be electrically connected to the conductive region of the first semiconductor substrate **110** or the second semiconductor substrate **210**. Each of the first semiconductor device **112** and the second semiconductor device **212** may further include a conductive wiring or plug, which electrically connects the individual devices or at least two individual devices to the conductive region of the first semiconductor substrate **110** or the second semiconductor substrate **210**. Each of the individual devices may be electrically isolated from other individual devices by an insulation film.

[0031] At least one of the first and second semiconductor chips **100** and **200** may include a memory semiconductor chip. In some embodiments, the first semiconductor chip **100** may include a buffer chip, which includes a serial-parallel conversion circuit and controls the second semiconductor chips **200**, and the second semiconductor chips **200** may include a memory chip including memory cells. For example, the semiconductor package **1000** including the first semiconductor chip **100** and the second semiconductor chips **200** may correspond to high-bandwidth memory (HBM). The first semiconductor chip **100** may be referred to as an HBM controller die, and each of the second semiconductor chips **200** may be referred to as a dynamic random access memory (DRAM) die.

[0032] The first wiring structure **130** may include a plurality of first wiring patterns **132**, a plurality of first wiring vias **134** respectively connected to the first wiring patterns **132**, and a first inter-wiring insulation layer **136** at least partially surrounding the first wiring patterns **132** and the first wiring vias **134**. In some embodiments, the first wiring patterns **132** may have a thickness of about 0.5 μm or less. In some embodiments, the first wiring structure **130** may have a multi-layer wiring structure including the first wiring patterns **132** and the first wiring vias **134** at different vertical levels.

[0033] The second wiring structure **230** may include a plurality of second wiring patterns **232**, a plurality of second wiring vias **234** respectively connected to the second wiring patterns **232**, and a second inter-wiring insulation layer **236** at least partially surrounding the second wiring patterns **232** and the second wiring vias **234**. In some embodiments, the second wiring patterns **232** may have a thickness of about 0.5 μm or less. In some embodiments, the second wiring structure **230** may have a multi-layer wiring structure including the second wiring patterns **232** and the second wiring vias **234** at different vertical levels.

[0034] For example, the first wiring patterns 132, the first wiring vias 134, the second wiring patterns 232, and the second wiring vias 234 may include a metal material such as aluminum, copper, or tungsten. In some embodiments, the first wiring patterns 132, the first wiring vias 134, the second wiring patterns 232, and the second wiring vias 234 may include a wiring barrier film and a wiring metal layer. The wiring barrier film may include a metal, a metal nitride, or an alloy. The wiring metal layer may include tungsten (W), aluminum (Al), titanium (Ti), tantalum (Ta), ruthenium (Ru), manganese (Mn), and/or copper (Cu).

[0035] When the first wiring structure 130 and the second wiring structure 230 have a multi-layer wiring structure, each of the first inter-wiring insulation layer 136 and the second inter-wiring insulation layer 236 may have a multi-layer structure, in which a plurality of insulation layers are stacked in correspondence to the multi-layer wiring structure of the first or second wiring structure 130 or 230. For example, the first inter-wiring insulation layer 136 and the second inter-wiring insulation layer 236 may each include silicon oxide, silicon nitride, silicon oxynitride, an insulation material having a lower permittivity than silicon oxide, or a combination thereof. In some embodiments, the first inter-wiring insulation layer 136 and the second inter-wiring insulation layer 236 may each include a tetraethyl orthosilicate (TEOS) film or an ultra low-k (ULK) film having an ultra-low dielectric constant of about 2.2 to about 2.4. The ULK film may include an SiOC film or an SiCOH film.

[0036] The first through electrodes 120 and the second through electrodes 220 may include a through silicon via (TSV). Each of the first through electrodes 120 may include a conductive plug, which passes through the first semiconductor substrate 110, and a conductive barrier film at least partially surrounding the conductive plug. Each of the second through electrodes 220 may include a conductive plug, which passes through the second semiconductor substrate 210, and a conductive barrier film at least partially surrounding the conductive plug. The conductive plug may have a pillar shape, and the conductive barrier film may have a cylindrical shape at least partially surrounding the side wall of the conductive plug. A via insulation film may be disposed between each first through electrode 120 and the first semiconductor substrate 110 may at least partially surround the side wall of the first through electrode 120. A via insulation film may be disposed between each second through electrode 220 and the second semiconductor substrate 210 and may at least partially surround the side wall of the second through electrode 220. The first through electrode 120 and the second through electrode 220 may have a via-first structure, a via-middle structure, or a via-last structure.

[0037] The first semiconductor chip 100 may have a first horizontal width W1 and a first vertical height H1. Each of the second semiconductor chips 200 may have a second horizontal width W2 and a second vertical height H2. In some embodiments, the first horizontal width W1 may be greater than the second horizontal width W2. In some embodiments, the first vertical height H1 may be substantially equal to the second vertical height H2. For example, the first vertical height H1 and the second vertical height H2 may be about 50 μm to about 90 μm .

[0038] A plurality of coupling pads 320 may electrically connect the second wiring patterns 232 and/or the second

wiring vias 234 of the second wiring structure 230 to the first through electrodes 120 or the second through electrodes 220 therebelow.

[0039] For example, the second wiring patterns 232 and/or the second wiring vias 234 of the second wiring structure 230 of the lowermost second semiconductor chip 200L may be electrically connected to the first through electrodes 120 of the first semiconductor chip 100 therebelow through a plurality of coupling pads 320, i.e., a plurality of first coupling pads. The second wiring patterns 232 and/or the second wiring vias 234 of the second wiring structure 230 of each of the second semiconductor chips 200, except for the lowermost second semiconductor chip 200L, may be electrically connected to the second through electrodes 220 of each second semiconductor chip 200 therebelow through a plurality of coupling pads 320, i.e., a plurality of second coupling pads.

[0040] The coupling pads 320 disposed between the first semiconductor chip 100 and the lowermost second semiconductor chip 200L or between two adjacent second semiconductor chips 200 may be at least partially surrounded by a chip coupling insulation layer 300. The coupling pads 320 may pass through the chip coupling insulation layer 300. A plurality of chip coupling insulation layers 300 may be respectively disposed between the first semiconductor chip 100 and the lowermost second semiconductor chip 200L and between two adjacent second semiconductor chips 200.

[0041] Each of the coupling pads 320 may be formed by respectively forming conductive material layers, e.g., a top chip connection pad 322 and a bottom chip connection pad 324 in FIG. 5A, on facing surfaces of respective two adjacent semiconductor chips among the first semiconductor chip 100 and the second semiconductor chips 200 and performing diffusion bonding such that the conductive material layers facing each other are expanded by heat to contact each other and integrated with each other through the diffusion of metal atoms.

[0042] Each of the chip coupling insulation layers 300 may be formed by respectively forming insulation material layers, e.g., a top chip coupling insulation material layer 302 and a bottom chip coupling insulation material layer 304 in FIGS. 5A to 5C, on facing surfaces of respective two adjacent semiconductor chips among the first semiconductor chip 100 and the second semiconductor chips 200 and performing diffusion bonding, during the formation of the coupling pads 320, such that the insulation material layers facing each other are expanded by heat to contact each other and are then integrated with each other through the diffusion of atoms.

[0043] Among the chip coupling insulation layers 300, a lowermost chip coupling insulation layer 300L disposed between the first semiconductor chip 100 and the lowermost second semiconductor chip 200L may be formed when an insulation material layer, e.g., a lowermost top chip coupling insulation material layer 302L in FIG. 5A, covering the top surface of the first semiconductor chip 100 and an insulation material layer, e.g., the bottom chip coupling insulation material layer 304, covering the bottom surface of the lowermost second semiconductor chip 200L are diffusion bonded.

[0044] The lowermost chip coupling insulation layer 300L may have a first recess 300R in an upper portion such that the thickness of a portion of the lowermost chip coupling insulation layer 300L, which overlaps with the lowermost

second semiconductor chip 200L in the vertical direction, is greater than the thickness of a portion of the lowermost chip coupling insulation layer 300L, which does not overlap with the lowermost second semiconductor chip 200L. The first recess 300R may be disposed in the portion of the lowermost chip coupling insulation layer 300L, which does not overlap with the lowermost second semiconductor chip 200L in the vertical direction. A central portion of the lowermost chip coupling insulation layer 300L, i.e., the portion of the lowermost chip coupling insulation layer 300L which overlaps with the lowermost second semiconductor chip 200L in the vertical direction, may protrude upwards from an edge portion of the lowermost chip coupling insulation layer 300L, i.e., the portion of the lowermost chip coupling insulation layer 300L which does not overlap with the lowermost second semiconductor chip 200L in the vertical direction, and the lowermost chip coupling insulation layer 300L may have a flat/planar bottom surface.

[0045] The lowermost chip coupling insulation layer 300L may completely cover a top surface of the first semiconductor chip 100, which does not overlap with the lowermost second semiconductor chip 200L in the vertical direction. A portion of the top surface of the first semiconductor chip 100, which overlaps with the lowermost second semiconductor chip 200L in the vertical direction, and a portion of the bottom surface of the lowermost second semiconductor chip 200L may be covered with the coupling pads 320, and the other portions thereof may be covered with the lowermost chip coupling insulation layer 300L.

[0046] Each of the chip coupling insulation layers 300, except for the lowermost chip coupling insulation layer 300L and the coupling pads 320, may completely cover the top and bottom surfaces of respective two adjacent second semiconductor chips 200, wherein the top and bottom surfaces thereof face each other. The chip coupling insulation layers 300, except for the lowermost chip coupling insulation layer 300L, may have flat top and bottom surfaces and may thus have substantially the same thickness as each other.

[0047] A supporting dummy substrate 400 may be stacked on the uppermost second semiconductor chip 200H. For example, the supporting dummy substrate 400 may include a semiconductor material such as silicon (Si). In some embodiments, the supporting dummy substrate 400 may include only a semiconductor material. For example, the supporting dummy substrate 400 may be a part of a bare wafer.

[0048] The supporting dummy substrate 400 may have a third horizontal width W3 and a third vertical height H3. In some embodiments, the third horizontal width W3 may be less than each of the first and second horizontal widths W1 and W2. In some embodiments, the third vertical height H3 may be greater than each of the first and second vertical heights H1 and H2. For example, the third vertical height H3 may be about 100 μm to about 500 μm .

[0049] A supporting coupling insulation layer 350 may be disposed between the uppermost second semiconductor chip 200H and the supporting dummy substrate 400. The supporting coupling insulation layer 350 may be formed by respectively forming insulation material layers, e.g., the top chip coupling insulation material layer 302 and a bottom dummy coupling insulation material layer 364 in FIG. 5D, on the top surface of the uppermost second semiconductor chip 200H and the bottom surface of the supporting dummy

substrate 400, which face each other, and performing diffusion bonding such that the insulation material layers facing each other are expanded by heat to contact each other and are integrated with each other through the diffusion of atoms.

[0050] Only a semiconductor material may be exposed on the bottom surface of the supporting dummy substrate 400. Accordingly, the top surface of the supporting coupling insulation layer 350 may be in contact with only the semiconductor material. The supporting coupling insulation layer 350 may completely cover the bottom surface of the supporting dummy substrate 400. In some embodiments, when a plurality of top chip connection pads 322 are arranged on the top surface of the uppermost second semiconductor chip 200H, the supporting coupling insulation layer 350 may at least partially surround the top chip connection pads 322. For example, the supporting coupling insulation layer 350 may cover the top surface, i.e., the inactive surface, of the second semiconductor substrate 210 of the uppermost second semiconductor chip 200H and the side and top surfaces of the top chip connection pads 322. The top chip connection pads 322 may have the supporting coupling insulation layer 350 therebetween and may be separated from the supporting dummy substrate 400. In some embodiments, when the top chip connection pads 322 are omitted from the top surface of the uppermost second semiconductor chip 200H, the supporting coupling insulation layer 350 may cover the top surface of the second semiconductor substrate 210 of the uppermost second semiconductor chip 200H and a plurality of second through electrodes 220 exposed on the top surface of the second semiconductor substrate 210 of the uppermost second semiconductor chip 200H.

[0051] The supporting coupling insulation layer 350 may have a second recess 350R in an upper portion thereof such that the thickness of a portion of the supporting coupling insulation layer 350, which overlaps with the supporting dummy substrate 400 in the vertical direction, is greater than the thickness of a portion of the supporting coupling insulation layer 350, which does not overlap with the supporting dummy substrate 400 in the vertical direction. The second recess 350R may be disposed in the portion of the supporting coupling insulation layer 350, which does not overlap with the supporting dummy substrate 400 in the vertical direction. A central portion of the supporting coupling insulation layer 350, i.e., the portion of the supporting coupling insulation layer 350 which overlaps with the supporting dummy substrate 400 in the vertical direction, may protrude upwards from an edge portion of the supporting coupling insulation layer 350, i.e., the portion of the supporting coupling insulation layer 350 which does not overlap with the supporting dummy substrate 400 in the vertical direction, and the supporting coupling insulation layer 350 may have a flat bottom surface.

[0052] The lowermost chip coupling insulation layer 300L may have the first horizontal width W1. The chip coupling insulation layers 300, except for the lowermost chip coupling insulation layer 300L, may have the second horizontal width W2. The supporting coupling insulation layer 350 may have the second horizontal width W2. The chip coupling insulation layers 300, except for the lowermost chip coupling insulation layer 300L, may overlap with the second semiconductor chips 200 in the vertical direction. Side surfaces of the chip coupling insulation layers 300, except for the lowermost chip coupling insulation layer 300L, may

be aligned and coplanar in the vertical direction with side surfaces of the second semiconductor chips **200**.

[0053] The chip coupling insulation layers **300** and the supporting coupling insulation layer **350** may include SiO, SiN, SiCN, SiCO, or a polymeric material. The polymeric material may include benzocyclobutene (BCB), polyimide (PI), polybenzoxazole (PBO), Si, acrylate, or epoxy. For example, the chip coupling insulation layers **300** and the supporting coupling insulation layer **350** may include silicon oxide. In some embodiments, the chip coupling insulation layers **300** and the supporting coupling insulation layer **350** may include the same material as each other. For example, the chip coupling insulation layers **300** and the supporting coupling insulation layer **350** may have a thickness of about 100 nm to about 1 μ m.

[0054] The semiconductor package **1000** may further include a package molding layer **500** disposed on the first semiconductor chip **100**. The package molding layer **500** may cover the top surface of the first semiconductor chip **100** and at least partially surround the side surfaces of the second semiconductor chips **200** and the side surfaces of the supporting dummy substrate **400**. For example, the package molding layer **500** may include an epoxy mold compound (EMC). In some embodiments, the package molding layer **500** may cover the top surface of the supporting dummy substrate **400**. In some embodiments, the package molding layer **500** might not cover the top surface of the supporting dummy substrate **400**. For example, a heat dissipation unit may be attached to the supporting dummy substrate **400** with a thermal interface material (TIM) disposed between the heat dissipation unit and the supporting dummy substrate **400**.

[0055] In some embodiments, the semiconductor package **1000** may further include a base redistribution layer **600** disposed on the bottom surface of the first semiconductor chip **100**. The base redistribution layer **600** may include a plurality of package redistribution line patterns **620**, a plurality of package redistribution vias **640**, and a package redistribution insulation layer **660**. In some embodiments, a plurality of package redistribution insulation layers **660** may be stacked. For example, the package redistribution insulation layer **660** may be formed from photo imageable dielectric (PID) or photosensitive polyimide (PSPI). For example, the package redistribution line patterns **620** and the package redistribution vias **640** may include a metal, such as copper (Cu), aluminum (Al), tungsten (W), titanium (Ti), tantalum (Ta), indium (In), molybdenum (Mo), manganese (Mn), cobalt (Co), tin (Sn), nickel (Ni), magnesium (Mg), rhenium (Re), beryllium (Be), gallium (Ga), or ruthenium (Ru), or an alloy thereof but are not necessarily limited thereto. In some embodiments, the package redistribution line patterns **620** and the package redistribution vias **640** may be formed by stacking a metal or a metal alloy on a seed layer including titanium, titanium nitride, or titanium tungsten.

[0056] The package redistribution line patterns **620** may be disposed on at least one of the top and bottom surfaces of the package redistribution insulation layer **660**. The package redistribution vias **640** may pass through the package redistribution insulation layer **660** and may be in contact with some of the package redistribution line patterns **620**. In some embodiments, at least some of the package redistribution line patterns **620** may be formed together and integrated with some of the package redistribution vias **640**. For example, a package redistribution line pattern **620** may be

integrated with a package redistribution via **640** that is in contact with the top surface of the package redistribution line pattern **620**. The package redistribution insulation layer **660** may at least partially surround the package redistribution line patterns **620** and the package redistribution vias **640**.

[0057] The package redistribution line patterns **620** and the package redistribution vias **640** may be electrically connected to the chip pads **150**. In some embodiments, at least some of the package redistribution vias **640** may be in contact with the chip pads **150**. For example, when the base redistribution layer **600** includes a stack of a plurality of package redistribution insulation layers **660**, a package redistribution via **640** that passes through an uppermost package redistribution insulation layer **660** may be in contact with and electrically connected to a chip pad **150**.

[0058] In some embodiments, the package redistribution vias **640** may have a tapered shape having a horizontal width decreasing upwards. For example, the horizontal width of the package redistribution vias **640** may increase away from the first semiconductor chip **100**.

[0059] Among the package redistribution line patterns **620**, package redistribution line patterns **620** on the bottom surface of the base redistribution layer **600** may be referred to as package pads **650**. A plurality of package connection terminals **700** may be respectively attached to the package pads **650**. For example, the package connection terminals **700** may include a solder ball or a bump.

[0060] In some embodiments, the semiconductor package **1000** might not include the base redistribution layer **600**. For example, the package connection terminals **700** may be respectively attached to the chip pads **150**.

[0061] The horizontal width and area of the base redistribution layer **600** may be equal to the horizontal width and area of the first semiconductor chip **100**. The base redistribution layer **600** may overlap with the first semiconductor chip **100** in the vertical direction.

[0062] For example, the base redistribution layer **600**, the first semiconductor chip **100**, and the package molding layer **500** may have substantially the same horizontal width and area as one another. The side surfaces of the base redistribution layer **600**, the first semiconductor chip **100**, and the package molding layer **500** may be aligned and coplanar with one another in the vertical direction.

[0063] Due to hybrid bonding in which the coupling pads **320** and the chip coupling insulation layers **300** are formed using diffusion bonding, the first semiconductor chip **100** and the second semiconductor chips **200** may be stacked in the semiconductor package **1000**. Because of the first semiconductor chip **100** and the second semiconductor chips **200** have substantially the same thickness (vertical height) and are relatively thin, there may be flexible bending during hybrid bonding, and therefore, a bonding failure may be prevented from occurring among the first semiconductor chip **100** and the second semiconductor chips **200**, and stress is prevented from concentrating on the first semiconductor chip **100** and the second semiconductor chips **200**, which are bonded to one another, during a subsequent thermal process.

[0064] Because the thickness (vertical height) of the supporting dummy substrate **400** of the semiconductor package **1000** is relatively great, the structural reliability of the semiconductor package **1000** may be increased, and heat may be effectively released to the outside of the semiconductor package **1000** through the supporting dummy sub-

strate **400**. The supporting dummy substrate **400** may be bonded to the uppermost second semiconductor chip **200H** by the supporting coupling insulation layer **350**. The top and bottom surfaces of the supporting coupling insulation layer **350** may be in contact with only semiconductor materials of the supporting dummy substrate **400** and the uppermost second semiconductor chip **200H**. Accordingly, even when a bonding failure occurs between the supporting dummy substrate **400** and the uppermost second semiconductor chip **200H** because the supporting dummy substrate **400** is sufficiently thick to resist bending, the bonding failure might not affect the operational reliability of the semiconductor package **1000**.

[0065] The second semiconductor chips **200** of the semiconductor package **1000** may be formed by the same process, and accordingly, manufacturing processes may be simplified and manufacturing cost may be reduced.

[0066] Referring to FIG. 1B, a semiconductor package **1000a** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** may have the first horizontal width **W1** and the first vertical height **H1**, and each of the second semiconductor chips **200** may have the second horizontal width **W2** and the second vertical height **H2**. A supporting dummy substrate **400a** may be stacked on the uppermost second semiconductor chip **200H**. For example, the supporting dummy substrate **400a** may include a semiconductor material such as silicon (Si). In some embodiments, the supporting dummy substrate **400a** may include only a semiconductor material. For example, the supporting dummy substrate **400a** may be a part of a bare wafer.

[0067] The supporting dummy substrate **400a** may have a third horizontal width **W3a** and the third vertical height **H3**. In some embodiments, the third horizontal width **W3a** may be substantially equal to the second horizontal width **W2**. In some embodiments, the third vertical height **H3** may be greater than each of the first and second vertical heights **H1** and **H2**.

[0068] A supporting coupling insulation layer **350a** may be disposed between the uppermost second semiconductor chip **200H** and the supporting dummy substrate **400a**. The supporting coupling insulation layer **350a** may be formed by respectively forming insulation material layers on the top surface of the uppermost second semiconductor chip **200H** and the bottom surface of the supporting dummy substrate **400a**, which face each other, and performing diffusion bonding such that the insulation material layers facing each other are expanded by heat to contact each other and integrated with each other through the diffusion of atoms.

[0069] Only a semiconductor material may be exposed on the bottom surface of the supporting dummy substrate **400a**. Accordingly, the top surface of the supporting coupling insulation layer **350a** may be in contact with only the semiconductor material. The supporting coupling insulation layer **350a** may completely cover the bottom surface of the supporting dummy substrate **400a**. Because the top chip connection pads **322** are arranged on the top surface of the uppermost second semiconductor chip **200H**, the supporting coupling insulation layer **350a** may at least partially surround the top chip connection pads **322**. For example, the supporting coupling insulation layer **350a** may cover the top surface, i.e., the inactive surface, of the second semicon-

ductor substrate **210** of the uppermost second semiconductor chip **200H** and the side and top surfaces of the top chip connection pads **322**.

[0070] The lowermost chip coupling insulation layer **300L** may have the first horizontal width **W1**. The chip coupling insulation layers **300**, except for the lowermost chip coupling insulation layer **300L**, may have the second horizontal width **W2**. The supporting coupling insulation layer **350a** may have the second horizontal width **W2**. The chip coupling insulation layers **300**, except for the lowermost chip coupling insulation layer **300L**, may overlap with the second semiconductor chips **200**, the supporting coupling insulation layer **350a**, and the supporting dummy substrate **400a** in the vertical direction. Side surfaces of the chip coupling insulation layers **300**, except for the lowermost chip coupling insulation layer **300L**, side surfaces of the second semiconductor chips **200**, a side surface of the supporting coupling insulation layer **350a**, and a side surface of the supporting dummy substrate **400a** may be aligned and coplanar with one another in the vertical direction.

[0071] The semiconductor package **1000a** may further include a package molding layer **500** on the first semiconductor chip **100**. The package molding layer **500** may cover the top surface of the first semiconductor chip **100** and at least partially surround the side surfaces of the second semiconductor chips **200** and the side surfaces of the supporting dummy substrate **400a**. In some embodiments, the package molding layer **500** may cover the top surface of the supporting dummy substrate **400a**. In some embodiments, the package molding layer **500** might not cover the top surface of the supporting dummy substrate **400a**.

[0072] Referring to FIG. 1C, a semiconductor package **1000b** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The semiconductor package **1000b** may include a plurality of thermal coupling pads **320a** instead of the top chip connection pads **322** of the semiconductor package **1000** of FIG. 1A. The thermal coupling pads **320a** may be disposed between the top surface of the uppermost second semiconductor chip **200H** and the bottom surface of the supporting dummy substrate **400**. Each of the thermal coupling pads **320a** may be formed by respectively forming conductive material layers, e.g., a top chip connection pad **322** and a bottom dummy pad **328** in FIG. 7A, on respective facing surfaces of the uppermost second semiconductor chip **200H** and the supporting dummy substrate **400** and performing diffusion bonding such that the conductive material layers facing each other are expanded by heat to contact each other and integrated with each other through diffusion of metal atoms. For example, the thermal coupling pads **320a** may include a material including copper (Cu).

[0073] The top surface of each of the thermal coupling pads **320a** may be in contact with the bottom surface of the supporting dummy substrate **400**, and the bottom surface of each of the thermal coupling pads **320a** may be in contact with a second through electrode **220** of the uppermost second semiconductor chip **200H**. In some embodiments, the bottom surface of each of the thermal coupling pads **320a** may cover the top surface of a second through electrode **220** of the uppermost second semiconductor chip **200H** and a portion of the top surface of the second semiconductor substrate **210** of the uppermost second semiconductor chip **200H**, which is adjacent to the top surface of the second through electrode **220**. The supporting coupling

insulation layer 350 may at least partially surround the thermal coupling pads 320a. For example, the supporting coupling insulation layer 350 may cover the side surfaces of the thermal coupling pads 320a and may fill a space between the uppermost second semiconductor chip 200H and the supporting dummy substrate 400.

[0074] Because the semiconductor package 1000b includes the thermal coupling pads 320a and the supporting coupling insulation layer 350 disposed between the uppermost second semiconductor chip 200H and the supporting dummy substrate 400, an adhesive strength between the uppermost second semiconductor chip 200H and the supporting dummy substrate 400 may increase, thereby increasing structural reliability, and heat transmission from the uppermost second semiconductor chip 200H to the supporting dummy substrate 400 may increase, thereby increasing the ability of the semiconductor package 1000b to release heat.

[0075] Referring to FIG. 1D, a semiconductor package 1000c may include the first semiconductor chip 100 and the second semiconductor chips 200. The semiconductor package 1000c may include a plurality of thermal coupling pads 320a instead of the top chip connection pads 322 of the semiconductor package 1000a of FIG. 1B. The supporting coupling insulation layer 350a may at least partially surround the thermal coupling pads 320a. For example, the supporting coupling insulation layer 350a may cover the side surfaces of the thermal coupling pads 320a and may fill a space between the uppermost second semiconductor chip 200H and the supporting dummy substrate 400.

[0076] Referring to FIG. 2A, a semiconductor package 1002 may include the first semiconductor chip 100 and the second semiconductor chips 200. The first semiconductor chip 100 and the second semiconductor chips 200 of the semiconductor package 1002 may be electrically connected to each other through the coupling pads 320 and may thus exchange signals with each other and provide power and ground.

[0077] The first semiconductor chip 100 may have the first horizontal width W1 and the first vertical height H1, and each of the second semiconductor chips 200 may have the second horizontal width W2 and the second vertical height H2.

[0078] A plurality of supporting dummy substrates 402 may be sequentially stacked on the uppermost second semiconductor chip 200H. For example, the supporting dummy substrates 402 may include a semiconductor material such as silicon (Si). In some embodiments, the supporting dummy substrates 402 may include only a semiconductor material. For example, each of the supporting dummy substrates 402 may be a part of a bare wafer. The total vertical height of the stack of the supporting dummy substrates 402 may be greater than the second vertical height H2. For example, the total vertical height of the stack of the supporting dummy substrates 402 may be about 100 μm to about 500 μm .

[0079] Each of the supporting dummy substrates 402 may have the third horizontal width W3 and a third vertical height H3a. In some embodiments, the third horizontal width W3 may be less than each of the first and second horizontal widths W1 and W2. In some embodiments, the third vertical height H3a may be substantially equal to each of the first and second vertical heights H1 and H2. For example, the third vertical height H3a may be about 50 μm to about 90 μm . In some embodiments, the third vertical

height H3a may be less than each of the first and second vertical heights H1 and H2. For example, the third vertical height H3a may be substantially equal to the vertical height of each of the first and second semiconductor substrates 110 and 210 and may be several μm less than each of the first and second vertical heights H1 and H2.

[0080] In some embodiments, the vertical height of an uppermost supporting dummy substrate 402H among the supporting dummy substrates 402 may be substantially equal to the vertical height of the other supporting dummy substrates 402. In some embodiments, the vertical height of an uppermost supporting dummy substrate 402H among the supporting dummy substrates 402 may be less than the vertical height of the other supporting dummy substrates 402.

[0081] A supporting coupling insulation layer 352 may be disposed between the uppermost second semiconductor chip 200H and a lowermost supporting dummy substrate 402 at the bottom of the stack of the supporting dummy substrates 402, and a supporting coupling insulation layer 352 may be disposed between two adjacent supporting dummy substrates 402. Among a plurality of supporting coupling insulation layers 352, a supporting coupling insulation layer 352 disposed between the uppermost second semiconductor chip 200H and the lowermost supporting dummy substrate 402 may be referred to as a lowermost supporting coupling insulation layer 352L.

[0082] The lowermost supporting coupling insulation layer 352L may have a second recess 352R in an upper portion thereof such that the thickness of a portion of the lowermost supporting coupling insulation layer 352L, which overlaps with the supporting dummy substrates 402 in the vertical direction, is greater than the thickness of a portion of the lowermost supporting coupling insulation layer 352L, which does not overlap with the supporting dummy substrates 402 in the vertical direction. The second recess 352R may be disposed in the portion of the lowermost supporting coupling insulation layer 352L, which does not overlap with the supporting dummy substrates 402 in the vertical direction. A central portion of the lowermost supporting coupling insulation layer 352L, i.e., the portion of the lowermost supporting coupling insulation layer 352L which overlaps with the supporting dummy substrates 402 in the vertical direction, may protrude upwards from an edge portion of the lowermost supporting coupling insulation layer 352L, i.e., the portion of the lowermost supporting coupling insulation layer 352L which does not overlap with the supporting dummy substrates 402 in the vertical direction, and the lowermost supporting coupling insulation layer 352L may have a flat bottom surface.

[0083] The top surface of the lowermost supporting coupling insulation layer 352L among the supporting coupling insulation layers 352 may be in contact with only a semiconductor material. The lowermost supporting coupling insulation layer 352L may completely cover the bottom surface of the lowermost supporting dummy substrate 402. Because the top chip connection pads 322 are arranged on the top surface of the uppermost second semiconductor chip 200H, the lowermost supporting coupling insulation layer 352L may at least partially surround the top chip connection pads 322. For example, the lowermost supporting coupling insulation layer 352L may cover the top surface, i.e., the inactive surface, of the second semiconductor substrate 210

of the uppermost second semiconductor chip 200H and the side and top surfaces of the top chip connection pads 322.

[0084] Only a semiconductor material may be exposed on the bottom and top surfaces of each of the supporting dummy substrates 402. Each of the supporting coupling insulation layers 352, except for the lowermost supporting coupling insulation layer 352L, may completely cover the top and bottom surfaces of respective two adjacent supporting dummy substrates 402, wherein the top and bottom surfaces thereof face each other. Accordingly, the top and bottom surfaces of each of the supporting coupling insulation layers 352, except for the lowermost supporting coupling insulation layer 352L, may be in contact with only a semiconductor material. The supporting coupling insulation layers 352, except for the lowermost supporting coupling insulation layer 352L, may have flat top and bottom surfaces and may thus have substantially the same thickness as each other.

[0085] Referring to FIG. 2B, a semiconductor package 1002a may include the first semiconductor chip 100 and the second semiconductor chips 200. The first semiconductor chip 100 and the second semiconductor chips 200 of the semiconductor package 1002a may be electrically connected to each other through the coupling pads 320 and may thus exchange signals with each other and provide power and ground.

[0086] The first semiconductor chip 100 may have the first horizontal width W1 and the first vertical height H1, and each of the second semiconductor chips 200 may have the second horizontal width W2 and the second vertical height H2.

[0087] A plurality of supporting dummy substrates 402a may be sequentially stacked on the uppermost second semiconductor chip 200H. For example, the supporting dummy substrates 402a may include a semiconductor material such as silicon (Si). In some embodiments, the supporting dummy substrates 402a may include only a semiconductor material. For example, each of the supporting dummy substrates 402a may be a part of a bare wafer. Each of the supporting dummy substrate 402a may have the third horizontal width W3a and a third vertical height H3a. In some embodiments, the third horizontal width W3a may be substantially equal to the second horizontal width W2.

[0088] A supporting coupling insulation layer 352a may be disposed between the uppermost second semiconductor chip 200H and a lowermost supporting dummy substrate 402a at the bottom of the stack of the supporting dummy substrates 402a, and a supporting coupling insulation layer 352a may be disposed between two adjacent supporting dummy substrates 402a.

[0089] Each of the supporting coupling insulation layers 352a may completely cover the bottom surface of the lowermost supporting dummy substrate 402a or the top and bottom surfaces of respective two adjacent supporting dummy substrates 402a, wherein the top and bottom surfaces thereof face each other. The supporting coupling insulation layers 352a may have flat top and bottom surfaces and may thus have substantially the same thickness as each other.

[0090] Among the supporting coupling insulation layer 352a, a lowermost supporting coupling insulation layer 352a may at least partially surround the top chip connection pads 322. For example, the lowermost supporting coupling insulation layer 352a may cover the top surface, i.e., the

inactive surface, of the second semiconductor substrate 210 of the uppermost second semiconductor chip 200H and the side and top surfaces of the top chip connection pads 322.

[0091] The supporting coupling insulation layers 352a may have the third horizontal width W3a. The chip coupling insulation layers 300, except for the lowermost chip coupling insulation layer 300L, may overlap with the second semiconductor chips 200, the supporting coupling insulation layers 352a, and the supporting dummy substrates 402a in the vertical direction. Side surfaces of the chip coupling insulation layers 300, except for the lowermost chip coupling insulation layer 300L, side surfaces of the second semiconductor chips 200, side surfaces of the supporting coupling insulation layers 352a, and side surfaces of the supporting dummy substrates 402a may be aligned and coplanar with one another in the vertical direction.

[0092] Referring to FIG. 2C, a semiconductor package 1002b may include the first semiconductor chip 100 and the second semiconductor chips 200. The semiconductor package 1002b may include a plurality of thermal coupling pads 320a instead of the top chip connection pads 322 of the semiconductor package 1002 of FIG. 2A. The thermal coupling pads 320a may be disposed between the top surface of the uppermost second semiconductor chip 200H and the bottom surface of the lowermost supporting dummy substrate 402.

[0093] Because the thermal coupling pads 320a are disposed between the uppermost second semiconductor chip 200H and the lowermost supporting dummy substrate 402, the lowermost supporting coupling insulation layer 352L may at least partially surround the thermal coupling pads 320a. For example, the lowermost supporting coupling insulation layer 352L may cover the side surfaces of the thermal coupling pads 320a and may fill a space between the uppermost second semiconductor chip 200H and the lowermost supporting dummy substrate 402.

[0094] Referring to FIG. 2D, a semiconductor package 1002c may include the first semiconductor chip 100 and the second semiconductor chips 200. The semiconductor package 1002c may include a plurality of thermal coupling pads 320a instead of the top chip connection pads 322 of the semiconductor package 1002a of FIG. 2B. The lowermost supporting coupling insulation layer 352a may at least partially surround the thermal coupling pads 320a. For example, the lowermost supporting coupling insulation layer 352a may cover the side surfaces of the thermal coupling pads 320a and may fill a space between the uppermost second semiconductor chip 200H and the lowermost supporting dummy substrate 402a.

[0095] Referring to FIG. 2E, a semiconductor package 1002d may include the first semiconductor chip 100 and the second semiconductor chips 200. Compared to the semiconductor package 1002b of FIG. 2C, the semiconductor package 1002d may further include a plurality of dummy coupling pads 320b.

[0096] The dummy coupling pads 320b may be disposed between two adjacent supporting dummy substrates 402. The top and bottom surfaces of each of the dummy coupling pads 320b may be in contact with the bottom and top surfaces of respective two adjacent lowermost supporting dummy substrates 402, which face each other. Each of the dummy coupling pads 320b may be formed by respectively forming conductive material layers, e.g., a top dummy pad 326 and a bottom dummy pad 328 in FIGS. 10A to 10C, on

respective facing top and bottom surfaces of the two adjacent supporting dummy substrates **402** and performing diffusion bonding such that the conductive material layers facing each other are expanded by heat to contact each other and are then integrated with each other through the diffusion of metal atoms. For example, the dummy coupling pads **320b** may include a material including Copper (Cu).

[0097] Each of the supporting coupling insulation layers **352**, except for the lowermost supporting coupling insulation layer **352L**, may at least partially surround the dummy coupling pads **320b**. For example, each of the supporting coupling insulation layers **352**, except for the lowermost supporting coupling insulation layer **352L**, may cover the side surfaces of the dummy coupling pads **320b** and may fill a space between two adjacent supporting dummy substrates **402**.

[0098] Because the semiconductor package **1002d** includes the dummy coupling pads **320b** disposed between two adjacent supporting dummy substrates **402**, an adhesive strength between the supporting dummy substrates **402** may increase, thereby increasing structural reliability, and heat transmission through the supporting dummy substrates **402** may increase, thereby increasing the ability of the semiconductor package **1002d** to release heat.

[0099] Referring to FIG. 2F, a semiconductor package **1002e** may include the first semiconductor chip **100** and the second semiconductor chips **200**. Compared to the semiconductor package **1002c** of FIG. 2D, the semiconductor package **1002e** may further include a plurality of dummy coupling pads **320b**. The supporting coupling insulation layers **352a**, except for the lowermost supporting coupling insulation layer **352a**, may at least partially surround the dummy coupling pads **320b**. For example, each of the supporting coupling insulation layers **352a**, except for the lowermost supporting coupling insulation layer **352a**, may cover the side surfaces of the dummy coupling pads **320b** and may fill a space between two adjacent supporting dummy substrates **402a**.

[0100] Referring to FIG. 2G, a semiconductor package **1002f** may include the first semiconductor chip **100** and the second semiconductor chips **200**. Compared to the semiconductor package **1002d** of FIG. 2E, the semiconductor package **1002f** may further include a plurality of top dummy pads **326**.

[0101] The top dummy pads **326** may be disposed on the top surface of the uppermost supporting dummy substrate **402H**. In some embodiments, the package molding layer **500** may at least partially surround the top dummy pads **326**. For example, the package molding layer **500** may cover the top surface of the uppermost supporting dummy substrate **402H** and the side surfaces of the top dummy pads **326** and may expose the top surfaces of the top dummy pads **326**.

[0102] Because the semiconductor package **1002f** includes the top dummy pads **326** on the top surface of the uppermost supporting dummy substrate **402H**, the ability of the semiconductor package **1002f** to release heat may be increased.

[0103] Referring to FIG. 2H, a semiconductor package **1002g** may include the first semiconductor chip **100** and the second semiconductor chips **200**. Compared to the semiconductor package **1002e** of FIG. 2F, the semiconductor package **1002g** may further include a plurality of top dummy pads **326**.

[0104] The top dummy pads **326** may be disposed on the top surface of an uppermost supporting dummy substrate

402aH. In some embodiments, the package molding layer **500** may at least partially surround the top dummy pads **326**. For example, the package molding layer **500** may cover the top surface of the uppermost supporting dummy substrate **402aH** and the side surfaces of the top dummy pads **326** and may expose the top surfaces of the top dummy pads **326**.

[0105] Referring to FIG. 3A, a semiconductor package **1004** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1004** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1004** may include a plurality of supporting dummy substrates **402b** instead of the supporting dummy substrates **402** of the semiconductor package **1002** of FIG. 2A.

[0106] The supporting dummy substrates **402b** may be stacked on the uppermost second semiconductor chip **200H**. For example, the supporting dummy substrates **402b** may include a semiconductor material such as silicon (Si). In some embodiments, the supporting dummy substrates **402b** may include only a semiconductor material. For example, each of the supporting dummy substrates **402b** may be a part of a bare wafer. The total vertical height of the stack of the supporting dummy substrates **402b** may be greater than the second vertical height **H2**. For example, the total vertical height of the stack of the supporting dummy substrates **402b** may be about 100 μm to about 500 μm .

[0107] Each of the supporting dummy substrates **402b**, except for an uppermost supporting dummy substrate **402bH**, may have the third horizontal width **W3** and the third vertical height **H3a**. The uppermost supporting dummy substrate **402bH** may have the third horizontal width **W3** and a fourth vertical height **H4**. The fourth vertical height **H4** may be less than the third vertical height **H3a**. For example, the fourth vertical height **H4** may be several μm less than the third vertical height **H3a**.

[0108] Referring to FIG. 3B, a semiconductor package **1004a** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1004a** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1004a** may include a plurality of supporting dummy substrates **402c** instead of the supporting dummy substrates **402a** of the semiconductor package **1002a** of FIG. 2B.

[0109] The supporting dummy substrates **402c** may be stacked on the uppermost second semiconductor chip **200H**. For example, the supporting dummy substrates **402c** may include a semiconductor material such as silicon (Si). In some embodiments, the supporting dummy substrates **402c** may include only a semiconductor material. For example, each of the supporting dummy substrates **402c** may be a part of a bare wafer. A supporting dummy substrate **402c** at the top among the supporting dummy substrates **402c** may be referred to as an uppermost supporting dummy substrate **402cH**.

[0110] The supporting dummy substrates **402c**, except for the uppermost supporting dummy substrate **402cH**, may have the third horizontal width **W3a** and the third vertical height **H3a**. The uppermost supporting dummy substrate

402cH may have the third horizontal width **W3a** and the fourth vertical height **H4**. The fourth vertical height **H4** may be less than the third vertical height **H3a**. For example, the fourth vertical height **H4** may be several μm less than the third vertical height **H3a**.

[0111] Referring to FIG. 3C, a semiconductor package **1004b** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1004b** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1004b** may include a plurality of supporting dummy substrates **402b** instead of the supporting dummy substrates **402** of the semiconductor package **1002b** of FIG. 2C.

[0112] Referring to FIG. 3D, a semiconductor package **1004c** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1004c** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1004c** may include a plurality of supporting dummy substrates **402c** instead of the supporting dummy substrates **402a** of the semiconductor package **1002c** of FIG. 2D.

[0113] Referring to FIG. 3E, a semiconductor package **1004d** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1004d** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1004d** may include a plurality of supporting dummy substrates **402b** instead of the supporting dummy substrates **402** of the semiconductor package **1002d** of FIG. 2E.

[0114] Referring to FIG. 3F, a semiconductor package **1004e** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1004e** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1004e** may include a plurality of supporting dummy substrates **402c** instead of the supporting dummy substrates **402a** of the semiconductor package **1002e** of FIG. 2F.

[0115] Referring to FIG. 4A, a semiconductor package **1006** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006** may include a plurality of supporting dummy substrates **404** instead of the supporting dummy substrates **402** of the semiconductor package **1002** of FIG. 2A.

[0116] The supporting dummy substrates **404** may be stacked on the uppermost second semiconductor chip **200H**. For example, the supporting dummy substrates **404** may include a semiconductor material such as silicon (Si). In

some embodiments, the supporting dummy substrates **404** may include only a semiconductor material. For example, each of the supporting dummy substrates **404** may be a part of a bare wafer. A supporting dummy substrate **404** at the top among the supporting dummy substrates **404** may be referred to as an uppermost supporting dummy substrate **404H**. The total vertical height of the stack of the supporting dummy substrates **404** may be greater than the second vertical height **H2**. For example, the total vertical height of the stack of the supporting dummy substrates **404** may be about $100\text{ }\mu\text{m}$ to about $500\text{ }\mu\text{m}$.

[0117] The supporting dummy substrates **404**, except for the uppermost supporting dummy substrate **404H**, may have the third horizontal width **W3** and the third vertical height **H3a**. The uppermost supporting dummy substrate **404H** may have the third horizontal width **W3** and a fourth vertical height **H4a**. The fourth vertical height **H4a** may be greater than the third vertical height **H3a**. For example, the fourth vertical height **H4a** may be several tens to hundreds of μm greater than the third vertical height **H3a**.

[0118] Referring to FIG. 4B, a semiconductor package **1006a** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006a** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006a** may include a plurality of supporting dummy substrates **402a** instead of the supporting dummy substrates **402a** of the semiconductor package **1002a** of FIG. 2B.

[0119] The supporting dummy substrates **404a** may be stacked on the uppermost second semiconductor chip **200H**. For example, the supporting dummy substrates **404a** may include a semiconductor material such as silicon (Si). In some embodiments, the supporting dummy substrates **404a** may include only a semiconductor material. For example, each of the supporting dummy substrates **404a** may be a part of a bare wafer. A supporting dummy substrate **404a** at the top among the supporting dummy substrates **404a** may be referred to as an uppermost supporting dummy substrate **404aH**.

[0120] The supporting dummy substrates **404a**, except for the uppermost supporting dummy substrate **404aH**, may have the third horizontal width **W3a** and the third vertical height **H3a**. The uppermost supporting dummy substrate **404aH** may have the third horizontal width **W3a** and the fourth vertical height **H4a**. The fourth vertical height **H4a** may be greater than the third vertical height **H3a**. For example, the fourth vertical height **H4a** may be several tens to hundreds of μm greater than the third vertical height **H3a**.

[0121] Referring to FIG. 4C, a semiconductor package **1006b** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006b** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006b** may include a plurality of supporting dummy substrates **404** instead of the supporting dummy substrates **402** of the semiconductor package **1002b** of FIG. 2C.

[0122] Referring to FIG. 4D, a semiconductor package **1006c** may include the first semiconductor chip **100** and the

second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006c** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006c** may include a plurality of supporting dummy substrates **404a** instead of the supporting dummy substrates **402a** of the semiconductor package **1002c** of FIG. 2D.

[0123] Referring to FIG. 4E, a semiconductor package **1006d** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006d** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006d** may include a plurality of supporting dummy substrates **404** instead of the supporting dummy substrates **402** of the semiconductor package **1002d** of FIG. 2E.

[0124] Referring to FIG. 4F, a semiconductor package **1006e** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006e** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006e** may include a plurality of supporting dummy substrates **404a** instead of the supporting dummy substrates **402a** of the semiconductor package **1002e** of FIG. 2F.

[0125] Referring to FIG. 4G, a semiconductor package **1006f** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006f** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006f** may include a plurality of supporting dummy substrates **404** instead of the supporting dummy substrates **402** of the semiconductor package **1002f** of FIG. 2G.

[0126] Referring to FIG. 4H, a semiconductor package **1006g** may include the first semiconductor chip **100** and the second semiconductor chips **200**. The first semiconductor chip **100** and the second semiconductor chips **200** of the semiconductor package **1006g** may be electrically connected to each other through the coupling pads **320** and may thus exchange signals with each other and provide power and ground. The semiconductor package **1006g** may include a plurality of supporting dummy substrates **404a** instead of the supporting dummy substrates **402a** of the semiconductor package **1002g** of FIG. 2H.

[0127] FIGS. 5A to 5I are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. For example, FIGS. 5A to 5I are cross-sectional views of a method of manufacturing the semiconductor package **1000** of FIG. 1A, and to the extent that a detailed description of an element has been omitted, it may be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0128] Referring to FIG. 5A, a plurality of top chip connection pads **322** and the top chip coupling insulation

material layer **302** are formed on the top surface of the first semiconductor chip **100**. The top chip connection pads **322** may be disposed on the top surface, i.e., the inactive surface, of the first semiconductor chip **100**. The top chip connection pads **322** may be disposed on the top surface of the first semiconductor chip **100** and may be connected to a plurality of first through electrodes **120**. The top chip coupling insulation material layer **302** may be formed on the top surface, i.e., the inactive surface, of the first semiconductor chip **100** to at least partially surround the side surfaces of the top chip connection pads **322**. The top chip coupling insulation material layer **302** may cover the top surface of the first semiconductor chip **100** and the side surfaces of the top chip connection pads **322** and may expose the top surfaces of the top chip connection pads **322**.

[0129] The first semiconductor chip **100** having formed there on the top chip connection pads **322** and the top chip coupling insulation material layer **302** is attached to a first supporting substrate **10**. After a first release film **20** is attached to the top surface of the first supporting substrate **10**, the first semiconductor chip **100** may be attached to the first release film **20**. The first semiconductor chip **100** may be attached to the first release film **20** such that the first wiring structure **130** faces the first supporting substrate **10**.

[0130] A plurality of top chip connection pads **322** and a top chip coupling insulation material layer **302** are formed on the top surface of the second semiconductor chip **200**. The top chip connection pads **322** may be disposed on the top surface, i.e., the inactive surface, of the second semiconductor chip **200**. The top chip connection pads **322** may be disposed on the top surface of the second semiconductor chip **200** and may be connected to a plurality of second through electrodes **220**. The top chip coupling insulation material layer **302** may be formed on the top surface, i.e., the inactive surface, of the second semiconductor chip **200** to at least partially surround the side surfaces of the top chip connection pads **322**. The top chip coupling insulation material layer **302** may cover the top surface of the second semiconductor chip **200** and the side surfaces of the top chip connection pads **322** and may expose the top surfaces of the top chip connection pads **322**.

[0131] A plurality of bottom chip connection pads **324** and the bottom chip coupling insulation material layer **304** are formed on the bottom surface of the second semiconductor chip **200**. The bottom chip connection pads **324** may be disposed on the bottom surface of the second semiconductor chip **200**, i.e., the bottom surface of the second wiring structure **230**. Each of the bottom chip connection pads **324** may be disposed on the bottom surface of the second semiconductor chip **200** and may be connected to a second wiring pattern **232** and/or a second wiring via **234**. The bottom chip coupling insulation material layer **304** may be formed on the bottom surface of the second semiconductor chip **200** to at least partially surround the side surfaces of the bottom chip connection pads **324**. The bottom chip coupling insulation material layer **304** may cover the bottom surface of the second semiconductor chip **200** and the side surfaces of the bottom chip connection pads **324** and may expose the bottom surfaces of the bottom chip connection pads **324**.

[0132] Among the top chip coupling insulation material layers **302** respectively formed on the top surfaces of the first and second semiconductor chips **100** and **200**, the top chip coupling insulation material layer **302** on the top surface of

the first semiconductor chip **100** may be separately referred to as the lowermost top chip coupling insulation material layer **302L**.

[0133] The second semiconductor chip **200** is disposed on the first semiconductor chip **100**. The second semiconductor chip **200** may correspond to the lowermost second semiconductor chip **200L** in FIG. 1A. The lowermost second semiconductor chip **200L** may be disposed on the first semiconductor chip **100** such that the second wiring structure **230** faces the first semiconductor chip **100**. The lowermost second semiconductor chip **200L** may be disposed on the first semiconductor chip **100** such that the bottom chip connection pads **324** on the bottom surface of the lowermost second semiconductor chip **200L** may respectively correspond to the top chip connection pads **322** on the top surface of the first semiconductor chip **100**. For example, the top chip connection pads **322** and the bottom chip connection pads **324** may include a material including copper (Cu).

[0134] Referring to FIGS. 5A and 5B, by applying heat and/or pressure in a process of disposing the lowermost second semiconductor chip **200L** on the first semiconductor chip **100**, the top chip connection pads **322** may be bonded to the bottom chip connection pads **324**, and the top chip coupling insulation material layer **302** may be bonded to the bottom chip coupling insulation material layer **304**. In some embodiments, the top chip connection pads **322** may be covalently bonded to the bottom chip connection pads **324**, and the top chip coupling insulation material layer **302** may be covalently bonded to the bottom chip coupling insulation material layer **304**. For example, heat at a first temperature may be applied in the process of disposing the second semiconductor chip **200** on the first semiconductor chip **100**.

[0135] Thereafter, heat at a second temperature that is higher than the first temperature is applied such that the top chip connection pads **322** and the bottom chip connection pads **324** combine to form a plurality of coupling pads **320** and the top chip coupling insulation material layer **302** and the bottom chip coupling insulation material layer **304** combine to form a chip coupling insulation layer **300**. The chip coupling insulation layer **300** disposed between the first semiconductor chip **100** and the lowermost second semiconductor chip **200L** may be separately referred to as the lowermost chip coupling insulation layer **300L**. The top chip connection pads **322** and the bottom chip connection pads **324** respectively corresponding to the top chip connection pads **322** may be expanded by heat to contact each other and diffusion bonded to each other and may be integrated with each other through diffusion of metal atoms, thereby forming the coupling pads **320**.

[0136] Referring to FIG. 5C, a plurality of second semiconductor chips **200** may be sequentially disposed on the lowermost second semiconductor chip **200L**. A plurality of bottom chip connection pads **324** and a bottom chip coupling insulation material layer **304** may be formed on the bottom surface of each of the second semiconductor chips **200** sequentially disposed on the lowermost second semiconductor chip **200L**, and the top chip connection pads **322** and a top chip coupling insulation material layer **302** may be formed on the top surface of each second semiconductor chip **200**.

[0137] Thereafter, in a similar manner to that described above with reference to FIG. 5B, the top chip connection pads **322** and the bottom chip connection pads **324** combine to form a plurality of coupling pads **320** and the top chip

coupling insulation material layer **302** and the bottom chip coupling insulation material layer **304** combine to form a chip coupling insulation layer **300**, disposed between two adjacent second semiconductor chips **200**, and accordingly, the second semiconductor chips **200** may be sequentially attached to the first semiconductor chip **100**.

[0138] Referring to FIG. 5D, after the bottom dummy coupling insulation material layer **364** is formed on the bottom surface of the supporting dummy substrate **400**, the supporting dummy substrate **400** is disposed on the uppermost second semiconductor chip **200H**.

[0139] The second semiconductor chips **200** may have the second horizontal width **W2**, and the supporting dummy substrate **400** may have the third horizontal width **W3** that is less than the second horizontal width **W2**. In some embodiments, the third horizontal width **W3** may be several tens to hundreds of μm less than the second horizontal width **W2**.

[0140] The supporting dummy substrate **400** may be disposed on the uppermost second semiconductor chip **200H** using the edge of the uppermost second semiconductor chip **200H** as an align key.

[0141] Referring to FIG. 5E, in a similar manner to that described above with reference to FIG. 5B, the top chip coupling insulation material layer **302** and the bottom dummy coupling insulation material layer **364** combine to form the supporting coupling insulation layer **350** disposed between the uppermost second semiconductor chip **200H** and the supporting dummy substrate **400**, and accordingly, the supporting dummy substrate **400** may be attached to the uppermost second semiconductor chip **200H**.

[0142] Referring to FIG. 5F, the package molding layer **500** may be formed on the first semiconductor chip **100** to cover the top surface of the first semiconductor chip **100** and at least partially surround the side surfaces of the supporting dummy substrate **400** and the second semiconductor chips **200**.

[0143] After the package molding layer **500** is formed, the first supporting substrate **10** having the first release film **20** attached thereto may be separated from the first semiconductor chip **100**.

[0144] Referring to FIG. 5G, the resultant structure of FIG. 5F is turned upside down and attached to a second supporting substrate **12**. After a second release film **22** is attached to the top surface of the second supporting substrate **12**, the resultant structure of FIG. 5F that has been turned upside down may be attached to the second release film **22**. The supporting dummy substrate **400** and the package molding layer **500** may be in contact with the second release film **22**.

[0145] Referring to FIG. 5H, the base redistribution layer **600** is formed on the first wiring structure **130** of the first semiconductor chip **100**. The base redistribution layer **600** may include the package redistribution line patterns **620**, the package redistribution vias **640**, and the package redistribution insulation layer **660**. At least some of the package redistribution vias **640** or at least some of the package redistribution line patterns **620** may be in contact with the chip pads **150**. Among the package redistribution line patterns **620**, package redistribution line patterns **620** on the top surface of the base redistribution layer **600** may be referred to as package pads **650**.

[0146] In some embodiments, the package redistribution vias **640** may have a tapered shape having a horizontal width

decreasing downwards. For example, the horizontal width of the package redistribution vias **640** may increase away from the first semiconductor chip **100**.

[0147] Referring to FIG. **5I**, the package connection terminals **700** are respectively attached to the package pads **650**.

[0148] Thereafter, the second supporting substrate **12** having the second release film **22** attached thereto may be separated from the supporting dummy substrate **400** and the package molding layer **500**, and then the resultant structure is turned upside down. As a result, the semiconductor package **1000** of FIG. **1A** may be formed.

[0149] FIGS. **6A** and **6B** are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. For example, FIGS. **6A** and **6B** are cross-sectional views of a method of manufacturing the semiconductor package **1000a** of FIG. **1B**, and to the extent that a detailed description of an element has been omitted, it may be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0150] Referring to FIG. **6A**, after the bottom dummy coupling insulation material layer **364** is formed on the bottom surface of the supporting dummy substrate **400a**, the supporting dummy substrate **400a** is disposed on the uppermost second semiconductor chip **200H** of the resultant structure of FIG. **5C**.

[0151] The second semiconductor chips **200** may have the second horizontal width **W2**, and the supporting dummy substrate **400a** may have the third horizontal width **W3a**. In some embodiments, the third horizontal width **W3a** may be substantially equal to the second horizontal width **W2**.

[0152] The supporting dummy substrate **400a** may be disposed on the uppermost second semiconductor chip **200H** such that the edge of the supporting dummy substrate **400a** is aligned with the edge of the uppermost second semiconductor chip **200H**.

[0153] Referring to FIGS. **6A** and **6B**, in a similar manner to that described above with reference to FIG. **5B**, the top chip coupling insulation material layer **302** and a bottom dummy coupling insulation material layer **364a** combine to form the supporting coupling insulation layer **350a** disposed between the uppermost second semiconductor chip **200H** and the supporting dummy substrate **400a**, and accordingly, the supporting dummy substrate **400a** may be attached to the uppermost second semiconductor chip **200H**.

[0154] Thereafter, the semiconductor package **1000a** of FIG. **1B** may be formed using the method described with reference to FIGS. **5F** to **5I**.

[0155] FIGS. **7A** and **7B** are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. For example, FIGS. **7A** and **7B** are cross-sectional views of a method of manufacturing the semiconductor package **1000b** of FIG. **1C**, and to the extent that a detailed description of an element has been omitted, it may be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0156] Referring to FIG. **7A**, after a plurality of bottom dummy pads **328** and the bottom dummy coupling insulation material layer **364** at least partially surrounding the side surfaces of the bottom dummy pads **328** are formed on the bottom surface of the supporting dummy substrate **400**, the supporting dummy substrate **400** is disposed on the upper-

most second semiconductor chip **200H** of the resultant structure of FIG. **5C**. For example, the bottom dummy pads **328** may include a material including copper (Cu).

[0157] Referring to FIGS. **7A** and **7B**, when heat and/or pressure is applied in a process of disposing the supporting dummy substrate **400** on the uppermost second semiconductor chip **200H**, the top chip connection pads **322** and the bottom dummy pads **328** may combine to form the thermal coupling pads **320a**, and the top chip coupling insulation material layer **302** and the bottom dummy coupling insulation material layer **364** may combine to form the supporting coupling insulation layer **350**.

[0158] Thereafter, the semiconductor package **1000b** of FIG. **1C** may be formed using the method described with reference to FIGS. **5F** to **5I**.

[0159] FIGS. **8A** to **8D** are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. For example, FIGS. **8A** to **8D** are cross-sectional views of a method of manufacturing the semiconductor package **1002** of FIG. **2A**, and to the extent that a detailed description of an element has been omitted, it may be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0160] Referring to FIG. **8A**, after the bottom dummy coupling insulation material layer **364** is formed on the bottom surface of a supporting dummy substrate **402** and a top dummy coupling insulation material layer **362** is formed on the top surface of the supporting dummy substrate **402**, the supporting dummy substrate **402** is disposed on the uppermost second semiconductor chip **200H** of the resultant structure of FIG. **5C**.

[0161] The second semiconductor chips **200** may have the second horizontal width **W2** and the second vertical height **H2**, and the supporting dummy substrate **402** may have the third horizontal width **W3** and the third vertical height **H3a**. In some embodiments, the third horizontal width **W3** may be less than each of the first and second horizontal widths **W1** and **W2**. In some embodiments, the third vertical height **H3a** may be substantially equal to each of the first and second heights **H1** and **H2**. The supporting dummy substrate **402** may be disposed on the uppermost second semiconductor chip **200H** using the edge of the uppermost second semiconductor chip **200H** as an align key.

[0162] Referring to FIGS. **8A** and **8B**, in a similar manner to that described above with reference to FIG. **5B**, the top chip coupling insulation material layer **302** and the bottom dummy coupling insulation material layer **364** combine to form the lowermost supporting coupling insulation layer **352L** disposed between the uppermost second semiconductor chip **200H** and the supporting dummy substrate **402**, and accordingly, the supporting dummy substrate **402** may be attached to the uppermost second semiconductor chip **200H**.

[0163] Referring to FIGS. **8C** and **8D**, another supporting dummy substrate **402**, on the bottom and top surfaces of which the bottom dummy coupling insulation material layer **364** and the top dummy coupling insulation material layer **362** are respectively formed, may be disposed on the supporting dummy substrate **402** such that a plurality of supporting dummy substrates **402** are stacked on the uppermost second semiconductor chip **200H**. In some embodiments, the top dummy coupling insulation material layer **362** might not be formed on the top surface of the uppermost supporting dummy substrate **402**.

[0164] In a similar manner to that described above with reference to FIG. 5B, the top dummy coupling insulation material layer 362 and the bottom dummy coupling insulation material layer 364, which are respectively formed on the top and bottom surfaces of respective supporting dummy substrates 402 facing each other, combine to form a supporting coupling insulation layer 352, and accordingly, a plurality of supporting dummy substrates 402 may be attached to the uppermost second semiconductor chip 200H.

[0165] Thereafter, the semiconductor package 1002 of FIG. 2A may be formed using the method described with reference to FIGS. 5F to 5I.

[0166] FIGS. 9A and 9B are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. For example, FIGS. 9A and 9B are cross-sectional views of a method of manufacturing the semiconductor package 1002b of FIG. 2C, and to the extent that a detailed description of an element has been omitted, it may be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0167] Referring to FIG. 9A, after the bottom dummy pads 328 and the bottom dummy coupling insulation material layer 364 at least partially surrounding the side surfaces of the bottom dummy pads 328 are formed on the bottom surface of a supporting dummy substrate 402 and the top dummy coupling insulation material layer 362 is formed on the top surface of the supporting dummy substrate 402, the supporting dummy substrate 402 is disposed on the uppermost second semiconductor chip 200H of the resultant structure of FIG. 5C.

[0168] Referring to FIGS. 9A and 9B, in a similar manner to that described above with reference to FIG. 5B, the top chip connection pads 322 and the bottom dummy pads 328 combine to form the thermal coupling pads 320a and the top chip coupling insulation material layer 302 and the bottom dummy coupling insulation material layer 364 combine to form the lowermost supporting coupling insulation layer 352L, disposed between the uppermost second semiconductor chip 200H and the supporting dummy substrate 402, and accordingly, the supporting dummy substrate 402 may be attached to the uppermost second semiconductor chip 200H.

[0169] Thereafter, the semiconductor package 1002b of FIG. 2C may be formed using the method described with reference to FIGS. 8C, 8D, and 5F to 5I.

[0170] FIGS. 10A to 10C are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. For example, FIGS. 10A to 10D are cross-sectional sectional views of a method of manufacturing the semiconductor package 1002d of FIG. 2E or the semiconductor package 1002f of FIG. 2G, and to the extent that a detailed description of an element has been omitted, it may be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0171] Referring to FIG. 10A, after a plurality of top dummy pads 326 are formed on the top surface of the supporting dummy substrate 402 in FIG. 9A such that the side surfaces of the top dummy pads 326 are at least partially surrounded by a top dummy coupling insulation material layer 362, the supporting dummy substrate 402 may be disposed on the uppermost second semiconductor chip 200H of the resultant structure of FIG. 5C. Thereafter, a plurality of top chip connection pads 322 and a plurality of bottom

dummy pads 328 combine to form a thermal coupling pads 320a and a top chip coupling insulation material layer 302 and a bottom dummy coupling insulation material layer 364 combine to form the lowermost supporting coupling insulation layer 352L, disposed between the uppermost second semiconductor chip 200H and the supporting dummy substrate 402, and accordingly, the supporting dummy substrate 402 may be attached to the uppermost second semiconductor chip 200H. For example, the top dummy pads 326 may include a material including copper (Cu).

[0172] Referring to FIG. 10B, a plurality of bottom dummy pads 328 and a bottom dummy coupling insulation material layer 364 at least partially surrounding the side surfaces of the bottom dummy pads 328 may be formed on the bottom surface of another supporting dummy substrate 402, and a plurality of top dummy pads 326 and a top dummy coupling insulation material layer 362 at least partially surrounding the side surfaces of the top dummy pads 326 may be formed on the top surface of the supporting dummy substrate 402. Thereafter, the supporting dummy substrate 402 may be disposed on the resultant structure of FIG. 10A, i.e., the lowermost supporting dummy substrate 402. Accordingly, a plurality of supporting dummy substrates 402 may be stacked on the uppermost second semiconductor chip 200H.

[0173] Referring to FIGS. 10B and 10C, the top dummy coupling insulation material layer 362 and the bottom dummy coupling insulation material layer 364, which are respectively formed on the top and bottom surfaces of the respective supporting dummy substrates 402 facing each other, combine to form a supporting coupling insulation layer 352, and the top dummy pads 326 and the bottom dummy pads 328, which are respectively formed on the top and bottom surfaces of the respective supporting dummy substrates 402 facing each other, combine to form a plurality of dummy coupling pads 320b. Accordingly, a plurality of supporting dummy substrates 402 may be attached to the uppermost second semiconductor chip 200H.

[0174] In some embodiments, as shown in FIG. 2E, a plurality of top dummy pads 326 and a top dummy coupling insulation material layer 362 might not be formed on the top surface of the uppermost supporting dummy substrate 402H. In some embodiments, as shown in FIG. 2G, a plurality of top dummy pads 326 may be formed on the top surface of the uppermost supporting dummy substrate 402H.

[0175] FIGS. 11A and 11B are cross-sectional views of a method of manufacturing a semiconductor package, according to embodiments. For example, FIGS. 11A and 11B are cross-sectional views of a method of manufacturing the semiconductor package 1004 of FIG. 3A, and to the extent that a detailed description of an element has been omitted, it may be assumed that the element is at least similar to corresponding elements that have been described elsewhere within the specification.

[0176] Referring to FIG. 11A, a plurality of supporting dummy substrates 402b may be attached to the uppermost second semiconductor chip 200H using the method described with reference to FIGS. 8A to 8D. The supporting dummy substrates 402b in FIG. 11A may be substantially the same as those in FIG. 7D.

[0177] Thereafter, a preliminary package molding layer 500P may be formed on the first semiconductor chip 100 to cover the top surface of the first semiconductor chip 100 and at least partially surrounding the side surfaces of the second

semiconductor chips **200** and the side and top surfaces of the supporting dummy substrates **402b**.

[0178] Referring to FIGS. 11A and 11B, the package molding layer **500** is formed by removing an upper portion of the preliminary package molding layer **500P**. The package molding layer **500** may be formed by removing an upper portion of the preliminary package molding layer **500P** using a grinding process.

[0179] In some embodiments, during the grinding process for the formation of the package molding layer **500**, an upper portion of the uppermost supporting dummy substrate **402bH** may also be removed. Accordingly, the fourth vertical height **H4** of the uppermost supporting dummy substrate **402bH** may be less than the third vertical height **H3a** of the other supporting dummy substrates **402b**.

[0180] Thereafter, the semiconductor package **1004** of FIG. 3A may be formed using the method described with reference to FIGS. 5G to 5I.

[0181] The semiconductor package **1006** of FIG. 4A may be formed using the method of FIGS. 8A to 8D such that the fourth vertical height **H4a** of the uppermost supporting dummy substrate **404H** among a plurality of supporting dummy substrates **404** is greater than the third vertical height **H3a** of the other supporting dummy substrates **404**. The semiconductor package **1006a** of FIG. 4B may be formed using the method of FIGS. 6A and 6B.

[0182] FIGS. 12 to 14 are conceptual cross-sectional views of a process of forming a coupling pad in a method of manufacturing a semiconductor package, according to an embodiment.

[0183] As shown in (a) of FIG. 12, a top chip connection pad **322** may have a different horizontal width than a bottom chip connection pad **324**. The conditions of a planarization process for the formation of the top chip connection pad **322** and the bottom chip connection pad **324** may be controlled such that the top surface of one of the top chip connection pad **322** and the bottom chip connection pad **324** is convex and the top surface of the other of the top chip connection pad **322** and the bottom chip connection pad **324** is concave. When heat at the first temperature is applied, a top chip coupling insulation material layer **302** may contact the bottom chip coupling insulation material layer **304**, as shown in (b) of FIG. 12. When heat at the second temperature is applied, the top chip connection pad **322** and the bottom chip connection pad **324** may expand to contact each other, as shown in (c) of FIG. 12, and may then be integrated into a coupling pad **320** through diffusion of metal atoms of the top and bottom chip connection pads **322** and **324**, as shown in (d) of FIG. 12.

[0184] The top chip coupling insulation material layer **302** and the bottom chip coupling insulation material layer **304** may be covalently bonded to each other and integrated into a chip coupling insulation layer **300**. A thermal coupling pad **320a** and a dummy coupling pad **320b** may be formed by a method that is substantially the same as the method of forming the coupling pad **320**, and supporting coupling insulation layers **350**, **350a**, **352**, and **352a** may be formed by a method that is substantially the same as the method of forming the chip coupling insulation layer **300**.

[0185] As shown in (a) of FIG. 13, the top chip connection pad **322** may have the same horizontal width as the bottom chip connection pad **324**. The conditions of a planarization process for the formation of the top chip connection pad **322** and the bottom chip connection pad **324** may be controlled

such that the top surface of one of the top chip connection pad **322** and the bottom chip connection pad **324** is convex and the top surface of the other of the top chip connection pad **322** and the bottom chip connection pad **324** is concave. When heat at the first temperature is applied, the top chip coupling insulation material layer **302** may contact the bottom chip coupling insulation material layer **304**, as shown in (b) of FIG. 13. When heat at the second temperature is applied, the top chip connection pad **322** and the bottom chip connection pad **324** may expand to contact each other, as shown in (c) of FIG. 13, and may then be integrated into the coupling pad **320** through the diffusion of metal atoms of the top and bottom chip connection pads **322** and **324**, as shown in (d) of FIG. 13.

[0186] The top chip coupling insulation material layer **302** and the bottom chip coupling insulation material layer **304** may be covalently bonded to each other and integrated into the chip coupling insulation layer **300**.

[0187] As shown in (a) of FIG. 14, the top surface of the top chip connection pad **322** may be coplanar with the top surface of the top chip coupling insulation material layer **302**, and the top surface of the bottom chip connection pad **324** may be coplanar with the top surface of the bottom chip coupling insulation material layer **304**. In some embodiments, the top chip connection pad **322** may have the same horizontal width as the bottom chip connection pad **324**. In some embodiments, the top chip connection pad **322** may have a different horizontal width than the bottom chip connection pad **324**. When heat at the first temperature is applied, the top chip coupling insulation material layer **302** may contact the bottom chip coupling insulation material layer **304**, as shown in (b) of FIG. 14. When heat at the second temperature is applied, the top chip connection pad **322** and the bottom chip connection pad **324** may be diffusion bonded to each other through diffusion of metal atoms of the top and bottom chip connection pads **322** and **324** and may thus be integrated into the coupling pad **320**, as shown in (c) of FIG. 14.

[0188] The top chip coupling insulation material layer **302** and the bottom chip coupling insulation material layer **304** may be covalently bonded to each other and integrated into the chip coupling insulation layer **300**.

[0189] FIG. 15 is a conceptual diagram of the shape of a semiconductor package according to an embodiment.

[0190] Referring to FIG. 15, a semiconductor package **1** may include the first semiconductor chip **100**, a plurality of second semiconductor chips **200** stacked on the first semiconductor chip **100**, and a supporting dummy substrate **400** attached to the uppermost second semiconductor chip **200H**. A chip coupling insulation layer **300** may be disposed between the first semiconductor chip **100** and the lowermost second semiconductor chip **200L** and a chip coupling insulation layer **300** may be disposed between two adjacent second semiconductor chips **200**. The supporting coupling insulation layer **350** may be disposed between the uppermost second semiconductor chip **200H** and the supporting dummy substrate **400**.

[0191] Instead of including the supporting dummy substrate **400**, the semiconductor package **1** may include the supporting dummy substrate **400a** in FIGS. 1B and 1D, a plurality of supporting dummy substrates **402** in FIGS. 2A, 2C, 2E, and 2G, a plurality of supporting dummy substrates **402a** in FIGS. 2B, 2D, 2F, and 2H, a plurality of supporting dummy substrates **402b** in FIGS. 3A, 3C, and 3E, a plurality

of supporting dummy substrates **402c** in FIGS. 3B, 3D, and 3F, a plurality of supporting dummy substrates **404** in FIGS. 4A, 4C, 4E, and 4G, or a plurality of supporting dummy substrates **404a** in FIGS. 4B, 4D, 4F, and 4H; and, instead of the supporting coupling insulation layer **350**, the supporting coupling insulation layer **350a** in FIGS. 1B and 1D, the supporting coupling insulation layer **352** in FIGS. 2A, 2C, 2E, 2G, 3A, 3C, 3E, 4A, 4C, 4E, and 4G or the supporting coupling insulation layer **352a** in FIGS. 2B, 2D, 2F, 2H, 3B, 3D, 3F, 4B, 4D, 4F, and 4H.

[0192] In the semiconductor package **1**, the first semiconductor chip **100** includes a plurality of first through electrodes **120**, each of all the second semiconductor chips **200** including the uppermost second semiconductor chip **200H** includes a plurality of second through electrodes **220** as shown in FIGS. 1A to 4H, a plurality of coupling pads **320** are arranged among the first semiconductor chip **100** and the second semiconductor chips **200**, and a plurality of thermal coupling pads **320a** similar to the coupling pads **320** are disposed between the uppermost second semiconductor chip **200H** and the supporting dummy substrate **400**. Accordingly, even when warpage occurs in the first semiconductor chip **100** and the second semiconductor chips **200**, the first semiconductor chip **100** and the second semiconductor chips **200** may have the same warpage shape bulging in one direction. For example, the first semiconductor chip **100** and the second semiconductor chips **200** may have substantially the same warpage shape bulging upwards, i.e., in a direction from the first semiconductor chip **100** to the second semiconductor chips **200**. Accordingly, connection reliability between the first semiconductor chip **100** and the second semiconductor chips **200** may be increased.

[0193] While the inventive concept has been particularly shown and described with reference to embodiments thereof, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure.

1-20. (canceled)

21. A semiconductor package comprising:

- a first semiconductor chip including a first substrate and a plurality of first through electrodes disposed in the first substrate;
- a first insulation layer disposed on the first semiconductor chip;
- a plurality of first coupling pads disposed in the first insulation layer;
- a second semiconductor chip disposed on the first insulation layer, and including a second substrate and a plurality of second through electrodes disposed in the second substrate;
- a second insulation layer disposed on the second semiconductor chip;
- a plurality of second coupling pads disposed in the second insulation layer;
- a third semiconductor chip disposed on the second insulation layer, and including a third substrate and a plurality of third through electrodes disposed in the third substrate;
- a third insulation layer disposed on the third semiconductor chip;
- a plurality of connection pads disposed in the third insulation layer; and
- a dummy substrate disposed on the third insulation layer,

wherein a vertical thickness of the second semiconductor chip is substantially the same as a vertical thickness of the third semiconductor chip.

22. The semiconductor package of claim 21, wherein the plurality of connection pads contact the plurality of third through electrodes.

23. The semiconductor package of claim 21, wherein the first semiconductor chip is a high-bandwidth memory (HBM) controller, and

each of the second semiconductor chip and the third semiconductor chip is a dynamic random access memory (DRAM).

24. The semiconductor package of claim 21, wherein the plurality of connection pads contact the dummy substrate.

25. The semiconductor package of claim 21, wherein the plurality of connection pads are spaced apart from the dummy substrate.

26. The semiconductor package of claim 21, wherein a horizontal width of the first semiconductor chip is greater than a horizontal width of the second semiconductor chip.

27. The semiconductor package of claim 21, wherein a horizontal width of the third semiconductor chip is different from a horizontal width of the dummy substrate.

28. The semiconductor package of claim 21, wherein each of the plurality of first coupling pads is formed by combining a first chip connection pad and a second chip connection pad.

29. The semiconductor package of claim 21, wherein a vertical thickness of the dummy substrate is greater than the vertical thickness of the third semiconductor chip.

30. A semiconductor package comprising:

- a first semiconductor chip including a first substrate and a plurality of first through electrodes disposed in the first substrate;
- a first insulation layer disposed on the first semiconductor chip;
- a plurality of first coupling pads disposed in the first insulation layer;
- a second semiconductor chip disposed on the first insulation layer, and including a second substrate and a plurality of second through electrodes disposed in the second substrate;
- a second insulation layer disposed on the second semiconductor chip;
- a plurality of second coupling pads disposed in the second insulation layer;
- a third semiconductor chip disposed on the second insulation layer, and including a third substrate and a plurality of third through electrodes disposed in the third substrate;
- a third insulation layer disposed on the third semiconductor chip;
- a plurality of connection pads disposed in the third insulation layer;
- a first dummy substrate disposed on the third insulation layer;
- a fourth insulation layer disposed on the first dummy substrate; and
- a second dummy substrate disposed on the fourth insulation layer,

wherein a vertical height of the second semiconductor chip is substantially the same as a vertical height of the third semiconductor chip, and

the first semiconductor chip is a controller, and each of the second semiconductor chip and the third semiconductor chip is a memory.

31. The semiconductor package of claim **30**, wherein each of the plurality of first coupling pads is formed by combining a first chip connection pad and a second chip connection pad.

32. The semiconductor package of claim **30**, wherein the second insulation layer and the plurality of second coupling pads are formed by a hybrid bonding.

33. The semiconductor package of claim **30**, wherein the second insulation layer includes SiO, SiN, SiCN, SiCO or a polymeric material.

34. The semiconductor package of claim **30**, wherein the first semiconductor chip is a high-bandwidth memory (HBM) controller, and

each of the second semiconductor chip and the third semiconductor chip is a dynamic random access memory (DRAM).

35. The semiconductor package of claim **30**, wherein a vertical height of the first dummy substrate is different from the vertical height of the third semiconductor chip.

36. The semiconductor package of claim **30**, wherein a vertical height of the first dummy substrate is substantially the same as the vertical height of the third semiconductor chip.

37. A semiconductor package comprising:

a redistribution layer;

a first semiconductor chip disposed on the redistribution layer, and including a first substrate and a plurality of first through electrodes disposed in the first substrate;

a first insulation layer disposed on the first semiconductor chip;

a plurality of first coupling pads disposed in the first insulation layer;

a second semiconductor chip disposed on the first insulation layer, and including a second substrate and a plurality of second through electrodes disposed in the second substrate;

a second insulation layer disposed on the second semiconductor chip;

a plurality of second coupling pads disposed in the second insulation layer;

a third semiconductor chip disposed on the second insulation layer, and including a third substrate and a plurality of third through electrodes disposed in the third substrate;

a third insulation layer disposed on the third semiconductor chip;

a plurality of connection pads disposed in the third insulation layer;

a dummy substrate disposed on the third insulation layer;

a first wiring structure disposed in the first semiconductor chip; and

a second wiring structure disposed in the second semiconductor chip,

wherein a vertical thickness of the second semiconductor chip is substantially the same as a vertical thickness of the third semiconductor chip.

38. The semiconductor package of claim **37**, wherein the redistribution layer includes a plurality of package redistribution line patterns, a plurality of package redistribution vias and a package redistribution insulation layer,

the first wiring structure includes a plurality of first wiring patterns, a plurality of first wiring vias connected to the plurality of first wiring patterns, and a first inter-wiring insulation layer at least partially surrounding the plurality of first wiring patterns and the plurality of first wiring vias, and

the second wiring structure includes a plurality of second wiring patterns, a plurality of second wiring vias connected to the plurality of second wiring patterns, and a second inter-wiring insulation layer at least partially surrounding the plurality of second wiring patterns and the plurality of second wiring vias.

39. The semiconductor package of claim **37**, wherein each of the plurality of first coupling pads is formed by combining a first chip connection pad and a second chip connection pad.

40. The semiconductor package of claim **37**, wherein a vertical thickness of the dummy substrate is different from the vertical thickness of the third semiconductor chip.

* * * * *